HOMOTOPICAL ADDITIVITY OF EQUIVARIANT LITTLE DISK OPERADS AND CONFIGURATION SPACES

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Abstract. Given a finite group G, a pair of orthogonal G-representations V and W, and a G-symmetric monoidal ∞-category $\mathcal{C}^{\otimes}$, we prove that the natural G-symmetric monoidal forgetful functor

$$\underline{\mathrm{Alg}}^{\otimes}_{\mathbb{E}_{V\oplus W}}(\mathcal{C}) \to \underline{\mathrm{Alg}}^{\otimes}_{\mathbb{E}_V}\underline{\mathrm{Alg}}^{\otimes}_{\mathbb{E}_W}(\mathcal{C})$$

is an equivalence. Moreover, we extend these results to equivariant little disk operads with (smooth) equivariant tangential structures. As an application, we lift Dwyer-Hess-Knudsen additivity of configuration spaces to the embedding right $\mathbb{E}_\Theta$-modules $E^\Theta_{(-)}$ of Θ-framed G-manifolds:

$$E^\Theta_M \boxtimes E^\Psi_N \simeq E^{\Theta\oplus\Psi}_{M\times N}.$$

As another application of our general techniques, we introduce an equivariant version of Voronov's Swiss cheese operads with faces up to codimension k, called $\mathbb{E}_{V,k}$, and construct equivalences

$$\mathbb{E}_{V,k}\otimes\mathbb{E}_{W,\ell} \simeq \mathbb{E}_{V\oplus W,k+\ell} \simeq \mathbb{E}_{V\oplus W}\otimes \mathrm{LMod}^{\otimes k+\ell}.$$

This affirms the common folklore claim that $\mathbb{E}_{n,1} \simeq \mathrm{SC}_{n+1}$ encodes (left) $\mathbb{E}_1$*-modules over* $\mathbb{E}_{n+1}$*-algebras.*

Contents

Introduction **2**
1 Morita theory and Boardman-Vogt tensor products **7**
1.1 Recollections on equivariant operads . . . 8
1.2 Weak approximations and reflective localizations . . . 16
1.3 Wreath products and the bookkeeping theorem . . . 19
2 Embeddings of orthogonal representations **25**
2.1 Rectilinear little V-polydisks . . . 25
2.2 Weiss covers and factorization . . . 29
2.3 Germs and tangential structures . . . 32
3 Little disk and prefactorization algebras and the additivity theorems **37**
3.1 Additivity of little disk G-operads . . . 37
3.2 Additivity with tangential structures . . . 41
3.3 Additivity for embedding calculus right-modules . . . 46
3.4 Additivity with corners . . . 48
References **55**

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Date: October 30, 2025.

Introduction

In [May72], the study of homotopy-coherent algebraic structures crystallized into a definition of *operads and algebras*, encapsulating Boardman-Vogt's little cubes action on iterated loop spaces as a factorization

$$\begin{array}{ccccc} & & & & \mathrm{Alg}_{\mathbb{E}_n}(\mathcal{S}_*) \\ & \overset{\exists}{\nearrow} & & & \downarrow U \\ \mathcal{S}_* & \longrightarrow & \Omega^n & \longrightarrow & \mathcal{S}_* \end{array}$$

Moreover, the *recognition principle* of [BV73; May72; Seg73] essentially proved that the restriction of Ω^n to $(n+1)$-connected pointed spaces is fully faithful with essential image the connected $\mathbb{E}_n$-spaces (see [HA] for this formulation). This is evidently additive in n; indeed, after Stasheff's $\mathbb{E}_1 = \mathbb{A}_\infty$-recognition principle [Sta63], this recognition principle is easily seen to be *equivalent* to the fact that the forgetful functor

$$\mathrm{Alg}_{\mathbb{E}_{n+m}}(\mathcal{S}_{*,\geq 1}) \xrightarrow{\sim} \mathrm{Alg}_{\mathbb{E}_n}\mathrm{Alg}^{\otimes}_{\mathbb{E}_m}(\mathcal{S}_{*,\geq 1})$$

is an equivalence for all $n, m \in \mathbb{N}$, where $\mathrm{Alg}^{\otimes}_{\mathbb{E}_n}(-)$ denotes the *pointwise* symmetric monoidal structure.

Naturally, one wonders about additivity for algebras in more general symmetric monoidal ∞-categories. If we 1-categorically model $\mathbb{E}_n^{\otimes}$ via the *little cubes topological operad* $C_n \in \mathsf{AlgFun}^{\circ}(\coprod_{n\in\mathbb{N}} B\Sigma_n, \mathsf{Top})$, Boardman and Vogt [BV73] constructed a point-set tensor product on topological operads with mapping property

$$\mathrm{Alg}^{1-\mathrm{cat}}_{\mathcal{O}\otimes\mathcal{P}}(\mathcal{C}) \xrightarrow{\sim} \mathrm{Alg}^{1-\mathrm{cat}}_{\mathcal{O}}\mathrm{Alg}^{1-\mathrm{cat},\otimes}_{\mathcal{P}}(\mathcal{C})$$

for a pointwise symmetric monoidal structure on $\mathrm{Alg}_{\mathcal{P}}(\mathcal{C})$. The evident operadic conjecture then demands an equivalence $C_n \otimes C_m \simeq C_{n+m}$; this was supplied by Dunn [Dun88].

Unfortunately, in making the corresponding additivity statement for algebras homotopical, we encounter difficulty: the point-set Boardman-Vogt tensor product is not known to admit a left derived functor, let alone satisfy a derived mapping property. One potential remedy in the 1-categorical setting (due to [FV13]) begins by verifying by hand that $\otimes$ is homotopical on weak equivalences between cofibrant operads which are additionally equivalent to C_k, but to the authors knowledge, further headway has not been made.

Instead, Lurie [Lur09] defined a manifestly homotopical tensor product $\overset{\mathrm{BV}}{\otimes}$ on the ∞-category of operads Op directly in terms of its mapping property, then constructed equivalences $\mathbb{E}_n^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathbb{E}_m^{\otimes} \simeq \mathbb{E}_{n+m}^{\otimes}$ and

$$\mathrm{Alg}_{\mathbb{E}_{n+m}}(\mathcal{C}) \xrightarrow{\sim} \mathrm{Alg}_{\mathbb{E}_n}\mathrm{Alg}^{\otimes}_{\mathbb{E}_m}(\mathcal{C}),$$

the latter natural in the symmetric monoidal ∞-category $\mathcal{C}$. We'll call this result *Dunn-Lurie additivity*.

In the years since, Dunn-Lurie additivity has become a basic tool of homotopy theory and algebraic K-theory; for instance, it underlies the $\mathbb{E}_{n-1}$-monoidal structure on left modules over an $\mathbb{E}_n$-algebra, as well as the $\mathbb{E}_{n-1}$-structure on the value of various $\mathbb{E}_1$-invariants on $\mathbb{E}_n$-algebras, such as topological Hochschild homology, algebraic K-theory, and various approximations therebetween. Crucially, Dunn-Lurie additivity is an indispensable tool in studying $\mathbb{E}_n$-algebras, as it often inductively reduces questions about $\mathbb{E}_n^{\otimes} \simeq \mathbb{E}_1^{\otimes n}$ to questions about $\mathbb{E}_1^{\otimes}$, whose algebras are simply presented as associative algebra objects.[1]

We are interested a version of this result which is genuine-equivariant over a finite group G; the definition of $\mathbb{E}_n^{\otimes}$ naturally extends to a *little V-disks G-operad* $\mathbb{E}_V^{\otimes}$ for any orthogonal G-representation V, which satisfy an approximation theorem and recognition principle for S^V-loop spaces [GM11; Hau80; Jur25; RS00; SW03]. Dunn's result has been lifted to G-operads in [Szc24] for a tensor product of operads in $s\mathsf{Set}_G$ with no known homotopical properties.

Now, Lurie's theory of *∞-operads and Boardman-Vogt tensor products* was extended to the G-equivariant setting in [NS22; Ste25b; Ste25d], whose notation we adopt. We show homotopical additivity in this setting.

Theorem A. *The natural G-operad maps $\mathbb{E}_V^{\otimes}, \mathbb{E}_W^{\otimes} \to \mathbb{E}_{V\oplus W}^{\otimes}$ extend to an equivalence*

$$\mathbb{E}_V^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathbb{E}_W^{\otimes} \xrightarrow{\sim} \mathbb{E}_{V\oplus W}^{\otimes};$$

in particular, for all G-symmetric monoidal ∞-categories $\mathcal{C}$, the forgetful functor yields an equivalence

$$\mathrm{Alg}_{\mathbb{E}_{V\oplus W}}(\mathcal{C}) \xrightarrow{\sim} \mathrm{Alg}_{\mathbb{E}_V}\underline{\mathrm{Alg}}^{\otimes}_{\mathbb{E}_W}(\mathcal{C}).$$

[1] This can be interpreted ∞-categorically or via Schwede-Shipley's right-transferred model structure on strict associative algebra structures; see e.g. [HA, Thm. 4.1.8.4].

For instance, by results of [Ste25d], this constructs an $\mathbb{E}_V$-algebra structure on the Real topological Hochschild homology and Real cyclic homology of an $\mathbb{E}_{V\oplus\sigma}$-algebra and a natural $\mathbb{E}_V$-monoidal structure on right-modules over an $\mathbb{E}_{V\oplus 1}$-algebra, as well as strongly suggesting that this is the universal such structure.

In general, when $G = C_2$, this reduces questions about $\mathbb{E}_V$-algebras into questions about $\mathbb{E}_1$-algebras and $\mathbb{E}_\sigma$-algebras, which are relatively well understood [Hil22], as their structure spaces are discrete (indeed, when V is 0-dimensional, its S-ary structure space is either empty or an $\mathrm{Aut}_H S$-torsor [Ste25c]).

Now, we describe a variation of Theorem A in the following topological settings.

Definition. Let $B_G O(n)$ be the classifying G-space for equivariant (orthogonal) vector bundles. A *G-tangential structure* is a map $\Theta\colon B_\Theta \to B_G O(n)$. A *$\Theta$-framed G-manifold* is a smooth n-manifold with G-action together with an (equivariant) lift of the classifying map for its tangent bundle along Θ:

$$\begin{array}{ccc} & & B_\Theta \\ & \overset{\tau^\Theta M}{\nearrow} & \downarrow{\scriptstyle \Theta} \\ M & \xrightarrow[\tau M]{} & B_G O(n) \end{array}$$

A Θ-framed embedding $M \hookrightarrow N$ is the data of a smooth equivariant embedding ι as well as a homotopy $\tau^\Theta N \circ \iota \sim \tau^\Theta M$. In the case that $B_\Theta = *$, so that $\Theta = \overline{V}$ classifies a representation V, we refer to Θ-framed G-manifolds as *V-framed G-manifolds*, and similarly for V-framed embeddings. ◁

Examples of this include the embedding categories of *oriented* G-manifolds and G-manifolds with equivariant almost-complex structures.

Generalizing [Lur09], we will define a G-operad of *Θ-framed little disk embeddings* $\mathbb{E}^\otimes_\Theta$, which will turn out to embed as a full G-suboperad of Horev's G-symmetric monoidal ∞-category of Θ-framed G-manifolds [Hor19] and Θ-framed embeddings spanned by those Θ-framed G-manifolds whose underlying G-manifold is equivariantly diffeomorphic to a (unit disk in an) orthogonal G-representation.

This is essentially characterized by a fully faithful embedding of its G-symmetric monoidal envelope

$$\mathrm{Env}_G(\mathbb{E}_\Theta)^\otimes \simeq \underline{\mathrm{Disk}}_G^{\Theta,\sqcup} \hookrightarrow \underline{\mathrm{Mfld}}_G^{\Theta,\sqcup},$$

with image the total spaces of Θ-framed equivariant vector bundles over finite G-sets (i.e. equivariant disjoint unions of representation disks); the *arity* G-symmetric monoidal functor $\underline{\mathrm{Disk}}_G^{\Theta,\sqcup} \to \underline{\mathbb{F}}_G^\sqcup = \mathrm{Env}_G(\mathrm{Comm}_G)^\otimes$ takes a bundle $D \to S$ to S. As an upshot, restriction along $\mathbb{E}^\otimes_\Theta \hookrightarrow \underline{\mathrm{Disk}}_G^{\Theta,\sqcup}$ yields a natural equivalence

$$\mathrm{Alg}_{\mathbb{E}_\Theta}(\mathcal{C}) \simeq \mathrm{Fun}^\otimes_G\left(\underline{\mathrm{Disk}}_G^{\Theta,\sqcup}, \mathcal{C}^\otimes\right).$$

Now, the disintegration and assembly procedure of [Ste25d] constructs another natural equivalence

$$\underline{\mathrm{Alg}}_{\mathbb{E}_\Theta}(\mathcal{C}) \xrightarrow{\sim} \underline{\lim}_{x\in B_\Theta} \underline{\mathrm{Alg}}_{\mathbb{E}_{\Theta(x)}}(\mathcal{C}).$$

Now, given a pair of G-tangential structures $\Theta\colon B_\Theta \to B_G O(k)$ and $\Psi\colon B_\Psi \to B_G O(k')$ we write

$$\Theta\oplus\Psi\colon B_\Theta \times B_\Psi \xrightarrow{\Theta\times\Psi} B_G O(k)\times B_G O(k') \xrightarrow{\oplus} B_G O(k+k')$$

for the *direct sum structure*. We bootstrap Theorem A to an additivity phenomenon for this structure.

Corollary B. *The natural G-operad maps $\mathbb{E}^\otimes_\Theta, \mathbb{E}^\otimes_\Psi \to \mathbb{E}^\otimes_{\Theta\oplus\Psi}$ extend to an equivalence*

$$\mathbb{E}^\otimes_\Theta \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_\Psi \xrightarrow{\sim} \mathbb{E}^\otimes_{\Theta\oplus\Psi}.$$

In particular, the box product of G-symmetric monoidal ∞-categories acts additively on disks:

$$\times\colon \underline{\mathrm{Disk}}_G^\Theta \square \underline{\mathrm{Disk}}_G^\Psi \xrightarrow{\sim} \underline{\mathrm{Disk}}_G^{\Theta\oplus\Psi}.$$

For an application, in [DHLSW25] the (nonequivariant) equivalence $\mathbb{E}^\otimes_{BU(1)} \overset{\mathrm{BV}}{\otimes} \mathbb{E}_n \simeq \mathbb{E}^\otimes_{BU(1)+n}$ was explicitly used to construct a natural $\mathbb{E}_n$-S^1-equivariant map $\mathrm{THH}(A) \to A$ when A is an $\mathbb{E}_{\mathrm{BU}(1)+n}$-algebra, therein attributed to Horev; the author expects their techniques and Corollary B to lift *mutatis mutandis* to a

construction of a C_2-twisted (Borel) $\mathbb{T}$-equivariant $\mathbb{E}_V$-map $\mathrm{THR}(A) \to A$ whenever A is an $\mathbb{E}_{V\times B_G\mathbb{T}_\sigma}$-algebra.[2] We leave this open for simplicity's sake.

For another application, this allows for a G-∞-categorical lift of Dwyer-Hess-Knudsen-Szczesny's (potentially still conjectural) result on structured configuration spaces in products [DHK18; Szc24]. In particular, following Krannich-Kupers [KK24], we may *define* right-modules over a G-operad $\mathcal{O}$ as G-presheaves on its G-symmetric monoidal envelope $\mathrm{Env}_G(\mathcal{O})^\otimes$, in which case the restricted Yoneda embedding yields a Θ-*framed embedding calculus module functor*

$$E^\Theta_{(-)}\colon \mathrm{Mfld}^\Theta_G \xrightarrow{\text{よ}} \mathrm{Psh}^G\left(\underline{\mathrm{Mfld}}^\Theta_G\right) \xrightarrow{\iota^*} \mathrm{Psh}^G\left(\underline{\mathrm{Disk}}^\Theta_G\right) \simeq \mathrm{RMod}_{\mathbb{E}_\Theta}.$$

Now, following Dwyer-Hess-Knudsen [DHK18], we define a *lifted Boardman-Vogt tensor product* functor by left Kan-extending the external tensor product of presheaves along the Box product structure map:

$$\boxtimes\colon \mathrm{RMod}_{\mathcal{O}}\times\mathrm{RMod}_{\mathcal{P}} \xrightarrow{\boxtimes} \mathrm{Psh}^G\left(\mathrm{Env}_G(\mathcal{O})\times\mathrm{Env}_G(\mathcal{P})\right) \xrightarrow{\ell_!} \mathrm{Psh}^G\left(\mathrm{Env}_G(\mathcal{O})\square\mathrm{Env}_G(\mathcal{P})\right) \simeq \mathrm{RMod}_{\mathcal{O}\overset{\mathrm{BV}}{\otimes}\mathcal{P}}$$

We verify an equivariant lift of Dwyer-Hess-Knudsen's (conjectural) theorem.

Corollary C. *Under the equivalence* $\mathbb{E}^\otimes_\Theta \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_\Psi \simeq \mathbb{E}^\otimes_{\Theta\oplus\Psi}$*, embedding calculus modules are additive:*

$$E^\Theta_M \boxtimes E^\Psi_N \simeq E^{\Theta\oplus\Psi}_{M\times N}.$$

Generalizing Voronov [Vor98], we go on to introduce a (V,d)-*swiss cheese* G-*operad* $\mathbb{E}^\otimes_{V,d}$; topologically this represents embeddings of $V\oplus\mathbb{R}^d$-framed G-manifolds with trivially equivariant corners of codimension $\le d$), and algebraically this is meant to represent diagrams of little disk algebras and modules; for instance, an $\mathbb{E}^\otimes_{V,1}$-algebra may be thought of as a pair of an $\mathbb{E}_{V+1}$-algebra A as well as an $\mathbb{E}_V$-algebra M in the category of right A-modules. To make this precise, we extend Theorem A to swiss cheese G-operads.

Theorem D. *The natural maps* $\mathbb{E}^\otimes_{V,d_V}, \mathbb{E}^\otimes_{W,d_W} \to \mathbb{E}^\otimes_{V\oplus W, d_V+d_W}$ *extend to equivalences*

$$\mathbb{E}^\otimes_{V,d_V} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_{W,d_W} \simeq \mathbb{E}^\otimes_{V\oplus W,d_V+d_W} \simeq \mathbb{E}^\otimes_{V\oplus W} \overset{\mathrm{BV}}{\otimes} \mathrm{RMod}^{\otimes d_V+d_W}.$$

In the case $G=e$, this is proved independently by Carmona-Svraka in [Car].

The strategy. One key insight to our strategy is that $\overset{\mathrm{BV}}{\otimes}$ is homotopical, so we may custom-fit topological models for the little V-disks operad to this problem, and prove *independently* that our custom-fit topological models are equivalent to $\mathbb{E}^\otimes_V$.

To do so, we will choose subspaces $E^H_V(S)\subset \mathrm{Emb}^H(S\cdot V, V)$ under the strong C^∞-topology whose image map $\mathsf{im}\colon E^H_V(S)\to \mathrm{Open}^H(V)$ to the G-invariant open subsets of V is particularly well behaved. Writing $\mathcal{U}=\mathsf{im}(E^H_V(S))$, we will use this to define topological models for the $\mathcal{U}$-*prefactorization and little* V-*disk* G-operads $\mathbb{P}^t_\mathcal{U}\to\mathbb{E}^t_\mathcal{U}$ as well as Dunn additivity equivalences on both levels. This will require the following:

(i) $E^H_V(S)$ will need to be suitably closed under restriction and composition in order to define G-operads.
(ii) $E^H_V(S)$ will need to be closed under "external products" to define the additivity map φ for $\mathbb{E}^t_\mathcal{U}$.
(iii) The center map $E^H_V(S)\to \mathrm{Conf}^H_S(V)$ must be an equivalence in order to compare with little V-disks.
(iv) We will want im to be injective in order to adequately relate composition and factorization.
(v) $\mathcal{U}$ must be suitably nice (as we will explain, an *equidiameterizable* G-*cover*) so that $\mathbb{P}^t_\mathcal{U}\to\mathbb{E}^t_\mathcal{U}$ is a weak approximation, embedding $\mathbb{E}_\mathcal{U}$-algebras as locally constant $\mathbb{P}_\mathcal{U}$-algebras.
(vi) $\mathcal{U}$ will need to satisfy a *factorization decomposability property* with respect to φ in order to prove prefactorization additivity.
(vii) $E^H_V(S)$ must arise as the homotopy fiber of the *transformation map* $\mathrm{Emb}^H(S\cdot V,V)\to\prod_{[H/K]\subset S} O^K(V)$ in order to realize the additivity map as the disintegration of a product functor on smooth G-manifolds.

[2] The only structural aspect of factorization homology which is missing to replicate this argument in the equivariant setting is the *trivial interval Fubini property* $\int_M(-)\simeq\int_{M\times(0,1)}(-)$. But indeed this follows in the relevant case $M=\rho_{C_2}-\{0\}$ by $\otimes$-excision for the "thickening" of the usual C_2-collar decomposition $S^\sigma\simeq D^1\sqcup_{\mathrm{Ind}_e^{C_2}D^1}D^1$:

$$\int_{\rho_{C_2}-\{0\}}A\simeq\int_{I\times D^1}A\otimes_{\int_{I\times\mathrm{Ind}_e^{C_2}A}}\int_{I\times D^1}A\simeq A\otimes_{N_e^{C_2}A}A\simeq\mathrm{THR}(A).$$

To fulfill Condition (iv), we chose a *sub-affine* model; to fulfill Conditions (iii) and (vii), we will have to chose affine transformations whose underlying linear transformation lies in some fixed contractible subgroup. Unfortunately, to fulfill Condition (ii), the center of $\mathrm{GL}(V)$ will not suffice, so we will need to choose a decomposition of our representation and a corresponding torus; that is, we impose *rectilinearity*.

Now, even under rectilinearity, *little disks* will not satisfy Condition (vi) (see Fig. 1). We can't generally choose *cubes* as we can't generally express orthogonal representations as products of 1-dimensional subrepresentations; we choose $E_V^H(S)$ as the space of *rectilinear embeddings of S-tuples of little representation polydisks*, which we will see satisfies these conditions. Using this, we construct a zigzag of equivalences

$$E_V^H(S) \simeq \mathrm{Emb}^{V,\text{equidiameter}}\left(S \cdot D^V, D^V\right)$$

compatibly with G-operadic composition, ensuring we've modelled $\mathbb{E}_V^\otimes$.

From here, we're forced to use strategies specific to a model for G-operads; writing

$$\mathrm{POp}_G := \mathrm{Cat}^{\text{Backwards–cocart}}_{/\mathrm{Span}_{f.i.}(\mathbb{F}_G)}$$

for the ∞-category of *G-preoperads*, we have a collection of functors:

$$\mathrm{Cat} \xleftarrow{\mathrm{Mon}_{(-)}(\mathcal{S})} \mathrm{Op}_G \underset{\perp}{\overset{L_{\mathrm{Op}_G}}{\rightleftarrows}} \mathrm{POp}_G \xrightarrow{\mathrm{Tot}} \mathrm{AlgPatt} \xrightarrow{\mathrm{Seg}_{(-)}(\mathcal{S})} \mathrm{Cat}$$

In [Ste25d] we constructed a natural equivalence $\mathrm{Mon}_{L_{\mathrm{Op}_G}\mathcal{O}}(\mathcal{S}) \simeq \mathrm{Seg}_{\mathrm{Tot}\mathcal{O}^\otimes}(\mathcal{S})$; since $\mathrm{Mon}_{(-)}(\mathcal{S})$ detects L_{Op_G}-equivalences [Ste25b], a map of G-preoperads is an L_{Op_G}-equivalence if and only if its map of patterns is a *Morita equivalence*, which can be verified using traditional higher category theory.

The heart of our strategy reduces to the Morita theory of G-preoperads for *prefactorization algebras*: given a well-chosen G-cover $\mathcal{U} \subset \underline{\mathsf{Open}}(V)$, we define a G-preoperad $\mathbb{P}_\mathcal{U}^\otimes$ for *fine $\mathcal{U}$-prefactorization algebras*, as well as a comparison map $\alpha_\mathcal{U} \colon \mathbb{P}_\mathcal{U}^\otimes \to \mathbb{E}_V^\otimes$. In Proposition 3.21, we show in good cases that $\alpha_\mathcal{U}$ is a *weak approximation*, i.e. it fully faithfully embeds $\mathbb{E}_V$-algebras as $\mathbb{P}_\mathcal{U}$-algebras with constant color G-functor $\mathcal{U} \to \mathcal{C}$ (called *locally constant* $\mathbb{P}_\mathcal{U}$-algebras); for instance, this holds for the collection of *all* equivariant disjoint unions of rectilinear V-polydisks, in which case we write $\mathbb{P}_V^\otimes := \mathbb{P}_\mathcal{U}^\otimes$.

What's left is to construct a convenient G-preoperad model for the Boardman-Vogt tensor product, construct an additivity map, and verify that it is a Morita equivalence.

For the former, we define a *wreath product* G-preoperad connected by a zigzag of natural Morita equivalences $\mathcal{O}^\otimes \overset{\mathrm{BV}}{\otimes} \mathcal{P}^\otimes \leftarrow \mathcal{O}^\otimes \odot \mathcal{P}^\otimes \to \mathcal{O}^\otimes \wr \mathcal{P}^\otimes$; this has the crucially convenient property that $\mathcal{O}^\otimes \wr \mathcal{P}^\otimes$ has n-truncated structure spaces whenever $\mathcal{O}^\otimes$ and $\mathcal{P}^\otimes$ do.

Using this description, we explicitly construct an additivity map $\varphi_\mathbb{E} \colon \mathbb{E}_V^\otimes \wr \mathbb{E}_W^\otimes \to \mathbb{E}_{V\oplus W}^\otimes$ and lift this to an additivity map $\varphi_\mathbb{P} \colon \mathbb{P}_V^\otimes \wr \mathbb{P}_W^\otimes \to \mathbb{P}_{V|W}^\otimes$; here, $\mathbb{P}_{V|W}^\otimes$ is the fine prefactorization G-preoperad for rectilinear $V \oplus W$-multi-polydisks which "lie in the image of $\varphi_\mathbb{E}$," i.e. a variant of Dunn's *decomposable little cubes*. We acquire a commutative diagram:

$$\begin{array}{ccccccc}
\mathbb{P}_V^\otimes \overset{\mathrm{BV}}{\otimes} \mathbb{P}_W^\otimes & \xleftarrow{L_\mathbb{P}} & \mathbb{P}_V^\otimes \odot \mathbb{P}_W^\otimes & \xrightarrow{M_\mathbb{P}} & \mathbb{P}_V^\otimes \wr \mathbb{P}_W^\otimes & \xrightarrow{\widetilde{\varphi}_\mathbb{P}} & \mathbb{P}_{V|W}^\otimes \\
\downarrow{\scriptstyle \alpha_V \otimes \alpha_W} & & \downarrow{\scriptstyle \alpha_V \times \alpha_W} & & \downarrow{\scriptstyle \alpha_V \wr \alpha_W} & & \downarrow{\scriptstyle \alpha_{V|W}} \\
\mathbb{E}_V^\otimes \overset{\mathrm{BV}}{\otimes} \mathbb{E}_W^\otimes & \xleftarrow[L_\mathbb{E}]{} & \mathbb{E}_V^\otimes \odot \mathbb{E}_W^\otimes & \xrightarrow[M_\mathbb{E}]{} & \mathbb{E}_V^\otimes \wr \mathbb{E}_W^\otimes & \xrightarrow[\varphi_\mathbb{E}]{} & \mathbb{E}_{V\oplus W}^\otimes
\end{array} \tag{1}$$

It suffices to verify that $\varphi_\mathbb{E} \circ M_\mathbb{E}$ is a Morita equivalence. We begin by using the 1-category theory of wreath products together with a specialization of Harpaz' theory of *strong approximations* [Har] to verify that $\widetilde{\varphi}_\mathbb{P}$ is a Morita equivalence; moreover, using an analog of Dunn's topology [Dun88], we verify that decomposable V-multi-polydisks form a suitably good G-Weiss cover so that $\alpha_{V|W}$ is also a weak approximation.[3] Under

[3] On the level of prefactorization algebras we may view this as saying that, absent a local constancy or cosheaf condition, additivity for V-disk prefactorization algebras may fail, as the prefactorization Dunn map only surjects onto the basis of *decomposable* elements; Dunn's argument [Dun88] essentially boils down to verifying that this basis is *coinitial* in the usual one, so the disparity disappears in presence of local constancy, and Berry's argument [Ber21] uses this to show that the disparity also disappears in presence of a cosheaf condition.

this, we find that pullback along $\varphi_{\mathbb{E}} \circ M_{\mathbb{E}}$ is an inclusion of full subcategories

$$\mathrm{Mon}_{\mathbb{E}_{V\oplus W}}(\mathcal{S}) \subset \mathrm{Mon}_{\mathbb{E}_V \odot \mathbb{E}_W}(\mathcal{S}) \subset \mathrm{Mon}_{\mathbb{P}_{V|W}}(\mathcal{S}),$$

each characterized by a local constancy condition. Some bookkeeping (recorded as Theorem 1.1) shows that these conditions agree for $\mathbb{E}^{\otimes}_{V\oplus W}$ and $\mathbb{E}^{\otimes}_V \odot \mathbb{E}^{\otimes}_W$, yielding Theorem A.

We prove Corollary B by reduction to Theorem A using the tensor product disintegration theorem of [Ste25d, Thm. 3.6]; indeed, we construct a G-natural transformation

$$\begin{array}{ccc} \underline{\mathcal{S}}_{G,/B_GO(k)} \times \underline{\mathcal{S}}_{G,/B_GO(k')} & \xrightarrow{\mathbb{E}^{\otimes}_{(-)} \times \mathbb{E}^{\otimes}_{(-)}} & \underline{\mathrm{Op}}_G \times \underline{\mathrm{Op}}_G \\ \downarrow{\scriptstyle \oplus} & \swarrow{\scriptstyle \varphi} & \downarrow{\scriptstyle \overset{\mathrm{BV}}{\otimes}} \\ \underline{\mathcal{S}}_{G,/B_GO(k+k')} & \xrightarrow[\mathbb{E}^{\otimes}_{(-)}]{} & \underline{\mathrm{Op}}_G \end{array}$$

which, on a pair of orbits $[G/H] \to B_GO(k)$ and $[G/H] \to B_GO(k')$ classifying representations V, W, specializes to the Dunn map $\varphi\colon \mathbb{E}^{\otimes}_V \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_W \to \mathbb{E}^{\otimes}_{V\oplus W}$. We use an equivalence $\mathrm{End}^G(V,V) \simeq \mathrm{Aut}^G(V)$ to verify that $U\mathbb{E}_\Theta \simeq B_\Theta$ and some elementary topology to construct, for each color $x \in B_\Theta$, an equivalence

$$\mathbb{E}^{\otimes}_{\Theta(x)} \xrightarrow{\sim} \mathrm{End}^{\mathrm{red}}_x \mathbb{E}^{\otimes}_\Theta;$$

the tensor disintegration theorem of [Ste25d, Cor. 3.6] then implies that φ is a G-natural equivalence.

Now, Corollary C is fairly simple; by computing the fibers of Θ-framed embedding spaces over configuration spaces, we may bootstrap the equivariant Weiss hypercodescent of equivariant configuration spaces from [Ste25c] to the spaces appearing in E^Θ_M, so an analogous hypercodescent argument to [DHK18] reduces to the case that M is in the Yoneda image of Disk^Θ_G. In this case, the result follows by unwinding definitions and applying $\mathbb{A}_2$-monoidality of $\mathbb{k}$ for an $\mathbb{A}_2$-monoidal structure on $\coprod_{n\in\mathbb{N},\, \Theta\in\pi_0\mathcal{S}_{G,/B_GO(n)}} \underline{\mathrm{Mfld}}^\Theta_G$ given by products with direct sum tangential structure.

Lastly, Theorem D follows *mutatis mutandis* from Theorem A.

Relationship to surrounding literature. This paper is the sixth installment in a series of papers [Ste24; Ste25a; Ste25b; Ste25c; Ste25d] which are intended to extend the parameterize and equivariant higher algebra of [BDGNS16; NS22] for use in equivariant homotopy theory and K-theory. As such, it is closely related to similar work in both different foundations and in the non-equivariant case. Moreover, the author intends to use it foundationally in further work concerning obstruction theory of $\mathbb{E}_V$-algebras.

As mentioned before, the proof of Theorem A is strongly inspired by the proofs of [Har; HA] for the case $G = e$. Additionally, the deduction of Corollary B from Theorem A is similar to that of Lurie [HA] assuming the statement of the disintegration and assembly theorem, though the proof of this theorem is quite different between [Ste25d] and [HA]. Moreover, the proof of Corollary 1.50 is *in essence*, a straightforward combination of work of Barkan and Harpaz [BHS22; Har].

In the equivariant case, the reader ought consider a comparison between our results and Sczczesny's point-set version [Szc24] to be a difficult open problem; indeed, to the author's knowledge, it is still an open problem whether the relevant tensor products are comparable in the *nonequivariant* case (c.f. [FV13]).

Moreover, to the author's knowledge, it is an open problem (in progress by other authors) to compare the ∞-categories of $\mathbb{E}_V$ algebras herein with the ∞-categories presented by the right-transferred model structure of e.g. [Hil22]. These are seen to be plausibly equivalent by the similarity between their associated monads (c.f. [GM17; Ste25b]).

The infinitary cases of Theorem A were already known by previous work of the author [Ste25d]; as mentioned before, the case $G = e$ of Theorem D appears in independent work of Carmona-Svraka [Car].

Notation and conventions. We broadly adopt the terminology of higher category theory from [HTT], equivariant higher category theory from [Sha22; Sha23], algebraic patterns from [CH21], and equivariant operads from [Ste25b; Ste25d]. Moreover, we will use the following notation:

- $\square$ will denote the box product of Mackey functors.
- $\boxtimes\colon \underline{\mathrm{Fun}}_G(\mathcal{C},\underline{\mathcal{S}}_G) \times \underline{\mathrm{Fun}}_G(\mathcal{D},\underline{\mathcal{S}}_G) \longrightarrow \underline{\mathrm{Fun}}_G(\mathcal{C}\times\mathcal{D},\underline{\mathcal{S}}_G)$ will denote the external tensor product.
- $\boxtimes$ is defined the following way: given two G-symmetric monoidal ∞-categories $\mathcal{C},\mathcal{D}$ and $\ell\colon : \mathcal{C}\times\mathcal{D} \to \mathcal{C}\square\mathcal{D}$ the lax structure map for $U\colon \mathrm{Cat}^{\otimes}_G \to \mathrm{Cat}_G$, we will always write $\mathcal{C}\boxtimes\mathcal{D} := \ell_!\mathcal{C}\boxtimes\mathcal{D}$.

- $\sqcup$ will always denote disjoint unions, which may not be categorical coproducts.
- $\coprod$ will always denote categorical coproducts.
- $\times$ will always denote products, which may not be categorical products.
- $\prod$ will always denote categorical products.
- $G \times_H M$ will always denote the indexed disjoint union, which may not be left adjoint to restriction.
- Ind will always be left adjoint to restriction.
- $\overset{\text{BV}}{\otimes}$ will always denote the Boardman-Vogt tensor product.
- $\wr$ will denote the wreath product of G-preoperads, and $\underline{\wr}$ of Γ-G-preoperads.
- $\odot$ will denote the operation taking G-preoperads $\mathcal{O}^{\otimes}, \mathcal{P}^{\otimes}$ to the relative pattern

$$\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes} \to \mathsf{Span}(\mathbb{F}_G)^2 \xrightarrow{\wedge} \mathsf{Span}(\mathbb{F}_G).$$

- $\underline{\odot}$ will denote the operation taking G-Γ-preoperads $\mathcal{O}^{\otimes}$, $\mathcal{P}^{\otimes}$ to the G-Γ-preoperad

$$\mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes} \to \underline{\mathbb{F}}^2_{\mathcal{T},*} \xrightarrow{\wedge} \underline{\mathbb{F}}_{\mathcal{T},*}.$$

- $\mathfrak{c}$ will denote a map taking a little disk embedding to its underlying configuration.
- $\mathfrak{t}$ will denote a map taking an affine little disk embedding to its underlying linear transformations.
- colim will denote equivariant ∞-categorical colimits; the terms "point-set" and "homotopy" colimits will appear textually when the distinction is relevant.
- Given a formal direct sum $V_\bullet = V_1 \oplus \cdots V_n$, the associated polydisk will be written $P^{V_\bullet} := \prod_{i=1}^n D^V$.

Moreover, "topological space" will always mean *compactly generated weakly-Hausdorff topological space*, and Top to the category of such topological spaces; all model-categorical aspects of Top will be defined with respect to the Quillen model structure. The word "space" will be reserved for objects in the ∞-category $\mathcal{S} \simeq \mathsf{Top}\left[WEQ^{-1}\right]$, or equivalently, objects in hoTop. If X, Y are topological spaces, $\mathsf{Map}(X,Y)$ will be denote the internal hom in Top, i.e. the compact-open topology; if M, N are G-manifolds, $\mathsf{Emb}^G(M,N)$ will denote the space of smooth equivariant maps under the strong C^∞-topology.

Acknowledgements. I would like to thank Thomas Blom, Mike Hopkins, Connor Malin, and Jan Steinebrunner for helpful conversations on this topic. In particular, I'm grateful to Connor Malin for pointing out the connection of the additivity theorem to embedding calculus, which ultimately led me to Corollary C. Additionally, I'm thankful to Maxime Ramzi for disabusing me of an early fatally flawed strategy for the proof of Theorem A.

I would be remiss to not mention the work which inspired this paper: I learned the main strategy categorical strategy for this paper by analogy to the beautiful proof of Yonatan Harpaz in [Har] and the topology from that of Gerald Dunn [Dun88]; also, the combinatorics only became tractable after the work of Shaul Barkan, Rune Haugseng, and Jan Steinebrunner in [BHS22] on *fibrous patterns*. Moreover, the Weiss G-hypercodcescent strategy for Corollary C is a simple and direct generalization of the strategy in [DHK18], though the use of $\mathcal{O}$-monoidality of the Yoneda embedding is new. Many of the manifold perspectives in this paper arose in forthoming work with Branko Juran, which will contain more careful analysis of equivariant manfiolds along these lines.

While developing this material, the author was partially supported by the NSF Grant No. DGE 2140743.

1. Morita theory and Boardman-Vogt tensor products

The contents of this section is of independent interest, so we temporarily replace $\mathcal{O}_G$ with an arbitrary atomic orbital 1-category $\mathcal{T}$; the reader is encouraged to revert $\mathcal{T}$ back to $\mathcal{O}_G$. We adopt the notation $\mathsf{POp}_{\mathcal{T}} := \mathsf{Cat}^{\text{int-cocart}}_{/\mathsf{Span}(\mathbb{F}_{\mathcal{T}})}$ for the *∞-category of $\mathcal{T}$-preoperads*.

We recall many relevant facts of these (originally proved in [BHS22; NS22; Ste25a; Ste25b; Ste25d]) in Section 1.1. In particular, we recall the *underlying algebraic pattern* functor therein; we use this to develop a theory of *weak and strong approximations of algebraic patterns* in Section 1.2.

Following this, in Section 1.3, we construct a *wreath product* model for the Boardman-Vogt tensor product together with a morphism of algebraic patterns $\mathcal{O}^{\otimes} \odot \mathcal{P}^{\otimes} \to \mathcal{O}^{\otimes} \wr \mathcal{O}^{\otimes}$, culminating in a procedure for pushing forward Boardman-Vogt tensor product computations along weak approximations. The following theorem manages the bookkeeping.

Theorem 1.1. *Fix algebraic patterns maps α_i, $\varphi_{\mathcal{E}}$, and $\varphi_{\mathcal{P}}$ over $\mathsf{Span}(\mathbb{F}_{\mathcal{T}})$ making the following commute*

$$\begin{array}{ccc} \mathcal{P}_1^{\otimes} \wr \mathcal{P}_2^{\otimes} & \xrightarrow{\varphi_{\mathcal{P}}} & \mathcal{P}_3^{\otimes} \\ {\scriptstyle \alpha_1 \wr \alpha_2}\downarrow & & \downarrow{\scriptstyle \alpha_3} \\ \mathcal{E}_1^{\otimes} \wr \mathcal{E}_2^{\otimes} & \xrightarrow{\varphi_{\mathcal{E}}} & \mathcal{E}_3^{\otimes} \end{array}$$

and satisfying the following conditions:
(BK-a) Each α_i is a weak approximation;
(BK-b) $\varphi_{\mathcal{P}}$ is a strong approximation; and
(BK-c) the induced $\mathcal{T}$-functor $U\varphi_{\mathcal{E}} \colon U\mathcal{E}_1 \times U\mathcal{E}_2 \to U\mathcal{E}_3$ is an equivalence.
The composite map $\mathcal{E}_1^{\otimes} \odot \mathcal{E}_2^{\otimes} \to \mathcal{E}_1^{\otimes} \wr \mathcal{E}_2^{\otimes} \to \mathcal{E}_3^{\otimes}$ induces an equivalence

$$\mathcal{E}_1^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathcal{E}_2^{\otimes} \xrightarrow{\sim} \mathcal{E}_3^{\otimes}.$$

We suggest to the reader that they lightly read Section 1.3 in order to gain a working understanding of wreath products and take Theorem 1.1 for granted, then return to this section after reading the remainder of the paper for the combinatorial details of wreath products and weak $\mathcal{T}$-preoperadic approximations.

1.1. Recollections on equivariant operads.

1.1.1. *The combinatorics of equivariant mutli-arity.*

Definition 1.2. A *$\mathcal{T}$-multi-arity* is a morphism $T \to S$ in $\mathbb{F}_{\mathcal{T}}$; a *$\mathcal{T}$-arity* is a multi-arity $T \to V$ where $V \in \mathcal{T}$, or equivalently, an element of the ∞-category $\mathrm{Tot}_{\simeq}\mathbb{F}_{\mathcal{T}}$. ◁

These are the *types* one inputs to an operation in the algebraic theory described by a *one-color $\mathcal{T}$-operad*; unfortunately, the main strategy of this paper shifts the data of these to $\mathcal{T}$-operads with more complicated colors, so we take careful account of the types involved in such theories. We will use the language of *multi-profiles* associated with a *coefficient system*.

Definition 1.3. Let $\mathcal{C}$ be an ∞-category. A *$\mathcal{T}$-coefficient system in $\mathcal{C}$* is a functor $\mathfrak{C} \colon \mathcal{T}^{\mathrm{op}} \to \mathcal{C}$; we refer to elements of $\mathfrak{C}^V$ as *V-colors* and, for $T \in \mathbb{F}_{\mathcal{T}}$, of $\mathfrak{C}^T = \prod_{V \in \mathrm{Orb}(T)} \mathfrak{C}^V$ as *T-colors*.

Given a $\mathcal{T}$-coefficient system of sets $\mathfrak{C} \colon \mathcal{T}^{\mathrm{op}} \to \mathsf{Set}$ and multi-arity $f \colon T \to S$, the set of *$\mathfrak{C}$-multi-profiles lying over f* is

$$\mathrm{Prf}_{\mathfrak{C}}^{f} \coloneqq \prod_{V \in S} \mathcal{C}^{T_V} \times \mathcal{C}^{V},$$

where $T_V \coloneqq T \times_S V$. We refer to a $\mathfrak{C}$-multi-profile as a *$\mathfrak{C}$-profile* if its underlying $\mathcal{T}$-multi-arity is a $\mathcal{T}$-arity. We will denote a generic $\mathfrak{C}$-multi-profile by $(\mathbf{C};\mathbf{D})$ (leaving f implicit) and a generic $\mathfrak{C}$-profile as $(\mathbf{C};D)$.

Given a $\mathfrak{C}$-multi-profile $(\mathbf{C};\mathbf{D})$ lying over $f \colon T \to S$, given an orbit $U \in \mathrm{Orb}(S)$, we write D_U for the corresponding U-color of $\mathbf{D}$ and we write $\mathbf{C}_U$ for the corresponding T_U-color of $\mathbf{C}$. Similarly, given $V \in \mathrm{Orb}(T)$, we write C_V for the corresponding V-color of $\mathbf{C}$. ◁

Construction 1.4. Let $T \xrightarrow{f} S \xrightarrow{g} R$ be composable multi-arities. By "forgetting the middle color" and composing multi-arities, we acquire an operation $\circ$ fitting into the following diagram:

$$\mathrm{Prf}_{\mathfrak{C}}^{f} \times_{\mathcal{C}_S} \mathrm{Prf}_{\mathfrak{C}}^{g} \xrightarrow{\circ} \mathrm{Prf}_{\mathfrak{C}}^{g \circ f}$$

We refer to the codomain as *composable pairs*. Moreover, given a pullback diagram

$$\begin{array}{ccc} h^*T & \xrightarrow{h'} & T \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ Q & \xrightarrow{h} & S \end{array}$$

we define a *pullback* operation $h^* \colon \mathrm{Prf}_{\mathfrak{C}}^{f} \to \mathrm{Prf}_{\mathfrak{C}}^{f'}$ by setting

$$f^*(\mathbf{C};\mathbf{D}) \coloneqq \Big(\big(C_{f'(W)} \mid W \in \mathrm{Orb}(f^*T)\big);\big(D_{f(W)} \mid W \in \mathrm{Orb}(Q)\big)\Big);$$

In particular, when $Q = W$ and $S = V$ are orbits, we write this as $\mathrm{Res}_V^W(\mathbf{C};\mathbf{D}) = \big(\mathrm{Res}_V^W \mathbf{C}; \mathrm{Res}_V^W D\big)$. ◁

Now, this allows for convenient description of $\mathfrak{C}$-*symmetric sequences* and the corresponding theory of $\mathfrak{C}$-*colored genuine* $\mathcal{T}$-*operads*. To describe this, we need the $\mathfrak{C}$-colored version of Σ, beginning with the following notion.

Definition 1.5. Given $(\mathbf{C}; D)$ an S-ary $\mathfrak{C}$-profile, the group of $\mathfrak{C}$-*colored automorphisms of* S is the subgroup

$$\mathrm{Aut}^V(\mathbf{C}) \subset \mathrm{Aut}^V(S)$$

consisting of automorphisms σ of $S \in \mathbb{F}_V$ such that, for all $U \in \mathrm{Orb}(S)$, $C_U = C_{\sigma U}$. ◁

In [Ste25b] we found a cocartesian fibration of $\mathcal{T}$-groupoids $F\colon \underline{\Sigma}_{\mathfrak{C}} \to \underline{\Sigma}_{\mathcal{T}} \coloneqq \underline{\mathbb{F}}^{\simeq}_{\mathcal{T}}$ with V-value groupoid

$$\Sigma_{\mathfrak{C},V} = \coprod_{\substack{S \in \mathbb{F}_V \\ (\mathbf{C};D) \in \mathfrak{C}^S \times \mathfrak{C}^V}} B\,\mathrm{Aut}^V(\mathbf{C}),$$

with restriction maps given by applying B to the evident maps $\mathrm{Aut}^V(\mathbf{C}) \to \mathrm{Aut}^U(\mathrm{Res}^V_U \mathbf{C})$ and with F given by applying B to the defining maps $\mathrm{Aut}^V(\mathbf{C}) \to \mathrm{Aut}^V(S)$.

Definition 1.6. Given $\mathcal{V}$ an ∞-category with finite products, the ∞-category of $\mathfrak{C}$-*symmetric sequences in* $\mathcal{V}$ is $\mathsf{Fun}(\mathrm{Tot}\underline{\Sigma}_{\mathfrak{C}}, \mathcal{V})$. Given a $\mathfrak{C}$-multi-profile $(\mathbf{C};\mathbf{D})$ lying over a multi-arity $f\colon T \to S$ and a $\mathfrak{C}$-symmetric sequence $\mathcal{O}$, we define the $\mathcal{V}$-*object of* $(\mathbf{C};\mathbf{D})$ *multi-operations* to be

$$\mathrm{Mul}^f_{\mathcal{O}}(\mathbf{C};\mathbf{D}) \coloneqq \prod_{U \in \mathrm{Orb}(S)} \mathcal{O}(\mathbf{C}_U; D_U). \qquad \triangleleft$$

Definition 1.7 ([NS22, Def. 2.5.4])**.** Let $\mathcal{V}$ be a 1-category with finite products. A $\mathfrak{C}$-*colored genuine* $\mathcal{T}$-*operad* $\mathcal{O}$ *in* $\mathcal{V}$ is the data of

(1) A $\mathfrak{C}$-symmetric sequence $\mathcal{O}$;
(2) For every color $C \in \mathfrak{C}$, an identity operation

$$1_C\colon *_{\mathcal{V}} \to \mathrm{Mul}^{\mathrm{id}}_{\mathcal{O}}(C;C)$$

(3) For every composable triple of multi-profiles $(\mathbf{B};\mathbf{C};D)$ with D a color, a composition map

$$\gamma\colon \mathcal{O}(\mathbf{C};D) \times \mathrm{Mul}^g_{\mathcal{O}}(\mathbf{B};\mathbf{C}) \longrightarrow \mathcal{O}(\mathbf{B};D)$$

subject to the following conditions:

(OP-a) (unitality) for every $\mathfrak{C}$-profile $(\mathbf{B};\mathbf{C})$, the following commutes

$$\begin{array}{ccc} \mathcal{O}(\mathbf{B};C) & \xrightarrow{\mathrm{id}\times\{1\}} & \mathcal{O}(\mathbf{B};C) \times \mathrm{Mul}^{\mathrm{id}}_{\mathcal{O}}(\mathbf{B};\mathbf{B}) \\ {\scriptstyle \{1\}\times\mathrm{id}}\downarrow & \searrow & \downarrow{\scriptstyle \gamma} \\ \mathcal{O}(C;C) \times \mathcal{O}(\mathbf{B};C) & \xrightarrow[\gamma]{} & \mathcal{O}(\mathbf{B};C) \end{array}$$

where γ is extended to

(OP-b) (associativity) for every composable triple $(\mathbf{A};\mathbf{B};\mathbf{C};D)$, the following commutes

$$\begin{array}{ccc} \mathcal{O}(\mathbf{C};D) \times \mathrm{Mul}^g_{\mathcal{O}}(\mathbf{B};\mathbf{C}) \times \mathrm{Mul}^h_{\mathcal{O}}(\mathbf{A};\mathbf{B}) & \longrightarrow & \mathcal{O}(\mathbf{B};D) \times \mathrm{Mul}^h_{\mathcal{O}}(\mathbf{A};\mathbf{B}) \\ \downarrow & & \downarrow \\ \mathcal{O}(\mathbf{C};D) \times \mathrm{Mul}^{h\circ g}_{\mathcal{O}}(\mathbf{A};\mathbf{C}) & \longrightarrow & \mathcal{O}(\mathbf{A};D) \end{array}$$

(OP-c) (restriction and units) The restriction map $\mathcal{O}(C;C) \to \mathcal{O}(\mathrm{Res}^V_U C; \mathrm{Res}^V_U C)$ takes 1_C to $1_{\mathrm{Res}^V_U C}$.

(OP-d) (restriction and composition) For every composable pair $(\mathbf{B};\mathbf{C};D)$ and orbit map $U \to V$, the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}(\mathbf{C};D) \times \mathrm{Mul}^g_{\mathcal{O}}(\mathbf{B};\mathbf{C}) & \xrightarrow{\gamma} & \mathcal{O}(\mathbf{B};D) \\ \downarrow{\scriptstyle (\mathrm{Res},\Delta)} & & \downarrow{\scriptstyle \mathrm{Res}} \\ \mathrm{Mul}^{\mathrm{Res}^V_U f}_{\mathcal{O}}\left(\mathrm{Res}^V_U \mathbf{C}; \mathrm{Res}^V_U D\right) \times \mathrm{Mul}^{\mathrm{Res}^V_U g}_{\mathcal{O}}\left(\mathrm{Res}^V_U \mathbf{B}; \mathrm{Res}^V_U \mathbf{C}\right) & \xrightarrow{\gamma} & \mathrm{Mul}^{\mathrm{Res}^V_U g\circ f}_{\mathcal{O}}\left(\mathrm{Res}^V_U \mathbf{B}; \mathrm{Res}^V_U D\right) \end{array}$$

where Δ here denotes the *restriction-twisted diagonal* operation

$$\prod_{V'\in \mathrm{Orb}(\pi\mathbf{C})} \mathcal{O}(\mathbf{B}_{V'};C_V) \xrightarrow{\Delta} \prod_{U'\in \mathrm{Orb}(\mathrm{Res}^V_U \pi\mathbf{C})} \mathcal{O}\big(\mathbf{B}_{o(U')};C_{o(U')}\big) \xrightarrow{\mathrm{Res}} \prod_{U'\in \mathrm{Orb}(\mathrm{Res}^V_U \pi\mathbf{C})} \mathcal{O}(\mathbf{B}_{U'};C_{U'})$$

for $o\colon \mathrm{Orb}(\mathrm{Res}^V_U S) \to \mathrm{Orb}(S)$ the evident *preimage* map (see [Bon19, Def. 3.22]).

(OP-e) ($\mathrm{Aut}_H S$-equivariance) For all composable pairs $(\mathbf{C};\mathbf{B};A)$, γ is Borel $\mathrm{Aut}^V(\mathbf{B}) \times \prod_{U\in\mathrm{Orb}(S)} \mathrm{Aut}^U(\mathbf{C}_U)$-equivariant.

A *morphism of $\mathfrak{C}$-colored genuine $\mathcal{T}$-operads in $\mathcal{V}$* is a morphism $\varphi\colon \mathcal{O}\to\mathcal{P}$ of $\mathfrak{C}$-symmetric sequences such that $\varphi(1_C) = 1_{\varphi(c)}$ and such that the following diagram commutes.

$$\begin{array}{ccc} \mathcal{O}(\mathbf{C};D)\times \mathrm{Mul}^g_{\mathcal{O}}(\mathbf{B};\mathbf{C}) & \xrightarrow{\varphi\times\varphi} & \mathcal{P}(\varphi\mathbf{C};\varphi D)\times \mathrm{Mul}^{\varphi g}_{\mathcal{P}}(\varphi\mathbf{B};\varphi\mathbf{C}) \\ \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \gamma} \\ \mathcal{O}(\mathbf{B};D) & \xrightarrow{\qquad\varphi\qquad} & \mathcal{P}(\varphi\mathbf{B};\varphi D) \end{array}$$

In the situation that $\mathcal{V}$ is a *homotopical category*, i.e. we've supplied a wide subcategory $\mathcal{V}^{\mathrm{WEQ}} \subset \mathcal{V}$ of "weak equivalences" satisfying two-out-of-six, a *weak equivalence of $\mathfrak{C}$-colored genuine $\mathcal{T}$-operads in $\mathcal{V}$* is a morphism $\mathfrak{O}\to\mathfrak{P}$ whose values $\mathcal{O}(\mathbf{C};D)\to\mathcal{P}(\varphi\mathbf{C};\varphi D)$ are each weak equivalences. In particular, when $\mathcal{V} = \mathsf{Top}$, a *weak equivalence of genuine topological $\mathcal{T}$-operads* is a weak equivalence with respect to pointwise π_*-isomorphisms. ◁

Construction 1.8. Given $f\colon \mathfrak{C}\to\mathfrak{D}$ a morphism of coefficient systems of sets, we acquire a $\mathcal{T}$-functor $\underline{\Sigma}_{\mathfrak{C}} \to \underline{\Sigma}_{\mathfrak{D}}$, and therefore a pullback functor

$$f^*\colon \mathsf{Fun}(\mathrm{Tot}\underline{\Sigma}_{\mathfrak{D}},\mathcal{V}) \to \mathsf{Fun}(\mathrm{Tot}\underline{\Sigma}_{\mathfrak{C}},\mathcal{V});$$

given $\mathcal{O}$ a $\mathfrak{D}$-colored genuine $\mathcal{T}$-operad, we define the *pullback $\mathfrak{C}$-colored genuine $\mathcal{T}$-operad* $f^*\mathcal{O}$ to have underlying $\mathfrak{C}$-symmetric sequence $f^*\mathcal{O}$, unit given by the map $1_C\colon 1_{\mathcal{V}} \to \mathcal{O}(\varphi C;\varphi C) \simeq f^*\mathcal{O}(C;C)$. The composition map is defined similarly. ◁

Definition 1.9. A *morphism of genuine $\mathcal{T}$-operads in $\mathcal{V}$* from a $\mathfrak{C}$-colored genuine $\mathcal{T}$-operad $\mathcal{P}$ to a $\mathfrak{D}$-colored genuine $\mathcal{T}$-operad $\mathcal{O}$ is a morphism of $\mathcal{T}$-coefficient systems $\varphi_c\colon \mathfrak{C}\to\mathfrak{D}$ together with a morphism of $\mathfrak{C}$-colored $\mathcal{T}$-operads $\varphi\colon \mathcal{O}\to\varphi_c^*\mathcal{P}$. We refer to the associated category as $g\mathrm{Op}_{\mathcal{T}}(\mathcal{V})$. ◁

In the case that $\mathcal{T} = \mathcal{O}_G$ is the orbit category of a finite group and $\mathcal{V} = \mathsf{sSet}$ under the Quillen homotopical structure, this agrees with the definition employed in [Bon19]. We will generally prefer the topological setting, but the distinction is easy to ignore by the following easy proposition, whose proof we omit.

Proposition 1.10. *Suppose $\mathcal{V}\to\mathcal{V}'$ is a functor of homotopical categories which reflects weak equivalences. Then, the corresponding functor $g\mathrm{Op}_{\mathcal{T}}(\mathcal{V}) \to g\mathrm{Op}_{\mathcal{T}}(\mathcal{V}')$ reflects weak equivalences.*

1.1.2. *More definitions.* Let $\mathsf{Span}(\mathbb{F}_{\mathcal{T}})$ be the *effective Burnside 2-category*, e.g. as in [HHLN23].

Definition 1.11. A *$\mathcal{T}$-preoperad* $\mathcal{O}^{\otimes}$ is a functor $\pi_{\mathcal{O}}\colon \mathrm{Tot}\mathcal{O}^{\otimes} \to \mathsf{Span}(\mathbb{F}_{\mathcal{T}})$ such that

(a) $\mathrm{Tot}\mathcal{O}^{\otimes}$ has π-cocartesian lifts for backwards maps in $\mathsf{Span}(\mathbb{F}_{\mathcal{T}})$.

We moreover say that $\mathcal{O}^{\otimes}$ is a *$\mathcal{T}$-operad* if it satisfies the following additional conditions:

(b) (Segal condition for colors) for every multi-profile $(\mathbf{C};\mathbf{D})$ lying over $f\colon T\to S$, cocartesian transport along the π-cocartesian lifts lying over the inclusions $(S \leftarrow U = U \mid U\in\mathrm{Orb}(S))$ together induce an equivalence

$$\mathcal{O}_S \simeq \prod_{U\in\mathrm{Orb}(S)} \mathcal{O}_U,$$

where $\mathcal{O}_S := \mathrm{Tot}\mathcal{O}^{\otimes} \times_{\mathsf{Span}(\mathbb{F}_{\mathcal{T}})} \{S\}$, and

(c) (Segal condition for multimorphisms) for every map of orbits $T\to S$ in I and pair of objects $(\mathbf{C},\mathbf{D}) \in \mathcal{O}_T\times\mathcal{O}_U$, postcomposition with the π-cocartesian lifts $\mathbf{D}\to D_U$ lying over the inclusions $(S\leftarrow U = U \mid U\in\mathrm{Orb}(S))$ induces an equivalence

$$\mathrm{Mul}^f_{\mathcal{O}^{\otimes}}(\mathbf{C},\mathbf{D}) \simeq \prod_{U\in\mathrm{Orb}(S)} \mathrm{Mul}^{T\leftarrow T_U \xrightarrow{f} U}_{\mathcal{O}^{\otimes}}(\mathbf{C},D_U).$$

where $T_U \coloneqq T \times_S U$ and $\mathrm{Mul}^f_{\mathcal{O}}(\mathbf{C};\mathbf{D}) \coloneqq \mathrm{Map}_{\mathrm{Tot}\mathcal{O}}(\mathbf{C},\mathbf{D}) \times_{\mathrm{Map}_{\mathrm{Span}(\mathbb{F}_{\mathcal{T}})}(\pi\mathbf{C},\mathbf{D})} \{f\}$.

We denote the ∞-category of $\mathcal{T}$-preoperads and functors over $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$ preserving backwards-cocartesian arrows, together with the full subcategory of $\mathcal{T}$-operads, as

$$\mathrm{Op}_{\mathcal{T}} \subset \mathrm{POp}_{\mathcal{T}} \coloneqq \mathrm{Cat}^{\mathrm{Backwards-cocart}}_{/\mathrm{Span}(\mathbb{F}_{\mathcal{T}})}.$$

We will refer to a morphism of $\mathcal{T}$-preoperads $\mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ as a *$\mathcal{O}$-algebra in $\mathcal{P}$*, and we let

$$\mathrm{Alg}_{\mathcal{O}}(\mathcal{P}) \coloneqq \mathrm{Fun}^{\mathrm{Backwards-cocart}}_{/\mathrm{Span}(\mathbb{F}_{\mathcal{T}})}(\mathcal{O}^{\otimes},\mathcal{P}^{\otimes}) \subset \mathrm{Fun}_{/\mathrm{Span}(\mathbb{F}_{\mathcal{T}})}(\mathcal{O}^{\otimes},\mathcal{P}^{\otimes})$$

be the full subcategory spanned by $\mathcal{O}$-algebras in $\mathcal{P}$. ◁

An important invariant captures *colors*.

Construction 1.12. Given a $\mathcal{T}$-preoperad $\mathcal{O}^{\otimes}$, its *$\mathcal{T}$-∞-category of colors* $U\mathcal{O}$ is the unstraightening of $\mathrm{Tot}U\mathcal{O}$:

$$\begin{array}{ccc} \mathrm{Tot}U\mathcal{O} & \longrightarrow & \mathrm{Tot}\mathcal{O}^{\otimes} \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{T}^{\mathrm{op}} & \longrightarrow & \mathrm{Span}(\mathbb{F}_{\mathcal{T}}) \end{array}$$

We say that $\mathcal{O}^{\otimes}$ *has one color* if $U\mathcal{O}$ is contractible. ◁

When f is an arity, so $D \coloneqq \mathbf{D}$ is a color, we define

$$\mathcal{O}(\mathbf{C};D) \coloneqq \mathrm{Mul}^f_{\mathcal{O}}(\mathbf{C};D).$$

In particular, when $\mathcal{O}^{\otimes}$ has at most one color, we write $\mathcal{O}(S) \coloneqq \mathcal{O}(S;V)$.

These invariants carry the essence of a $\mathcal{T}$-operad; for instance, they are jointly conservative.

Proposition 1.13 ([Ste25b, Prop. 2.64])**.** *If a morphism of $\mathcal{T}$-operads $\varphi\colon \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ induces an essential surjection $U\mathcal{O} \to U\mathcal{P}$ as well as equivalences $\mathcal{O}(\mathbf{C};D) \xrightarrow{\sim} \mathcal{P}(\varphi\mathbf{C};\varphi D)$, then φ is an equivalence.*

Now, these objects generally fit into an *algebraic pattern* framework in the sense of [CH21]:

Definition 1.14. An *algebraic pattern* is an ∞-category $\mathfrak{B}$ together with a factorization system $(\mathfrak{B}^{\mathrm{int}},\mathfrak{B}^{\mathrm{act}})$ and a full subcategory $\mathfrak{B}^{\mathrm{el}} \subset \mathfrak{B}^{\mathrm{int}}$; given $\mathcal{C}$ an ∞-category, a *segal $\mathfrak{B}$-object in $\mathcal{C}$* is a functor $\mathfrak{B} \to \mathcal{C}$ whose restricted functor $\mathfrak{B}^{\mathrm{int}} \to \mathcal{C}$ is right Kan-extended along the inclusion $\mathfrak{B}^{\mathrm{el}} \hookrightarrow \mathfrak{B}^{\mathrm{int}}$. ◁

Example 1.15. The *effective Burnside algebraic pattern* is $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$, together with the backwards-forwards factorization system $\mathbb{F}^{\mathrm{op}}_{\mathcal{T}},\mathbb{F}_{\mathcal{T}} \subset \mathrm{Span}(\mathbb{F}_{\mathcal{T}})$ of [HHLN23, Prop. 4.9] and elementary objects $\mathcal{T}^{\mathrm{op}} \subset \mathbb{F}^{\mathrm{op}}_{\mathcal{T}}$.

More generally, given $\mathcal{O}^{\otimes}$ a $\mathcal{T}$-preoperad, $\mathrm{Tot}\mathcal{O}^{\otimes}$ admits an algebraic pattern structure with inert arrows the cocartesian lifts of inert arrows of $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$, active arrows arbitrary lifts of active arrows, and elementary objects arbitrary lifts of elementary objects. ◁

In [Ste25b, Thm. A.1] we verified (following [BHS22]) that Segal $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$-objects in $\mathcal{C}$ are precisely product-preserving functors $\mathrm{Span}(\mathbb{F}_{\mathcal{T}}) \to \mathcal{C}$, i.e. *$\mathcal{T}$-commutative monoids in $\mathcal{C}$*, by proving a statement easily seen to be equivalent to the following.

Lemma 1.16 ([Ste25b, Lem. A.6])**.** $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})^{\mathrm{el}}_{/S}$ *splits as a disjoint union of ∞-categories, each of which has an terminal object of the form of an orbit inclusion $S \hookleftarrow U = U$.*

Now, [BHS22] developed a convenient framework of *fibrous patterns* (slightly modifying the earlier *weak Segal fibrations* of [CH21], which are higher categorical analogs of the fibrant objects of [HA, § B] and generalize the weak ∞-operads of [Har]); we verified in [Ste25b] that their argument extends to show that $\mathcal{T}$-operads are *precisely* fibrous $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$-patterns; it then follow from general category theory that the inclusion $\mathrm{Op}_{\mathcal{T}} \subset \mathrm{POp}_{\mathcal{T}}$ admits a left adjoint $L_{\mathrm{Op}_{\mathcal{T}}}\colon \mathrm{POp}_{\mathcal{T}} \to \mathrm{Op}_{\mathcal{T}}$ [BHS22], which we call *$\mathcal{T}$-operadic localization*.

Before continuing, we also require strict topological variants:

Variant 1.17. Let $\mathbf{Span}(\mathbb{F}_{\mathcal{T}})$ denote a *topological* category underlying $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$. A *topological $\mathcal{T}$-preoperad* is a functor $\mathcal{O}^t \to \mathbf{Span}(\mathbb{F}_{\mathcal{T}})$ whose homotopy coherent nerve is a $\mathcal{T}$-preoperad; morphisms of topological $\mathcal{T}$-preoperads are defined similarly. ◁

Given $\mathcal{O}$ a $\mathfrak{C}$-colored genuine topological $\mathcal{T}$-operad, we can define a topological category $N^t\mathcal{O}$ over $\mathbf{Span}(\mathbb{F}_{\mathcal{T}})$ to have objects

$$\mathrm{Ob}\,\pi^{-1}(\{S\}) \coloneqq \mathfrak{C}^S,$$

and morphism space lying over $S \xleftarrow{g} R \xrightarrow{f} T$ by

$$\mathrm{Map}^{S\xleftarrow{g}R\xrightarrow{f}T}_{N^{\otimes}\mathcal{O}}(\mathbf{C};\mathbf{D}) \coloneqq \mathrm{Mul}^f_{\mathcal{O}}(g^*\mathbf{C};\mathbf{D}),$$

with identity arrows and composition functions determined by $1_\bullet$ and γ. See [BP21; NS22] for details.

Proposition 1.18 ([NS22])**.** *$N^t\mathcal{O}$ is a topological $\mathcal{T}$-operad with $\underline{\pi}_0 UN^{\otimes}\mathcal{O} \simeq \mathfrak{C}$ and structure spaces*

$$\mathrm{Mul}^f_{N^{\otimes}\mathcal{O}}(\mathbf{C};\mathbf{D}) \simeq \mathrm{Mul}^f_{\mathcal{O}}(\mathbf{C};\mathbf{D}).$$

Moreover, if $\varphi\colon \mathcal{O} \to \mathcal{P}$ is a map of genuine topological $\mathcal{T}$-operads, then $N^t\varphi$ is a map of topological $\mathcal{T}$-operads; if φ is a weak equivalence of genuine topological $\mathcal{T}$-operads, then its homotopy coherent nerve $N^{\otimes}\varphi$ is an equivalence of $\mathcal{T}$-operads.

1.1.3. *The total $\mathcal{T}$-∞-category.* In this paper we will occasionally make use of yet another equivalent model for $\mathcal{T}$-operads, which we will refer to as the Γ-*space model*.

Construction 1.19 ([NS22, § 2.1] and [Ste25b, § A])**.** Consider the 2-category $\mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*} \simeq \mathrm{Span}_{s.i.,tdeg}(\underline{\mathbb{F}}^{\vee}_{\mathcal{T}})$, with its span factorization system as in [HHLN23]. This lifts to an algebraic pattern $\mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*}$ with elementary objects $\mathcal{T}^{\mathrm{op}}$, admitting a pattern map

$$s\colon \mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*} \hookrightarrow \mathrm{Span}_{all,tdeg}(\underline{\mathbb{F}}^{\vee}_{\mathcal{T}};\mathcal{T}^{\mathrm{op}}) \xrightarrow{\mathrm{ev}_0} \mathrm{Span}(\mathbb{F}_{\mathcal{T}}). \qquad \triangleleft$$

Proposition 1.20 ([BHS22, Prop. 5.2.14], generalized in [Ste25b, Thm. A.1])**.** *Pullback along the map $\mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*} \to \mathrm{Span}(\mathbb{F}_{\mathcal{T}})$ is an equivalence*

$$\mathrm{Op}_{\mathcal{T}} \simeq \mathrm{Fbrs}(\mathrm{Span}(\mathbb{F}_{\mathcal{T}})) \xrightarrow{\sim} \mathrm{Fbrs}(\mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*})$$

This Γ-space model recasts a version of $\mathcal{T}$-preoperad theory in the setting of $\mathcal{T}$-∞-*category theory*, which is occasionally useful; for instance, it allows for an *underlying* $\mathcal{T}$-∞-*category* as follows.

Construction 1.21. Given $\mathcal{O}^{\otimes}$ a $\mathcal{T}$-preoperad, we define its *total* $\mathcal{T}$-∞-*category* to have unstraightening the following horizontal composition:

$$\begin{array}{ccccc} \mathrm{TotTot}_{\mathcal{T}}\mathcal{O}^{\otimes} & \longrightarrow & \mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*} & \xrightarrow[\mathrm{ev}_1]{} & \mathcal{T}^{\mathrm{op}} \\ \downarrow & \lrcorner & \downarrow{\scriptstyle s} & & \\ \mathcal{O}^{\otimes} & \longrightarrow & \mathrm{Span}(\mathbb{F}_{\mathcal{T}}) & & \end{array}$$

That is, $\mathrm{Tot}_{\mathcal{T}}\mathcal{O}^{\otimes} \coloneqq \mathrm{St}_{\mathcal{T}^{\mathrm{op}}}\mathrm{TotTot}_{\mathcal{T}}\mathcal{O}^{\otimes}$. $\triangleleft$

1.1.4. *The Boardman-Vogt tensor product and pointwise tensor products of algebras.* We need the following.

Definition 1.22. A $\mathcal{T}$-*symmetric monoidal* ∞-*category* is a product preserving functor $\mathrm{Span}(\mathbb{F}_{\mathcal{T}}) \to \mathrm{Cat}$. A *symmetric monoidal* $\mathcal{T}$-∞-*category* is a product preserving functor $\mathcal{C}^{\otimes}\colon \mathrm{Span}(\mathbb{F}) \to \mathrm{Cat}_{\mathcal{T}}$; we refer to the associated $\mathcal{T}$-functor

$$-\otimes- = \mathcal{C}^{\otimes}\left(2\cdot *_{\mathcal{T}} = 2\cdot *_{\mathcal{T}} \xrightarrow{\nabla} *_{\mathcal{T}}\right)\colon \mathcal{C}^2 \longrightarrow \mathcal{C}$$

as the *tensor* $\mathcal{T}$-*bifunctor*. We say that $\mathcal{C}^{\otimes}$ is *presentably symmetric monoidal* if the underlying $\mathcal{T}$-∞-category $\mathcal{C} = \mathcal{C}^{\otimes} \times_{\mathrm{Span}(\mathbb{F}_{\mathcal{T}})} \mathcal{T}^{\mathrm{op}}$ is presentable and for each $C \in \mathcal{C}^{\mathcal{T}}$, the half-evaluated tensor $\mathcal{T}$-functor $C \otimes (-)\colon \mathcal{C} \to \mathcal{C}$ is a $\mathcal{T}$-left adjoint; equivalently, $\otimes$ preserves $\mathcal{T}$-colimits separately in each variable. $\triangleleft$

Now, in [Ste25b], whenever $\mathbb{F}_{\mathcal{T}}$ admitted products (such as $\mathcal{T} = \mathcal{O}_G$, or any other $\mathcal{T}$ with a terminal object), we defined the *Boardman-Vogt tensor product of* $\mathcal{T}$-*operads* $\overset{\mathrm{BV}}{\otimes}$ satisfying the following.

Proposition 1.23 ([Ste25d, Thm. E])**.** $\left(\mathrm{Op}_{\mathcal{T}}, \overset{\mathrm{BV}}{\otimes}\right)$ *underlies a presentably symmetric monoidal $\mathcal{T}$-category $\underline{\mathrm{Op}}^{\otimes}_{\mathcal{T}}$.*

We refer to the internal mapping object in $\underline{\mathrm{Op}}^{\otimes}_{\mathcal{T}}$ as the *pointwise $\mathcal{T}$-operad of $\mathcal{P}$-algebras in $\mathcal{Q}$*, so that

$$\mathrm{Alg}_{\mathcal{O}}\underline{\mathrm{Alg}}^{\otimes}_{\mathcal{P}}(\mathcal{Q}) \simeq \mathrm{Alg}_{\mathcal{O}\overset{\mathrm{BV}}{\otimes}\mathcal{P}}(\mathcal{Q}).$$

1.1.5. *(Co)cartesian structures, $\mathcal{O}$-(co)monoids, Morita theory.* We will need unitality:

Definition 1.24. A $\mathcal{T}$-operad $\mathcal{O}^{\otimes}$ is *unital* if, for all $\mathcal{O}$-colors C, the space $\mathcal{O}(\varnothing;C)$ is contractible; we say that $\mathcal{O}^{\otimes}$ is *reduced* if it additionally has one color. These span full subcategories $\mathrm{Op}_{\mathcal{T}}^{\mathrm{red}} \subset \mathrm{Op}_{\mathcal{T}}^{\mathrm{uni}} \subset \mathrm{Op}_{\mathcal{T}}$. ◁

Now, this allows us to state a universal property for a crucially important family of examples.

Proposition 1.25 ([Ste25d])**.** *The functor $U\colon \mathrm{Op}_{\mathcal{T}}^{\mathrm{uni}} \to \mathrm{Cat}_{\mathcal{T}}$ admits a fully faithful left adjoint $(-)^{\sqcup}$; for all unital $\mathcal{O}^{\otimes}$, the forgetful functor is an equivalence*

$$\underline{\mathrm{Alg}}_{\mathcal{O}}^{\otimes}\left(\mathcal{C}^{\sqcup}\right) \xrightarrow{\sim} \underline{\mathrm{Fun}}_{\mathcal{T}}\left(U\mathcal{O},\mathcal{C}\right)^{\sqcup}.$$

Moreover $\mathcal{C}^{\sqcup}$ is a $\mathcal{T}$-symmetric monoidal ∞-category if and only if $\mathcal{C}$ has finite $\mathcal{T}$-coproducts; a $\mathcal{T}$-symmetric monoidal ∞-category $\mathcal{C}^{\otimes}$ which is unital as a $\mathcal{T}$-operad has identity classifying an equivalence

$$\mathcal{C}^{\otimes} \xrightarrow{\sim} \mathcal{C}^{\sqcup}$$

if and only if its S-indexed tensor functor is equivalent to the S-indexed coproduct for all $S \in \underline{\mathbb{F}}_{\mathcal{T}}$.

For concreteness, we will painstakingly construct a topological model for this.

Construction 1.26. Let $\mathcal{C}$ be a topological $\mathcal{T}$-category, i.e. a functor $\mathcal{T}^{\mathrm{op}} \to \mathrm{Cat}_{\mathrm{Top}}$. We define a topological $\mathcal{T}$-operad $\mathrm{Tot}\mathcal{C}^{\sqcup} \longrightarrow \mathrm{Span}(\mathbb{F}_{\mathcal{T}})$ to have object fibers $\mathrm{Ob}\mathcal{C}^{\sqcup}_S \coloneqq \mathrm{Ob}\mathcal{C}^S$, and with topological mapping space over a span $R \xleftarrow{f} T \xrightarrow{g} S$ given by

$$\mathrm{Mul}^{R\xleftarrow{f}T\xrightarrow{g}S}_{\mathcal{C}^{\sqcup}}\left((X_U);(Y_V)\right) \coloneqq \prod_{W\in \mathrm{Orb}(T)} \mathrm{Map}_{\mathcal{C}}\left(X_{f(W)}, Y_{g(W)}\right),$$

with composition accomplished by composition of maps in $\mathcal{C}$; indeed, this is a topological presentation of the construction given the same name in [Ste25d, § A], i.e. this presents the cocartesian $\mathcal{T}$-operad structure on $\mathcal{C}$. ◁

An important property of these is the following, which follows by combining [Ste25d, Cor. 2.13] and [Ste25d, Prop. 2.18] with Proposition 1.25.

Corollary 1.27. *Given $\mathcal{T}$-∞-categories $\mathcal{C},\mathcal{D}$, there exists an equivalence $\mathcal{C}^{\sqcup} \overset{\mathrm{BV}}{\otimes} \mathcal{D}^{\sqcup} \simeq (\mathcal{C}\times\mathcal{D})^{\sqcup}$.*

Example 1.28. Proposition 1.25 shows $*^{\sqcup} \simeq \mathrm{Comm}_{\mathcal{T}}^{\otimes} = (\mathrm{Span}(\mathbb{F}_{\mathcal{T}}) = \mathrm{Span}(\mathbb{F}_{\mathcal{T}}))$; in particular, Corollary 1.27 shows that $\mathrm{Comm}_{\mathcal{T}}^{\otimes}$ is $\overset{\mathrm{BV}}{\otimes}$-idempotent. ◁

Dually, when $\mathcal{C}$ admits indexed products, it admits an essentially unique $\mathcal{T}$-symmetric monoidal structure $\mathcal{C}^{\times}$ whose indexed tensor products are indexed products. These also have explicitly describable algebras:

Proposition 1.29 ([Ste25d, Cor. 1.53])**.** *Given $\mathcal{O}^{\otimes}$ a $\mathcal{T}$-preoperad, the forgetful functor*

$$\mathrm{Alg}_{\mathcal{O}}\left(\underline{\mathrm{Coeff}}^{\mathcal{T}}\mathcal{C}^{\times}\right) \to \mathrm{Fun}(\mathrm{Tot}\mathcal{O}^{\otimes},\mathcal{C})$$

is fully faithful with image the Segal $\mathrm{Tot}\mathcal{O}^{\otimes}$-objects.

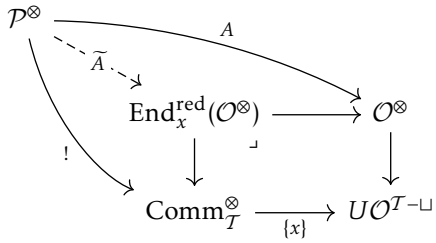
In particular, the atomic orbital counterpart to Elmendorf's theorem defines $\underline{\mathcal{S}}_{\mathcal{T}} \coloneqq \underline{\mathrm{Coeff}}^{\mathcal{T}}\mathcal{S}$, so this characterizes Segal $\mathrm{Tot}\mathcal{O}$-spaces as $\mathcal{O}$-algebras in $\mathcal{T}$-spaces.

1.1.6. *$\mathcal{T}$-operadic disintegration and assembly.* In [Ste25d, § 3.2], given a unital $\mathcal{T}$-operad $\mathcal{O}^{\otimes}$ and a V-object $x \in \mathcal{O}_V$, we defined the *reduced endomorphism V-operad* as the pullback

$$\begin{array}{ccc} \mathrm{End}_x^{\mathrm{red}}(\mathcal{O}^{\otimes}) & \longrightarrow & \mathrm{Res}_V^{\mathcal{T}}\mathcal{O}^{\otimes} \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Comm}_V^{\otimes} & \xrightarrow[\{x\}]{} & \left(\mathrm{Res}_V^{\mathcal{T}}U\mathcal{O}\right)^{V-\sqcup} \end{array}$$

where $\{x\}$ classifies X under Proposition 1.25 and Example 1.28. We will use the following terminology.

Construction 1.30. Given $\mathcal{P}^{\otimes}$ a reduced $\mathcal{T}$-operad, $\mathcal{O}^{\otimes}$ a unital $\mathcal{T}$-operad, and $A\colon \mathcal{P}^{\otimes} \to \mathcal{O}^{\otimes}$ an $\mathcal{O}$-algebra in $\mathcal{P}^{\otimes}$ with underlying $\mathcal{T}$-object $x \in \Gamma^{\mathcal{T}} U\mathcal{O}$, the *reduction of* A is the $\mathcal{P}$-algebra $\widetilde{A}$ defined as



This underlies an adjunction between *pointed* unital $\mathcal{T}$-operads and reduced $\mathcal{T}$-operads [Ste25a, Prop. 45]. ◁

These admit *disintegration and assembly*, i.e. an operadic form of straightening-unstraightening:

Proposition 1.31 ([Ste25d, Thm. 3.6])**.** *Suppose $\mathcal{O}^{\otimes}$ is a unital $\mathcal{T}$-operad such that $U\mathcal{O}$ is a $\mathcal{T}$-space. Then, the maps $\mathrm{End}_X^{\mathrm{red}}(\mathcal{O}^{\otimes}) \to \mathrm{Res}_V^{\mathcal{T}} \mathcal{O}^{\otimes}$ assemble to a $\mathcal{T}$-colimit diagram*

$$\underline{\mathrm{colim}}_{X \in U\mathcal{O}} \mathrm{End}_X^{\mathrm{red}}(\mathcal{O}^{\otimes}) \xrightarrow{\sim} \mathcal{O}^{\otimes}.$$

Since $\underline{\mathrm{Alg}}^{\otimes}_{(-)}(\mathcal{P}^{\otimes})$ is a $\mathcal{T}$-right adjoint out of $\underline{\mathrm{Op}}^{\mathrm{op}}_{\mathcal{T}}$, when $U\mathcal{O}$ is a $\mathcal{T}$-space, we acquire a $\mathcal{T}$-limit

$$\underline{\mathrm{Alg}}^{\otimes}_{\mathcal{O}}(\mathcal{P}) \simeq \lim_{x \in U\mathcal{O}} \underline{\mathrm{Alg}}_{\mathrm{End}_x^{\mathrm{red}}(\mathcal{O})}(\mathcal{P}).$$

More directly useful to us is the following tool for computing Boardman-Vogt tensor products.

Corollary 1.32 ([Ste25d, Cor. 3.9])**.** *Suppose $\varphi\colon \mathcal{O}^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathcal{P}^{\otimes} \to \mathcal{Q}^{\otimes}$ is a map of $\mathcal{T}$-operads inducing equivalences*

$$U\mathcal{O} \times U\mathcal{P} \xrightarrow{\sim} U\mathcal{Q}, \qquad \text{and}$$

$$\mathrm{End}_x^{\mathrm{red}}(\mathcal{O}^{\otimes}) \overset{\mathrm{BV}}{\otimes} \mathrm{End}_y^{\mathrm{red}}(\mathcal{P}^{\otimes}) \xrightarrow{\sim} \mathrm{End}_{(x,y)}^{\mathrm{red}}(\mathcal{Q}^{\otimes}),$$

and $U\mathcal{O}, U\mathcal{Q}$ are $\mathcal{T}$-spaces. Then, φ is an equivalence.

1.1.7. *$\mathcal{T}$-symmetric monoidal envelopes.* As mentioned before, [BHS22; NS22] constructed an adjunction

$$\mathrm{Env}\colon \mathrm{Op}_{\mathcal{T}} \rightleftarrows \mathrm{Cat}^{\otimes}_{\mathcal{T}}\colon U$$

in particular, in [BHS22, Prop. 4.2.1] they showed that the associated *sliced* left adjoint

$$\mathrm{Env}^{/}\colon \mathrm{Op}_{\mathcal{T}} \simeq \mathrm{Op}_{\mathcal{T},/\mathrm{Comm}^{\otimes}_{\mathcal{T}}} \hookrightarrow \mathrm{Cat}^{\otimes}_{\mathcal{T},/\underline{\mathbb{F}}^{\sqcup}_{\mathcal{T}}}$$

witnesses $\mathcal{T}$-operads as a colocalizing subcategory of $\mathcal{T}$-symmetric monoidal ∞-categories over $\underline{\mathbb{F}}^{\sqcup}_{\mathcal{T}}$, cut out by *equifiberation condition*. For our purposes, we restate the following characterization.

Proposition 1.33. *Suppose $\pi\colon \mathcal{C}^{\otimes} \to \underline{\mathbb{F}}^{\sqcup}_{\mathcal{T}}$ is a $\mathcal{T}$-symmetric monoidal functor such that the following squares are cartesian:*

$$\begin{array}{ccc} \mathcal{C}^V \times \mathcal{C}^V & \xrightarrow{\otimes} & \mathcal{C}^V \\ \downarrow & & \downarrow \\ \mathbb{F}_V \times \mathbb{F}_V & \xrightarrow{\sqcup} & \mathbb{F}_V \end{array} \qquad\qquad \begin{array}{ccc} \mathcal{C}^U & \xrightarrow{N_U^V} & \mathcal{C}^V \\ \downarrow & & \downarrow \\ \mathbb{F}_U & \xrightarrow{\mathrm{Ind}_U^V} & \mathbb{F}_V \end{array}$$

Let $\widetilde{\mathcal{C}}^{\otimes} \subset$ be the full $\mathcal{T}$-suboperad spanned by V-objects lying over $*_V$. Then, $\widetilde{\mathcal{C}}^{\otimes} \hookrightarrow \mathcal{C}^{\otimes}$ classifies an equivalence $\mathrm{Env}\big(\widetilde{\mathcal{C}}\big)^{\otimes} \xrightarrow{\sim} \mathcal{C}^{\otimes}$.

Proof. These conditions are equivalent to π being an equifibration by the same argument as [BHS22, Lem. 5.2.16]; the rest follows straightforwardly from [BHS22, Prop. 4.2.1]. □

We will use this to compute box products of $\mathcal{T}$-symmetric monoidal ∞-categories, where the Box product is the (localized) day convolution tensor product with respect to $\wedge$ on $\mathsf{Span}(\mathbb{F}_{\mathcal{T}})$ and $\times$ on Cat. Crucially, we use the following.

Proposition 1.34 ([Ste25b, Cor. 3.15])**.** *There exists a natural equivalence $\mathrm{Env}_{\mathcal{O}} \square \mathrm{Env}_{\mathcal{P}} \simeq \mathrm{Env}_{\mathcal{O} \overset{\mathrm{BV}}{\otimes} \mathcal{P}}$, and this uniquely characterizes $\overset{\mathrm{BV}}{\otimes}$.*

1.1.8. *$\mathcal{O}$-monoids, Morita equivalences.* With the above technology, we may reduce many computations of Boardman-Vogt tensor products to the reduced case. In this case, we will crucially use *Morita theory* in the following sense.

Definition 1.35. Let $\mathfrak{O}, \mathfrak{P}$ be algebraic patterns and $\varphi\colon \mathfrak{O} \to \mathfrak{P}$ a functor between their underlying ∞-categories. We say that φ is *compatible with Segal objects* if the pullback functor restricts to Segal objects:

$$\begin{array}{ccc} \mathsf{Fun}(\mathfrak{P},\mathcal{C}) & \xrightarrow{-\varphi^*} & \mathsf{Fun}(\mathfrak{O},\mathcal{C}) \\ \uparrow & & \uparrow \\ \mathsf{Seg}_{\mathfrak{P}}(\mathcal{C}) & \xrightarrow[\varphi^*_{\mathrm{Seg}}]{} & \mathsf{Seg}_{\mathfrak{O}}(\mathcal{C}) \end{array}$$

We additionally say that f is a *Morita equivalence* if φ^*_{Seg} is an equivalence. ◁

Example 1.36. A functor $\varphi\colon \mathfrak{O} \to \mathfrak{P}$ is a *strong Segal morphism* if, for all $O \in \mathfrak{O}$, the induced functor

$$\varphi^{\mathrm{el}}_{O/}\colon \mathfrak{O}^{\mathrm{el}}_{O/} \longrightarrow \mathfrak{P}^{\mathrm{el}}_{\varphi(O)/}$$

is initial. These are compatible with Segal objects (see [CH21, Rmk. 4.4]). ◁

Example 1.37 ([BHS22, Def. 4.11.1])**.** Given $\mathfrak{B}$ an algebraic pattern and $\mathfrak{O} \to \mathfrak{B}$ a functor admitting cocartesian lifts over inert morphisms, $\mathfrak{O}$ admits a canonical algebraic pattern structure whose inert morphisms are cocartesian lifts of inert arrows, whose active arrows are are arbitrary lifts of active arrows, and whose elementary objects are arbitrary lifts of elementary objects. In particular, given $\mathcal{O}^{\otimes} \to \mathsf{Span}(\mathbb{F}_{\mathcal{T}})$ a $\mathcal{T}$-preoperad, there is a canonical algebraic pattern structure on $\mathcal{O}^{\otimes}$. ◁

Now, by examining the *free Segal $\mathcal{O}$-space monad*, we have the following convenient property.

Proposition 1.38 ([Ste25b, Thm. 2.70])**.** *Suppose $\varphi\colon \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ is a map of $\mathcal{T}$-operads inducing an equivalence*

$$\mathrm{Alg}_{\mathcal{P}}(\underline{\mathcal{S}}_{\mathcal{T}}) \xrightarrow{\sim} \mathrm{Alg}_{\mathcal{O}}(\underline{\mathcal{S}}_{\mathcal{T}})$$

and an essential surjection $U\mathcal{O} \xrightarrow{\sim} U\mathcal{P}$; then, φ is an equivalence.

This allows us to identify Morita theory of $\mathcal{T}$-preoperads with a pre-completed $\mathcal{T}$-operad theory.

Corollary 1.39 ([Ste25d, Cor. A.26])**.** *If $\varphi\colon \mathcal{O}^{\otimes} \to \mathcal{P}^{\otimes}$ is a map of $\mathcal{T}$-preoperads inducing an essential surjection*

$$UL_{\mathrm{Op}_{\mathcal{T}}}\mathcal{O}^{\otimes} \longrightarrow UL_{\mathrm{Op}_{\mathcal{T}}}\mathcal{P}^{\otimes},$$

then φ is an $L_{\mathrm{Op}_{\mathcal{T}}}$-equivalence if and only if it's a Morita equivalence.

Now, we will move on to study this further via a theory of *weak approximations*, but first, we have some more discussion of point-set matters, as we will eventually use it to make various diagrams commute.

1.1.9. *Topological $\mathcal{T}$-subpreoperads.* We will, from time to time, enact the following evil.

Definition 1.40. A topological functor $F\colon \mathcal{C} \to \mathcal{D}$ is *point-set monic* if it is injective on objects and for each $(X,Y) \in \mathsf{Ob}(\mathcal{C})$ of hom-topological spaces $\underline{\mathrm{Hom}}(X,Y) \to \underline{\mathrm{Hom}}(FX,FY)$ is a monomorphism in Top. ◁

Evil is relatively common, per the following corollary of the axiom of choice.

Lemma 1.41. *Suppose $F\colon \mathcal{C} \to \mathcal{D}$ is **essentially** injective and each morphism of hom-topological spaces $\underline{\mathrm{Hom}}(X,Y) \to \underline{\mathrm{Hom}}(FX,FY)$ is monic. Then, there exists a diagram of topological functors*

$$\begin{array}{ccc} \mathcal{C}' & \xrightarrow{F'} & \mathcal{D}' \\ \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \beta'} \\ \mathcal{C} & \xrightarrow{F} & \mathcal{D} \end{array}$$

such that F' is point-set monic and β, β' are Dwyer-Kan equivalences.

Proof. Choose $\mathcal{C}', \mathcal{D}'$ to be the full topological subcategories spanned by skeleta for the underlying categories of $\mathcal{C}$ and $\mathcal{D}$; and F' has monic hom functions by cancellation and is injective on objects by skeletality. □

Now, this was designed for the sake of the following proposition.

Proposition 1.42. *Suppose we have a diagram of topological $\mathcal{T}$-preoperads specifying the solid arrows of*

$$\begin{array}{ccc} \mathcal{O}^{\otimes} & \overset{\varphi}{\dashrightarrow} & \mathcal{P}^{\otimes} \\ \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \iota'} \\ \mathcal{Q}^{\otimes} & \xrightarrow{\psi} & \mathcal{R}^{\otimes} \end{array}$$

where ι' is point-set monic. Then, there exists at most one topological functor φ filling in the dashed arrow such that the diagram strictly commutes, in which case φ presents a morphism of $\mathcal{T}$-preoperads. This filler exists if and only if the following conditions are satisfied:

(a) for all $\mathbf{C} \in \mathcal{O}^{\otimes}$, $\psi\iota(\mathbf{C})$ is in the image of ι', and

(b) for all multi-operations $\mu \in \mathrm{Mul}^{f}_{\mathcal{O}}(\mathbf{C};\mathbf{D})$, the element $\iota\psi(\mu)$ is in the image of ι'.

In this case, $\varphi(\mathbf{C}) = \iota'^{-1}\psi\iota(\mathbf{C})$ and $\varphi(\mu) = \iota'^{-1}\psi\iota(\mu)$.

Proof. Since ι' is point-set monic, if φ exists, then it is compatible with inert-cocartesian lifts, i.e. it's a morphism of $\mathcal{T}$-preoperads. Now, note that (a) and (b) are satisfied is φ exists, so instead assume that (a) and (b) are satisfied; then, we define the object function of φ by $\varphi(\mathbf{C}) \coloneqq \iota'^{-1}\psi\iota(\mathbf{C})$, which is well-defined and unique by assumption, and the morphism-function by $\varphi(\mu) = \iota'^{-1}\psi\iota(\mu)$. The fact that this is functorial follows by the fact that the composite $\iota' \circ \varphi = \psi \circ \iota$ is functorial and ι' is point-set monic. □

In the situation of Proposition 1.42, we say that ι' embeds $\mathcal{P}^{\otimes}$ as a *topological $\mathcal{T}$-subpreoperad*.

1.2. **Weak approximations and reflective localizations.** We now study Corollary 1.50.

1.2.1. *$\mathcal{T}$-algebraic patterns, Morita equivalences.* We begin with an auxiliary notion, which captures *Segal objects* in $\mathcal{T}$-categories which are not necessarily $\mathcal{T}$-categories of coefficient systems.

Definition 1.43. A *$\mathcal{T}$-algebraic pattern* is a functor $\mathcal{T}^{\mathrm{op}} \longrightarrow \mathrm{AlgPatt}$. Given a $\mathcal{T}$-algebraic pattern $\mathfrak{B}$ and a complete $\mathcal{T}$-category $\mathcal{C}$, we define the *$\mathcal{T}$-category of $\mathfrak{B}$-Segal objects in $\mathcal{C}$*

$$\underline{\mathrm{Seg}}_{\mathfrak{B}}(\mathcal{C}) \coloneqq \underline{\mathrm{Fun}}_{\mathcal{T}}(\mathfrak{B},\mathcal{C}) \times_{\underline{\mathrm{Fun}}_{\mathcal{T}}(\mathfrak{B}^{\mathrm{int}},\mathcal{C})} \underline{\mathrm{Fun}}_{\mathcal{T}}\left(\mathfrak{B}^{\mathrm{el}},\mathcal{C}\right);$$

we also write $\mathrm{Seg}_{\mathfrak{B}}(\mathcal{C}) \coloneqq \Gamma^{\mathcal{T}}\underline{\mathrm{Seg}}_{\mathfrak{B}}(\mathcal{C})$. We define *Morita equivalences* and *strong $\mathcal{T}$-Segal morphisms* similarly to the nonequivariant case. ◂

Proposition 1.44. *Postcomposition with fixed points furnishes an equivalence*

$$\mathrm{Seg}_{\mathfrak{B}}\left(\underline{\mathrm{Coeff}}^{\mathcal{T}}\mathcal{D}\right) \xrightarrow{\sim} \mathrm{Seg}_{\mathrm{Tot}\mathfrak{B}}(\mathcal{D})$$

Proof. In sight of [Sha23, Prop. 3.10], we're tasked with proving that the left diagram is a $\mathcal{T}$-right Kan extension if and only if the right is a right Kan extension:

$$\begin{array}{ccc} \mathfrak{B}^{\mathrm{el}} & \longrightarrow & \underline{\mathrm{Coeff}}^{\mathcal{T}}\mathcal{D} \\ \downarrow & \Nearrow & \nearrow \\ \mathfrak{B}^{\mathrm{int}} & & \end{array} \qquad \begin{array}{ccccc} \mathrm{Tot}\mathfrak{B}^{\mathrm{el}} & \longrightarrow & \mathrm{Coeff}^{\mathcal{T}}\mathcal{D} & \longrightarrow & \mathcal{D} \\ \downarrow & \Nearrow & & \nearrow & \\ \mathrm{Tot}\mathfrak{B}^{\mathrm{int}} & & & & \end{array}$$

Indeed, this follows by the same proof as [Sha23, Prop. 10.10], using [Sha23, Prop. 5.6] in place of [Sha23, Prop. 5.5]. □

This directly relates our notion to equivariant higher algebra.

Corollary 1.45. *If $\mathcal{O}^{\otimes}$ is a Γ-$\mathcal{T}$-preoperad, then the forgetful $\mathcal{T}$-functor*

$$\underline{\mathrm{Alg}}_{\mathcal{O}}(\mathcal{C}^{\times}) \longrightarrow \underline{\mathrm{Fun}}_{\mathcal{T}}\left(\mathrm{Tot}_{\mathcal{T}}\mathcal{O}^{\otimes},\mathcal{C}\right)$$

is fully faithful with image the Segal $\mathrm{Tot}_{\mathcal{T}}\mathcal{O}$-objects; in particular, Γ-$\mathcal{T}$-preoperad maps are $\mathcal{T}$-Morita equivalences if and only if they are $L_{\mathrm{Op}_{\mathcal{T}}}$-equivalences.

Proof. This follows by combining Proposition 1.44 with [Ste25d, Prop. 1.51]. □

1.2.2. *Weak approximations for $\mathcal{T}$-algebraic patterns.* We now approach weak approximations.

Definition 1.46. We say that a strong Segal morphism of $\mathcal{T}$-algebraic patterns $\alpha\colon \mathfrak{P} \to \mathfrak{E}$ is *BC in the sense of Barkan* if, for all $X \in \underline{\mathrm{Seg}}_{\mathcal{O}}(\mathcal{C})$ with $\mathcal{C}$ complete, the Beck-Chevalley map

$$(\alpha_* X)\,|_{\mathfrak{E}^{\mathrm{int}}} \longrightarrow \alpha_*^{\mathrm{int}}\left(X\,|_{\mathfrak{P}^{\mathrm{int}}}\right)$$

is an equivalence. ◁

The purpose of this definition is the following.

Proposition 1.47. *If α is BC in the sense of Barkan, then α_* preserves Segal $\mathcal{T}$-objects; if moreover α^{el} admits a fully faithful right adjoint β, then α^* is the inclusion of the coreflective $\mathcal{T}$-subcategory of Segal $\mathfrak{P}$-objects $Y\colon \mathfrak{P} \to \mathcal{C}$ whose elementary $\mathcal{T}$-functor $Y^{\mathrm{el}}\colon \mathfrak{P}^{\mathrm{el}} \to \mathcal{C}$ factors through β.*

We refer to elementary objects admitting such a factorization as *locally constant*.

Proof. See the proof of [CH21, Prop. 6.3] for compatibility of α_* with Segal objects, leading to a $\mathcal{T}$-adjunction

$$\alpha^*\colon \underline{\mathrm{Seg}}_{\mathfrak{E}}(\mathcal{C}) \rightleftarrows \underline{\mathrm{Seg}}_{\mathfrak{P}}(\mathcal{C})\colon \alpha_*.$$

The remaining statements are checked fiberwise, so we may replace $\underline{\mathrm{Seg}}$ with Seg. We must show that the unit of the adjunction on $\mathrm{Seg}_{\bullet}(\mathcal{C})$is always an equivalence, and the counit an equivalence precisely on locally constant Segal objects.

Now, note by the BC assumption that the unit of this adjunction restricts to elementaries as

$$\begin{aligned} X^{\mathrm{el}} &\longrightarrow (\alpha_*\alpha^* X)^{\mathrm{el}} \\ &\simeq \alpha_*^{\mathrm{el}}(\alpha^* X)^{\mathrm{el}} \\ &\simeq \alpha_*^{\mathrm{el}}\alpha^{\mathrm{el},*} X^{\mathrm{el}} \\ &\simeq \beta^*\alpha^{\mathrm{el},*} X^{\mathrm{el}} \\ &\simeq \left(\alpha^{\mathrm{el}} \circ \beta\right)^* X^{\mathrm{el}} \end{aligned}$$

with composite transformation given by functoriality of pullbacks along the counit of the adjunction $\alpha^{\mathrm{el}} \dashv \beta$; this is an equivalence by fully-faithfulness of β, and $(-)^{\mathrm{el}}$ is conservative, so the unit is an adjunction, implying that α^* is fully faithful.

Now, given Y a Segal $\mathfrak{P}$-object, a similar argument shows that the counit of the adjunction $\alpha^* \dashv \alpha_*$ restricts on elementaries to an arrow equivalent to the transformation $\left(\beta \circ \alpha^{\mathrm{el}}\right)^* Y^{\mathrm{el}} \to Y^{\mathrm{el}}$ given by pullback along the unit of the adjunction $\alpha^{\mathrm{el}} \dashv \beta$; this is an equivalence if and only if Y^{el} factors through β, as desired. □

Now, we may recognize these via the following condition on active arrow $\mathcal{T}$-categories.

Proposition 1.48. *Suppose $\alpha\colon \mathfrak{P} \to \mathfrak{E}$ is a strong Segal morphism of $\mathcal{T}$-algebraic patterns such that, for every $P \in \mathfrak{P}$, the homotopy fibers of*

$$\alpha^{\mathrm{act}}_{/P}\colon \mathfrak{P}^{\mathrm{act}}_{/P} \longrightarrow \mathfrak{E}^{\mathrm{act}}_{/\alpha P}$$

are weakly contractable, where $\mathfrak{P}^{\mathrm{act}}_{/P} = \mathrm{Ar}^{\mathrm{act}}(\mathfrak{P}) \times_{\mathfrak{P}} \{P\}$. Then α is BC in the sense of Barkan.

Proof. The proof of [Har, Prop. 4.2.18] works *mutatis mutandis*. □

Warning 1.49. There is a subtle difference between this and the condition that α possesses *unique lifting for active arrows* in the sense of [CH21, Def. 6.1], but to the author's knowledge, either condition only appears for the purpose of verifying that α is BC in the sense of Barkan. ◁

We immediately acquire the following corollary.

Corollary 1.50. *Suppose $\alpha\colon \mathfrak{P} \to \mathfrak{E}$ is a strong Segal morphism of $\mathcal{T}$-algebraic patterns such that*

(WA-a) for every $P \in \mathfrak{P}$, the homotopy fibers of

$$\alpha^{\mathrm{act}}_{/P}\colon \mathfrak{P}^{\mathrm{act}}_{/P} \longrightarrow \mathfrak{E}^{\mathrm{act}}_{/\alpha P}$$

are weakly contractable, and

(WA-b) $\alpha^{\mathrm{el}}\colon \mathfrak{P}^{\mathrm{el}} \to \mathfrak{E}^{\mathrm{el}}$ *is a reflective* $\mathcal{T}$*-localization.*

Then, for any complete $\mathcal{T}$*-*∞*-category* $\mathcal{C}$*, pullback yields a fully faithful embedding*

$$\alpha^*\colon \underline{\mathrm{Seg}}_{\mathfrak{E}}(\mathcal{C}) \hookrightarrow \underline{\mathrm{Seg}}_{\mathfrak{P}}(\mathcal{C})$$

with essential image the (colocalizing) $\mathcal{T}$*-subcategory of* α*-locally constant Segal* $\mathfrak{P}$*-objects. In particular, if* α^{el} *is an equivalence, then* α *is a Morita equivalence.*

In the former case, we say that α is a *weak* $\mathcal{T}$*-approximation*, and the latter a *strong* $\mathcal{T}$*-approximation*; this only mildly generalizes the homonymous concepts of [Har] in the case $\mathcal{T} = *$.

Now, we will want to transport these between patterns underlying the our two models for $\mathcal{T}$-operads, for which the following proposition is both useful and immediate.

Proposition 1.51. *Suppose we are given a pullback square of* $\mathcal{T}$*-algebraic patterns*

$$\begin{array}{ccc} \mathfrak{P}' & \longrightarrow & \mathfrak{P} \\ {\scriptstyle \alpha'}\downarrow & \lrcorner & \downarrow{\scriptstyle \alpha} \\ \mathfrak{E}' & \xrightarrow[s]{} & \mathfrak{E} \end{array}$$

satisfying the following conditions:

(a) for every $P \in \mathfrak{P}'$*, the* $\mathcal{T}$*-functor* $s^{\mathrm{act}}_{/P}\colon \mathfrak{P}^{\mathrm{act}\prime}_{/P} \longrightarrow \mathfrak{P}^{\mathrm{act}}_{/sP}$ *is an equivalence,*

(b) $s^{\mathrm{el}}\colon \mathfrak{P}^{\mathrm{el}\prime} \longrightarrow \mathfrak{P}^{\mathrm{el}}$ *is an equivalence, and*

(c) α *is a weak (resp. strong)* $\mathcal{T}$*-approximation.*

Then α' *is a weak (strong)* $\mathcal{T}$*-approximation.*

Corollary 1.52. *Suppose* $\alpha\colon \mathcal{P}^{\otimes} \to \mathcal{E}^{\otimes}$ *is a weak approximation of* $\mathcal{T}$*-preoperads; then* $s^*\alpha$ *is a weak approximation of* Γ*-*$\mathcal{T}$*-preoperads.*

Proof. The conditions of Proposition 1.51 were shown in [Ste25b, Thm. A.1] (as in [BHS22, Thm. 5.1.1]). □

1.2.3. *Reflective quotients, algebras over weak approximations.* For posterity, we will develop a version of Corollary 1.50 for algebras in a more general setting. This uses the following.

Definition 1.53. Given $\mathcal{P}^{\otimes}$ a $\mathcal{T}$-preoperad and $L\colon U\mathcal{P} \to \mathcal{C}$ a $\mathcal{T}$-localization, we define the corresponding *reflective quotient* as the local pushout $\mathrm{Refl}^{\otimes}_{\mathcal{C}}(\mathcal{P}) \coloneqq L_{\mathrm{Op}_{\mathcal{T}}}\mathcal{P}^{\otimes} \sqcup_{\mathrm{triv}(U\mathcal{P})^{\otimes}} \mathrm{triv}(\mathcal{C})^{\otimes}$. ◁

The following proposition follows immediately by unwinding definitions.

Proposition 1.54. *Pullback along the structure map* $\mathcal{P}^{\otimes} \to \mathrm{Refl}^{\otimes}_{\mathcal{C}}(\mathcal{P})$ *yields a filly faithful embedding*

$$\mathrm{Alg}_{\mathrm{Refl}_{\mathcal{C}}(\mathcal{P})}(\mathcal{Q}) \to \mathrm{Alg}_{\mathcal{P}}(\mathcal{Q})$$

with essential image spanned by those $\mathcal{P}$*-algebras whose underlying* $\mathcal{T}$*-funcor* $U\mathcal{P} \to U\mathcal{Q}$ *factors through* L*.*

This is closely related to weak approximations via the following.

Corollary 1.55. *If* $\mathcal{P}^{\otimes} \to \mathcal{E}^{\otimes}$ *is a weak approximation, then the induced map* $\varphi\colon \mathrm{Refl}^{\otimes}_{U\mathcal{E}}\mathcal{P} \to L_{\mathrm{Op}_{\mathcal{T}}}\mathcal{E}^{\otimes}$ *is an equivalence; in particular, for all* $\mathcal{T}$*-operads* $\mathcal{Q}$*, the pullback functor*

$$\mathrm{Alg}_{\mathcal{E}}(\mathcal{Q}) \longrightarrow \mathrm{Alg}_{\mathcal{P}}(\mathcal{Q})$$

is fully faithful with essential image the $\mathcal{P}$*-algebras whose color* $\mathcal{T}$*-functor factors as* $U\mathcal{P} \to U\mathcal{E} \to U\mathcal{Q}$*.*

Proof. Choosing $\mathcal{Q} \coloneqq \underline{\mathcal{S}}^{\times}_{\mathcal{T}}$, we acquire a diagram

$$\begin{array}{ccc} \mathrm{Seg}_{L_{\mathrm{Op}_{\mathcal{T}}}\mathcal{E}}(\mathcal{S}) & \xrightarrow{\sim} & \mathrm{Seg}_{\mathcal{E}}(\mathcal{S}) \\ \downarrow & & \downarrow \\ \mathrm{Seg}_{\mathrm{Refl}_{U\mathcal{E}}(\mathcal{P})}(\mathcal{S}) & \hookrightarrow & \mathrm{Seg}_{\mathcal{P}}(\mathcal{S}) \end{array}$$

so that the pullback functor $\mathrm{Seg}_{L\mathrm{Op}_{\mathcal{T}}\mathcal{E}}(\mathcal{S}) \to \mathrm{Seg}_{\mathrm{Refl}_{U\mathcal{E}}(\mathcal{P})}(\mathcal{S})$ is fully faithful by applying Corollary 1.50 and Propositions 1.29 and 1.54 and two-out-of-three. In fact, by Corollary 1.50 and Proposition 1.54 we find that it is essentially surjective, and hence an equivalence. By unwinding definitions, we find that φ is a U-equivalence, so Corollary 1.39 implies that it is an $L_{\mathrm{Op}_{\mathcal{T}}}$-equivalence. □

1.2.4. *A recognition result for weak approximations.* We will use a variation of [HA, Cor. 2.3.3.16]:

Proposition 1.56. *Suppose $\alpha\colon \mathcal{P}^{\otimes} \to \mathcal{E}^{\otimes}$ is a map of $\mathcal{T}$-preoperads over $\mathcal{C}^{\sqcup}$ such that $U\mathcal{E} \to \mathcal{C}$ is a cocartesian fibration whose fibers are spaces. Then, Condition (WA-a) of Corollary 1.50 is equivalent to the condition that, for all $P \in \mathrm{Tot}\mathcal{P}^{\otimes}$ and active $\mathcal{C}^{\sqcup}$-maps $f\colon D \to \pi_{\mathcal{P}}P$, the induced map of spaces*

$$\widetilde{\alpha}^{\mathrm{act},f}_{/P} : B\mathcal{P}^{\mathrm{act},f}_{/P} \longrightarrow \mathcal{E}^{\mathrm{act},f}_{/\alpha P}$$

is an equivalence, where $\mathcal{P}^{\mathrm{act},f}_{/P} := \mathcal{P}^{\mathrm{act}}_{/P} \times_{\mathcal{C}^{\sqcup,\mathrm{act}}_{/\pi_{\mathcal{P}}P}} \{f\}$. In particular, if $\mathcal{E}^{\otimes}$ is a $\mathcal{T}$-operad such that $U\mathcal{E} \to \mathcal{C}$ is an equivalence, then the above map has the form

$$\widetilde{\alpha}^{\mathrm{act},f}_{/P} : B\mathcal{P}^{\mathrm{act},f}_{/P} \longrightarrow \mathcal{E}(D; \pi_{\mathcal{P}}P)$$

Lemma 1.57. *In the situation of Proposition 1.56, the ∞-category $\mathcal{E}^{\mathrm{act},f}_{/\alpha P}$ is a space.*

Proof. The assumptions guarantee that the source functor

$$\mathcal{E}^{\mathrm{act},f}_{/\alpha P} \to \mathcal{E}_D$$

has codomain a space; moreover, the fibers are always spaces, as they are the relevant mapping spaces. Together these imply the result. □

Proof of Proposition 1.56. Fixing some $\tilde{f} \in \mathcal{E}^{\mathrm{act},f}_{/\alpha P}$ with underlying $\mathcal{C}^{\sqcup}$-map f, we draw the diagram

$$\begin{array}{ccccc}
F & \longrightarrow & \mathcal{P}^{\mathrm{act},f}_{/P} & \longrightarrow & \mathcal{P}^{\mathrm{act}}_{/P} \\
\downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\
\{\tilde{f}\} & \longrightarrow & \mathcal{E}^{\mathrm{act},f}_{/\alpha P} & \longrightarrow & \mathcal{E}^{\mathrm{act}}_{/\alpha P} \\
 & & \downarrow & & \downarrow \\
 & & \{f\} & \longrightarrow & \mathcal{C}^{\sqcup,\mathrm{act}}_{/\pi P}
\end{array}$$

We're tasked with verifying that F is weakly contractible. But we may apply the adjunction between B and $\mathcal{S} \hookrightarrow \mathsf{Cat}$ together with Lemma 1.57 to construct a fiber square

$$\begin{array}{ccc}
BF & \longrightarrow & B\mathcal{P}^{\mathrm{act},f}_{/\mathcal{P}} \\
\downarrow & \lrcorner & \downarrow \\
\{\tilde{f}\} & \longrightarrow & \mathcal{E}^{\mathrm{act},f}_{/\alpha P}
\end{array}$$

In particular, weak contractibility of F for all $\{\widetilde{f}\}$ is equivalent to contractibility of the homotopy fibers of $\widetilde{\alpha}^{\mathrm{act},f}_{/P}$ for all f, which is equivalent to $\widetilde{\alpha}^{\mathrm{act},f}_{/P}$ being an equivalence for all (f,P), as desired. □

1.3. **Wreath products and the bookkeeping theorem.** We now study our model for $\overset{\mathrm{BV}}{\otimes}$, *wreath products.*

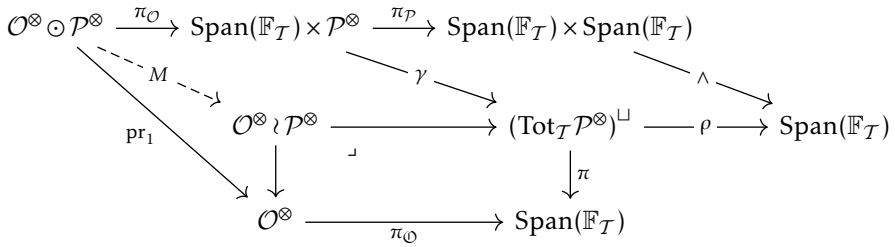
Construction 1.58. Given $\mathcal{O}^{\otimes}$ and $\mathcal{P}^{\otimes}$ a pair of $\mathcal{T}$-preoperads, in [Ste25d] we used the structure functor on $\mathrm{Tot}_{\mathcal{T}}\mathcal{P}^{\otimes}$ to construct an alternative structure functor $\rho\colon (\mathrm{Tot}_{\mathcal{T}}\mathcal{P}^{\otimes})^{\sqcup} \to \mathsf{Span}(\mathbb{F}_{\mathcal{T}})$; from this, we define the functor

$$\mathcal{O}^{\otimes} \wr \mathcal{P}^{\otimes} := \mathcal{O}^{\otimes} \times_{\mathsf{Span}(\mathbb{F}_{\mathcal{T}}),\pi} \left(\mathrm{Tot}_{\mathcal{T}}\mathcal{P}^{\otimes}\right)^{\sqcup} \xrightarrow{\rho} \mathsf{Span}(\mathbb{F}_{\mathcal{T}}).$$

Now, in the case that $\mathbb{F}_{\mathcal{T}}$ has products, we similarly define the functor

$$\mathcal{O}^{\otimes} \odot \mathcal{P}^{\otimes} \simeq \mathcal{O}^{\otimes} \times \mathcal{P}^{\otimes} \longrightarrow \mathsf{Span}(\mathbb{F}_{\mathcal{T}}) \times \mathsf{Span}(\mathbb{F}_{\mathcal{T}}) \xrightarrow{\wedge} \mathsf{Span}(\mathbb{F}_{\mathcal{T}}).$$

In [Ste25d] we extended this to a diagram



In particular, we acquire a map $M\colon \mathcal{O}^{\otimes}\odot\mathcal{P}^{\otimes}\longrightarrow\mathcal{O}^{\otimes}\wr\mathcal{P}^{\otimes}$ over $\mathrm{Span}(\mathbb{F}_{\mathcal{T}})$. We give $\mathcal{O}^{\otimes}\wr\mathcal{P}^{\otimes}$ the pullback pattern structure and $\mathcal{O}^{\otimes}\odot\mathcal{P}^{\otimes}$ the product pattern structure. ◁

We delay the proof of the following theorem until Page 24, focusing first on applications.

Theorem 1.59. *If $\mathcal{O}_1^{\otimes}\wr\mathcal{O}_2^{\otimes}\to\mathcal{O}_3^{\otimes}$ is a strong approximation, then it induces an equivalence*

$$\mathcal{O}_1^{\otimes}\overset{\mathrm{BV}}{\otimes}\mathcal{O}_2^{\otimes}\xrightarrow{\sim}\mathcal{O}_3^{\otimes}.$$

Now, to understand these better, note the following proposition.

Proposition 1.60. *Let $\mathcal{O}^{\otimes},\mathcal{P}^{\otimes}$ be $\mathcal{T}$-operads. Then, the fiber ∞-categories of $\mathcal{O}^{\otimes}\wr\mathcal{P}^{\otimes}$ are given by*

$$\left(\mathcal{O}^{\otimes}\wr\mathcal{P}^{\otimes}\right)_T\simeq\coprod_{T\to S}\mathcal{O}_S\times\mathcal{P}_T.$$

Moreover, given a multi-arity $g\colon T'\to T$, the multimorphism spaces of $\mathcal{O}^{\otimes}\wr\mathcal{P}^{\otimes}$ over g are

$$\mathrm{Mul}^{g}_{\mathcal{O}\wr\mathcal{P}}(h'\colon T'\to S',\mathbf{C}',\mathbf{D}';\ h\colon T\to S,\mathbf{C},\mathbf{D})\simeq\coprod_{\substack{f\colon S'\to S\\ fh=h'g}}\mathrm{Mul}^{f}_{\mathcal{O}}(\mathbf{C}';\mathbf{C})\times\mathrm{Mul}^{g}_{\mathcal{P}}(\mathbf{D}';\mathbf{D}).$$

1.3.1. *How do we use wreath products?* Now, we would like to describe a tractable part of maps out of wreath products. In particular, our main example will be understood primarily via its G-symmetric monoidal envelope, so we will want to understand some part of the associated G-symmetric monoidal functor

$$\mathrm{Env}(\mathcal{O})^{\otimes}\square\mathrm{Env}(\mathcal{P})^{\otimes}\to\mathrm{Env}(\mathcal{Q})^{\otimes}$$

associated with a map out of the topological wreath product. Now, the free functor $\mathrm{Cat}_G\to\mathrm{Cat}_G^{\otimes}$ admits an essentially unique symmetric monoidal structure by [Ste25b, Thm. 1.78], so the doctrinal adjunction yields a lax symmetric monoidal structure on the forgetful functor, which under composition yields a functor $\mathrm{Env}(\mathcal{O})\times\mathrm{Env}(\mathcal{P})\to\mathrm{Env}(\mathcal{Q})$. We can describe this functor.

Proposition 1.61. *Given unital topological G-operads $\mathcal{O}^t,\mathcal{P}^t,\mathcal{Q}^t$ and a topological relative pattern map*

$$\varphi^t\colon\mathcal{O}^t\wr\mathcal{P}^t\to\mathcal{Q}^t$$

define the topological G-functor $\psi\colon\mathrm{Env}^t_G(\mathcal{O})\times\mathrm{Env}^t_G(\mathcal{P})\to\mathrm{Env}^t_G(\mathcal{Q})$ on objects by

$$\psi((S,\mathbf{A},C),(T,\mathbf{B},D))=\left(S\times T,\varphi(\mathbf{A},\Delta^S\mathbf{B}),\varphi(C,\Delta^S D)\right)$$

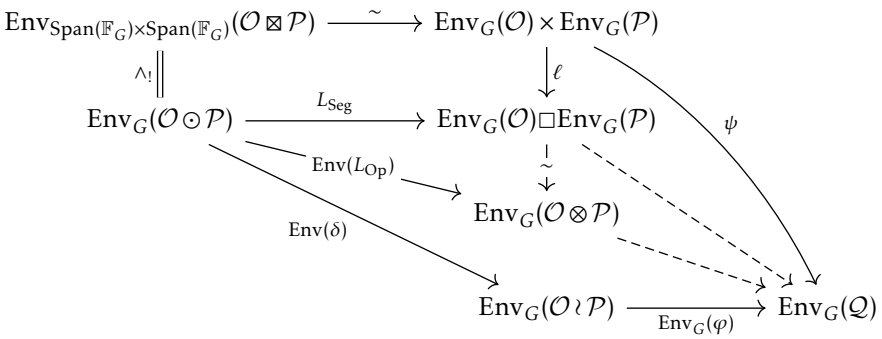
and on morphisms over $\mathbb{F}^t_G$ by

$$\begin{array}{ccccc}\mathrm{Map}^{g}_{\mathrm{Env}}((S,\mathbf{A},C),(T,\mathbf{B},D)) & & \xrightarrow{\quad\psi\quad} & & \mathrm{Map}^{g}_{\mathrm{Env}}\left(S\times T,\varphi(\mathbf{A},\Delta^S\mathbf{B}),\varphi(C,D)\right)\\ \| & & & & \|\\ \mathcal{O}(\mathbf{A};C)\times\mathcal{P}(\mathbf{B};D) & \xrightarrow{\Delta} & \mathcal{O}(\mathbf{A};C)\times\mathrm{Mul}^{g}_{\mathcal{P}}(\Delta^S\mathbf{B};\Delta^S D) & \xrightarrow{\varphi} & \mathcal{Q}(\varphi(\mathbf{A},\Delta^S\mathbf{B});\varphi(C,\Delta^S D))\end{array}$$

Then ψ presents a G-functor which is naturally equivalent to the composite

$$\mathrm{Env}_G(\mathcal{O})\times\mathrm{Env}_G(\mathcal{P})\xrightarrow{\ell}U\left(\mathrm{Env}_G(\mathcal{O})^{\otimes}\square\mathrm{Env}_G(\mathcal{P})^{\otimes}\right)\xrightarrow{\varphi}\mathrm{Env}_G(\mathcal{Q}),$$

Proof. We claim that subdiagram of solid arrows in the following commutes:



The universal properties of L_{Op} and L_{Seg} will then supply the dashed arrows making the whole diagram commute, and commutativity of the upper right triangle is the desired result.

The equivalence making the solid triangle commute is Proposition 1.38 and the commuativity of the solid square follows by unwinding definitions. We're left with identifying the two solid compositions $\mathrm{Env}_G(\mathcal{O})\times\mathrm{Env}_G(\mathcal{P})\to\mathrm{Env}_G(\mathcal{Q})$ agree with each other. Luckily, each occur as *topological* functors, so we may verify that they strictly agree on objects and arrows; this too can be seen by unwinding all of the definitions. □

The wreath product construction is of a nontrivial combinatorial nature; for simplicity's sake, we pre-macerate these combinatorics, yielding recognition results which will provide our interface for working with wreath products going forward. First, we describe how to compute wreath products of $\mathcal{T}$-operads with contractible structure spaces.

Proposition 1.62. *Suppose $\mathcal{O}^t,\mathcal{P}^t,\mathcal{Q}^t$ are topological $\mathcal{T}$-operads such that $\mathcal{Q}^t$ has contractible structure spaces. Then, a relative pattern map $\mathcal{O}^t\wr\mathcal{P}^t\to\mathcal{Q}^t$ is uniquely specified by a map of coefficient systems of sets*

$$\varphi\colon \coprod_{T\to S}\pi_0\mathcal{O}_S\times\pi_0\mathcal{P}_T\to\pi_0\mathcal{Q}_T$$

such that for all data

$$\begin{array}{ccc} T' & \xrightarrow{g} & T \\ \downarrow & & \downarrow \\ S' & \longrightarrow & V \end{array}; \qquad ((\mathbf{C}';C),(\mathbf{B}';\mathbf{B}))\in\mathcal{O}_{S'}\times\mathcal{O}_V\times\mathcal{P}_{T'}\times\mathcal{P}_T.$$

such that $\mathcal{O}(\mathbf{C}';C)\times\mathrm{Mul}^g_{\mathcal{P}}(\mathbf{B}';\mathbf{B})\neq\varnothing$, we have $\mathrm{Mul}^h_{\mathcal{Q}}(\mathbf{B}';\mathbf{B})\neq\varnothing$.

Moreover, the resulting map satisfies Condition (WA-b) if and only if φ is an equivalence; in this case, φ is a strong approximation if the converse to the above compositionality statement holds.

Proof. First note that φ is a map $\mathrm{Ob}(\mathcal{O}^t\wr\mathcal{P}^t)_T\to\mathcal{Q}_T$; by Proposition 1.60, to specify a topological functor $\mathcal{O}^t\wr\mathcal{P}^t\to\mathcal{Q}^t$ over $\mathbf{Span}(\mathbb{F}_T)$ it then suffices to define the data on mapping spaces. Indeed, we determine this canonically via the factorization

$$\begin{array}{ccc} \mathrm{Map}_{\mathcal{O}^\otimes\wr\mathcal{P}^\otimes}((\mathbf{C}',\mathbf{B}'),(\mathbf{B}',\mathbf{B})) & \longrightarrow & \mathrm{Map}_{\mathcal{Q}^\otimes}(\varphi(\mathbf{C}',\mathbf{B}',\varphi(\mathbf{B}';\mathbf{B})) \\ \uparrow & & \uparrow \\ \mathrm{Mul}^f_{\mathcal{O}^\otimes\wr\mathcal{P}^\otimes}((\mathbf{C}',\mathbf{B}');(\mathbf{C},\mathbf{B})) & \dashrightarrow & \mathrm{Mul}^f_{\mathcal{Q}^\otimes}(\varphi(\mathbf{C}',\mathbf{B}');\varphi(\mathbf{B}',\mathbf{B})) \end{array}$$

There exists an essentially unique filler as in the dashed arrow since whenever the domain is nonempty, the codomain is contractible by assumption. To verify functoriality, we must verify a that a collection of diagrams among products of multimorphism spaces of $\mathcal{Q}^\otimes$ commute; however, since $\mathcal{Q}^\otimes$ has contractible nonempty structure spaces, all terms in such diagrams are either empty or contractible, so all such diagrams commute.

The remaining statement follows by unwinding definitions to find that this is the relevant map between active arrow categories, and noting that the assumption implies *contractible* (not just weakly contractible) homotopy fibers. □

Now, we cover the case of maps to one-color $\mathcal{T}$-operads.

Proposition 1.63. *Suppose $\mathcal{O}^t, \mathcal{P}^t, \mathcal{Q}^t$ are the topological $\mathcal{T}$-operads such that $\mathcal{Q}^t$ has one color. A relative pattern map $\mathcal{O}^t \wr \mathcal{P}^t \to \mathcal{Q}^t$ is uniquely specified by the data of, for each composable arities $\mathrm{Ind}_V^{\mathcal{T}} T \to \mathrm{Ind}_V^{\mathcal{T}} S \to V$ and compatible $\mathcal{P}$ and $\mathcal{O}$ (multi-)profiles $(\mathbf{B}'; \mathbf{B})$ and $(\mathbf{C}'; C)$, a homotopy class of continuous map*

$$\varphi \colon \mathcal{O}(\mathbf{C}'; C) \times \mathrm{Mul}_{\mathcal{P}}^{g}(\mathbf{B}'; \mathbf{B}) \to \mathcal{Q}(T)$$

sending $\varphi 1 = 1$ and such that the following diagrams commute:

$$\begin{array}{ccc} \mathcal{O}(\mathbf{C}'; C) \times \mathrm{Mul}_{\mathcal{P}}^{g}(\mathbf{B}'; \mathbf{B}) & \xrightarrow{\quad\varphi\quad} & \mathcal{Q}(T) \\ \downarrow \mathrm{Res} & & \downarrow \mathrm{Res} \\ \mathcal{O}\left(\mathrm{Res}_U^V \mathbf{C}'; \mathrm{Res}_U^V C\right) \times \mathrm{Mul}_{\mathcal{P}}^{\mathrm{Res}_U^V g}\left(\mathrm{Res}_U^V \mathbf{B}'; \mathrm{Res}_U^V \mathbf{B}\right) & \xrightarrow{\varphi} & \mathcal{Q}\left(\mathrm{Res}_U^V T\right) \end{array} \tag{2}$$

$$\begin{array}{ccc} \left(\mathcal{O}(\mathbf{C}'; \mathbf{C}) \times \mathrm{Mul}_{\mathcal{O}}^{h}(\mathbf{C}''; \mathbf{C}')\right) \times \left(\mathrm{Mul}_{\mathcal{P}}^{g}(\mathbf{B}'; \mathbf{B}) \times \mathrm{Mul}_{\mathcal{P}}^{f}(\mathbf{B}''; \mathbf{B}')\right) & \xrightarrow{(\gamma,\gamma)} & \mathcal{O}(\mathbf{C}''; C) \times \mathrm{Mul}_{\mathcal{P}}^{f \circ g}(\mathbf{B}''; \mathbf{B}) \\ \wr & & \\ \left(\mathcal{O}(\mathbf{C}'; C) \times \mathrm{Mul}_{\mathcal{P}}^{f}(\mathbf{B}'; \mathbf{B})\right) \times \left(\mathrm{Mul}_{\mathcal{O}}^{h}(\mathbf{C}''; \mathbf{C}') \times \mathrm{Mul}_{\mathcal{P}}^{g}(\mathbf{B}''; \mathbf{B}')\right) & & \downarrow \varphi \\ \downarrow (\varphi,\varphi) & & \\ \mathcal{Q}(T) \times \mathrm{Mul}_{\mathcal{Q}}^{f \circ g}(T'; T) & \xrightarrow{\quad\gamma\quad} & \mathcal{Q}(T') \end{array} \tag{3}$$

The associated functor acts on active mapping spaces via φ.

Proof. Since $\mathcal{Q}^t \to \mathbf{Span}(\mathbb{F}_{\mathcal{T}})$ is bijective on objects, there is exactly one map $\mathrm{Ob}\left(\mathcal{O}^t \wr \mathcal{P}^t\right) \to \mathrm{Ob}\mathcal{Q}^t$ over $\mathrm{Ob}\,\mathbf{Span}(\mathbb{F}_{\mathcal{T}})$: $(\mathbf{C}, \mathbf{B}) \mapsto T'$. To construct such a functor which is compatible with inert arrows, we then must determine the data on mapping spaces, uniquely via the factorization

$$\begin{array}{ccccc} \mathrm{Map}_{\mathcal{O}^t \wr \mathcal{P}^t}((\mathbf{C}', \mathbf{B}'), (\mathbf{B}', \mathbf{B})) & \hookleftarrow & \mathrm{Mul}^{f}_{\mathcal{O}^t \wr \mathcal{P}^t}((\mathbf{C}', \mathbf{B}'); (\mathbf{C}, \mathbf{B})) & \longrightarrow & \mathcal{O}(\mathbf{C}'_K; C_K) \times \mathrm{Mul}^{g}_{\mathcal{P}}(\mathbf{B}'_K; \mathbf{B}_K) \\ \downarrow & & \downarrow & & \downarrow \varphi \\ \mathrm{Map}_{\mathcal{Q}^t}(T', T) & \longleftarrow & \mathrm{Mul}^{f}_{\mathcal{Q}^t}(T', T) & \longrightarrow & \mathcal{Q}(T'_K) \end{array}$$

We're left with functoriality; this follows by unwinding definitions. □

The proof of the following proposition is nearly identical, so we omit it.

Proposition 1.64. *Suppose $\mathcal{O}^t, \mathcal{P}^t, \mathcal{Q}^t$ are topological $\mathcal{T}$-operads and let P be a poset with $\mathrm{Infl}_e^{\mathcal{T}} P \simeq U\mathcal{Q}$ an equivalence. A relative pattern map map $\mathcal{O}^t \wr \mathcal{P}^t \to \mathcal{Q}^t$ is uniquely specified by a $\mathcal{T}$-functor $U\mathcal{O} \times U\mathcal{P} \to \mathrm{Infl}_e^{\mathcal{T}} P$ together with a homotopy class of continuous map*

$$\varphi \colon \mathcal{O}(\mathbf{C}'; C) \times \mathrm{Mul}_{\mathcal{P}}^{g}(\mathbf{B}'; \mathbf{B}) \to \mathcal{Q}(\varphi_P(\mathbf{C}', \mathbf{B}'); \varphi_P(C, \mathbf{B}))$$

sending $\varphi 1 = 1$ and such that the following diagrams commute:

$$\begin{array}{ccc} \mathcal{O}(\mathbf{C}'; C) \times \mathrm{Mul}_{\mathcal{P}}^{g}(\mathbf{B}'; \mathbf{B}) & \xrightarrow{\quad\varphi\quad} & \mathcal{Q}(\varphi_P(\mathbf{C}', \mathbf{B}'); \varphi_P(C, \mathbf{B})) \\ \downarrow \mathrm{Res} & & \downarrow \mathrm{Res} \\ \mathcal{O}\left(\mathrm{Res}_U^V \mathbf{C}'; \mathrm{Res}_U^V C\right) \times \mathrm{Mul}_{\mathcal{P}}^{\mathrm{Res}_U^V g}\left(\mathrm{Res}_U^V \mathbf{B}'; \mathrm{Res}_U^V \mathbf{B}\right) & \xrightarrow{\varphi} & \mathcal{Q}\left(\mathrm{Res}_U^V \varphi_P(\mathbf{C}', \mathbf{B}'); \mathrm{Res}_U^V \varphi_P(C, \mathbf{B})\right) \end{array} \tag{4}$$

$$
\begin{array}{ccc}
\Big(\mathcal{O}(\mathbf{C}';\mathbf{C})\times \mathrm{Mul}^{h}_{\mathcal{O}}(\mathbf{C}'';\mathbf{C}')\Big)\times\Big(\mathrm{Mul}^{g}_{\mathcal{P}}(\mathbf{B}';\mathbf{B})\times \mathrm{Mul}^{f}_{\mathcal{P}}(\mathbf{B}'';\mathbf{B}')\Big) & \xrightarrow{(\gamma,\gamma)} & \mathcal{O}(\mathbf{C}'';C)\times \mathrm{Mul}^{f\circ g}_{\mathcal{P}}(\mathbf{B}'';\mathbf{B}) \\
\wr\downarrow & & \\
\Big(\mathcal{O}(\mathbf{C}';C)\times \mathrm{Mul}^{f}_{\mathcal{P}}(\mathbf{B}';\mathbf{B})\Big)\times\Big(\mathrm{Mul}^{h}_{\mathcal{O}}(\mathbf{C}'';\mathbf{C}')\times \mathrm{Mul}^{g}_{\mathcal{P}}(\mathbf{B}'';\mathbf{B}')\Big) & & \varphi\downarrow \\
(\varphi,\varphi)\downarrow & & \\
\mathcal{Q}(\varphi(\mathbf{C}';\mathbf{B}');\varphi C;\mathbf{B})\times \mathrm{Mul}^{f\circ g}_{\mathcal{Q}}(\varphi(\mathbf{C}'';\mathbf{B}'');\varphi(\mathbf{C}';\mathbf{B}')) & \xrightarrow{\gamma} & \mathcal{Q}(\varphi(\mathbf{C}'';\mathbf{B}'');\varphi(C;\mathbf{B}))
\end{array}
\tag{5}
$$

The associated functor acts on active mapping spaces via φ.

1.3.2. *Wreath products vs tensor products.* We now compare $\wr$ with $\overset{\mathrm{BV}}{\otimes}$ via the following.

Construction 1.65. We define the algebraic pattern over $\mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*}$ as

$$\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes} := s^{*}\mathcal{O}^{\otimes}\wr\mathcal{P}^{\otimes};$$

this naturally underlies a $\mathcal{T}$-algebraic pattern over $\underline{\mathbb{F}}_{\mathcal{T},*}$. Similarly, we define the functor

$$\mathcal{O}_1^{\otimes}\underline{\odot}\mathcal{O}_2^{\otimes} := \mathcal{O}_1^{\otimes}\times_{\mathcal{T}^{\mathrm{op}}}\mathcal{O}_2^{\otimes} \longrightarrow \mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*}\times_{\mathcal{T}^{\mathrm{op}}}\mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*} \xrightarrow{\wedge} \mathrm{Tot}\underline{\mathbb{F}}_{\mathcal{T},*};$$

this naturally underlies a map of $\mathcal{T}$-algebraic patterns, and there is an evident map

$$\underline{M}\colon \mathcal{O}^{\otimes}\underline{\odot}\mathcal{P}^{\otimes}\to\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes}$$

extending the map of Construction 1.58.

Define the $\mathcal{T}$-algebraic pattern $\mathcal{P}^{c}$ over $\underline{\mathbb{F}}_{\mathcal{T},*}$ via the same $\mathcal{T}$-∞-category over $\underline{\mathbb{F}}_{\mathcal{T},*}$, with all morphisms inert and all objects elementary. We define the *mixed wreath product* $(\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes})^{\mathrm{mix}}$ of $\mathcal{O}^{\otimes}$ and $\mathcal{P}^{\otimes}$ as the same $\mathcal{T}$-∞-category over $\mathsf{Span}(\mathbb{F}_V)$ and inert arrows as $\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes}$ and the same elementary objects as $\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{c}$, i.e.

$$\Big(\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes}\Big)^{\mathrm{el}} = \mathcal{O}^{\mathrm{el}}\times\mathcal{P}^{\mathrm{el}}\subset\mathcal{O}^{\mathrm{el}}\times\mathcal{P}^{\mathrm{int}} = \Big(\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes}\Big)^{\mathrm{mix-el}}. \tag{6}$$

◂

There is a commutative diagram

$$
\begin{array}{ccccc}
\mathcal{O}^{\otimes}\underline{\odot}\mathcal{P}^{c} & & \xrightarrow{\quad t\quad} & & \mathcal{O}^{\otimes}\underline{\odot}\mathcal{P}^{\otimes} \\
{\scriptstyle \underline{M}^{\mathrm{coarse}}}\downarrow & & & & \downarrow{\scriptstyle \underline{M}} \\
\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{c} & \xrightarrow{\quad \underline{c}\quad} & (\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes})^{\mathrm{mix}} & \xrightarrow{\quad \underline{m}\quad} & \mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes}
\end{array}
$$

whose solid arrows are maps of patterns lying over $\underline{M}$, and whose dotted arrows lie over the identity. We want to verify that the top horizontal arrow is a Morita equivalence, which we will try to deduce from the fact that the bottom is a Morita equivalence, which is the following lemma.

Lemma 1.66. *$\underline{M}^{\mathrm{coarse}}$ is a strong approximation.*

Proof. All active arrows in $\mathcal{P}^{c}$ are equivalences, so including identities on the $\mathcal{P}^{c}$ induces an equivalence of arrows

$$
\begin{array}{ccc}
\Big(\mathcal{O}^{\otimes}\underline{\odot}\mathcal{P}^{c}\Big)^{\mathrm{act}}_{/(\mathbf{C};\mathbf{D})} & \longrightarrow & (\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{c})^{\mathrm{act}}_{/(\mathbf{C};\Delta^{S}\mathbf{D})} \\
\wr & & \wr \\
\mathcal{O}^{\mathrm{act}}_{/\mathbf{C}} & = & \mathcal{O}^{\mathrm{act}}_{/\mathbf{C}}
\end{array}
$$

confirming Condition (WA-a). We're left with verifying the strong form of Condition (WA-b), i.e. that

$$\mathcal{O}^{\mathrm{el}}\times\mathrm{Tot}_{\mathcal{T}}\mathcal{P}^{\otimes}\simeq\Big(\mathcal{O}^{\otimes}\underline{\odot}\mathcal{P}^{c}\Big)^{\mathrm{el}}\to\Big(\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{c}\Big)^{\mathrm{el}}$$

is an equivalence. This follows by unwinding definitions. □

Now, to make comparisons along vertical arrows, we use a version of a lemma due to Harpaz.

Lemma 1.67. *Suppose $(\mathcal{Q}^{\otimes},\mathcal{Q}^{\mathrm{int}},\mathcal{Q}^{\mathrm{act}},\mathcal{Q}^{\mathrm{el}})$ and $(\mathcal{Q}^{\otimes},\mathcal{Q}^{\mathrm{int}\prime},\mathcal{Q}^{\mathrm{act}\prime},\mathcal{Q}^{\mathrm{el}\prime})$ are two $\mathcal{T}$-pattern structures on the same ∞-category such that $\mathcal{Q}^{\mathrm{int}}\subset\mathcal{Q}^{\mathrm{int}\prime}$ and $\mathcal{Q}^{\mathrm{act}\prime}\subset\mathcal{Q}^{\mathrm{act}}$.*

(1) *(Unnecessary change of factorization system) Suppose* $\mathcal{Q}^{\mathrm{el}} = \mathcal{Q}^{\mathrm{el}\prime}$ *and for each* $Q \in \mathcal{Q}^{\mathrm{el}}$*, there exists a pair of initial wide subcategories* $\mathcal{Q}^{\mathrm{int}}_{X/,1} \subset \mathcal{Q}^{\mathrm{int}}_{X/}$ *and* $\mathcal{Q}^{\mathrm{int}\prime}_{X/,1} \subset \mathcal{Q}^{\mathrm{int}\prime}_{X/}$ *such that, for every path* $X \xrightarrow{f} Y \xrightarrow{g} Q$ *such that* $g \circ f$ *in* $\mathcal{Q}^{\mathrm{int}}$ *and* f *in* $\mathcal{Q}^{\mathrm{int}\prime}$*,* Y *is elementary. Then, the identity arrow on* $\mathcal{Q}^{\otimes}$ *lifts to a Morita equivalence between the two weak* ∞*-operad structures.*

(2) *(Change of elementaries) Suppose* $\mathcal{Q}^{\mathrm{int}} = \mathcal{Q}^{\mathrm{int}\prime}$ *and* $\mathcal{Q}^{\mathrm{act}} = \mathcal{Q}^{\mathrm{act}\prime}$*, and additionally* $\mathcal{Q}^{\mathrm{el}} \subset \mathcal{Q}^{\mathrm{el}\prime}$*. Then,* $F\colon \mathcal{Q}^{\otimes} \to \mathcal{C}$ *is a* $\left(\mathcal{Q}^{\otimes}, \mathcal{Q}^{\mathrm{int}}, \mathcal{Q}^{\mathrm{act}}, \mathcal{Q}^{\mathrm{el}}\right)$*-Segal object if and only if it satisfies the following conditions:*
   (a) F *is a* $\left(\mathcal{Q}^{\otimes}, \mathcal{Q}^{\mathrm{int}\prime}, \mathcal{Q}^{\mathrm{act}\prime}, \mathcal{Q}^{\mathrm{el}\prime}\right)$*-Segal object, and*
   (b) $F|_{\mathcal{Q}^{\mathrm{el}\prime}}$ *is right Kan extended from* $F|_{\mathcal{Q}^{\mathrm{el}}}$*.*

Proof. The proof of [Har, Lem. 4.2.21] extends mutatis mutandis. □

We separate out the unnecessary change of factorization system first.

Lemma 1.68. *c is a Morita equivalence.*

Proof. By Lemma 1.67, it suffices to verify that, for all coarse-inert composite arrows

$$f\colon (\mathbf{C}; (\mathbf{D}_U)_S) \xrightarrow{f'} (\mathbf{C}''; (\mathbf{D}''_{U''})_{S''}) \to (\mathbf{C}'; (\mathbf{D}'_{U'})_{S'})$$

with f' mixed-inert and $(\mathbf{C}'; (\mathbf{D}_{U'})_S)$ mixed-elementary *such that f and f' lie in the initial subcategory of inert morphisms with nonempty domain*, $(\mathbf{C}''; (\mathbf{D}''_{U''})_{S''})$ is coarse-elementary. The assumptions guarantee that $\mathbf{C}'$ lies over a transitive G-set, which implies the same of $\mathbf{C}$ by coarse inertness of f, and $\mathbf{C}''$ by mixed-inertness of f', which is the desired coarse-elementary property. □

Proof of Theorem 1.59. Lemmas 1.66 to 1.68 and two-out-of-three construct, for each complete $\mathcal{T}$-category $\mathcal{C}$, a commutative diagram of fully faithful functors

$$\begin{array}{ccccc}
\mathrm{Seg}_{\mathcal{O}}\mathrm{Seg}_{\mathcal{P}}(\mathcal{C}) & \simeq & \mathrm{Seg}_{\mathcal{O}\underline{\odot}\mathcal{P}}(\mathcal{C}) & \longleftarrow & \mathrm{Seg}_{\mathcal{O}\underline{\wr}\mathcal{P}}(\mathcal{C}) \\
\downarrow{\scriptstyle \iota_{\underline{\odot}}} & & \downarrow & & \downarrow{\scriptstyle \iota_{\underline{\wr}}} \\
\mathrm{Seg}_{\mathcal{O}}\mathrm{Fun}(\mathcal{P},\mathcal{C}) & \simeq & \mathrm{Seg}_{\mathcal{O}\underline{\odot}\mathcal{P}^c}(\mathcal{C}) & \simeq & \mathrm{Seg}_{\mathcal{O}\underline{\wr}\mathcal{P}^c}(\mathcal{C}) \simeq \mathrm{Seg}_{(\mathcal{O}\underline{\wr}\mathcal{P})^{\mathrm{mix}}}(\mathcal{C})
\end{array}$$

It suffices to verify that $\mathrm{im}\,\iota_{\underline{\wr}} \subset \mathrm{im}\,\iota_{\underline{\odot}}$. By Lemma 1.67, the image of $\underline{M}^{\mathrm{coarse}*}\iota_{\underline{\wr}}$ is those coarse Segal objects $F\colon \mathcal{O}^{\otimes}\underline{\odot}\mathcal{P}^{\otimes} \to \mathcal{C}$ such that

(a) $F|_{(\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes})^{\mathrm{mix-el}}}$ is right Kan extended from $F|_{(\mathcal{O}^{\otimes}\underline{\wr}\mathcal{P}^{\otimes})^{\mathrm{el}}}$,

and the image of $\iota_{\underline{\odot}}$ is hose F such that

(b) for all $O \in \mathcal{O}^{\otimes}$, the $\mathcal{T}$-functor $\{O\} \times \mathcal{P}^{\otimes} \to \mathcal{C}$ is a Segal object.

We're tasked with verifying that Conditions (a) and (b) are equivalent. Now, in sight of Eq. (6), this compiles down to verifying that a functor is right Kan extended along a base changed functor $\mathcal{C} \times_{\mathcal{D}} \mathcal{E} \to \mathcal{C} \times_{\mathcal{D}} \mathcal{E}'$ if and only if, for all $C \in \mathcal{C}$, the associated functor is right Kan extended along the functor $\{C\} \times \mathcal{E} \to \{C\} \times \mathcal{E}'$; but this follows from the fact that $\mathcal{T}$-limits in product $\mathcal{T}$-categories are products of $\mathcal{T}$-limits in the factors. □

1.3.3. *The bookkeeping theorem.* We now prove the main bookkeeping theorem, Theorem 1.1.

Proposition 1.69. *$\wedge_! \mathcal{O}_1^{\otimes} \underline{\odot} \mathcal{O}_2^{\otimes} \simeq \mathcal{O}_1^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathcal{O}_2^{\otimes}$; in particular, if $\varphi\colon \mathcal{O}_1^{\otimes} \underline{\wr} \mathcal{O}_2^{\otimes} \longrightarrow \mathcal{O}_3^{\otimes}$ is a Morita equivalence, then, the induced $\mathcal{T}$-operad map*

$$\mathcal{O}_1^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathcal{O}_2^{\otimes} \longrightarrow \mathcal{O}_3^{\otimes}$$

is an equivalence.

Proof. By the same proof as [Ste25b, Cor. 3.11] and [Ste25d, Cor. E] (which is itself a minor variation of [BS24, Thm. E]), we find that $L_{\mathrm{Fbrs}}\left(\mathcal{O}_1^{\otimes} \underline{\odot} \mathcal{O}_2^{\otimes}\right)$ provides a binary operation on $\mathrm{Op}_{\mathcal{T}}$ intertwining with $\square$ via Env, which forces it to agree with $\overset{\mathrm{BV}}{\otimes}$ by uniqueness of such an operation. □

Now, the following lemma follows by compatibility of products with pullbacks and equivalences.

Lemma 1.70. *In the situation of Theorem 1.1, the map $\alpha_1 \underline{\odot} \alpha_2$ is a weak approximation of $\mathcal{T}$-algebraic patterns.*

Proof of Theorem 1.1. We may pull back our data to construct the $\mathcal{T}$-algebraic pattern diagram

$$\begin{array}{ccccc} \mathcal{P}_1^{\otimes}\underline{\odot}\mathcal{P}_2^{\otimes} & \xrightarrow{\underline{M}} & \mathcal{P}_1^{\otimes}\underline{\wr}\mathcal{P}_2^{\otimes} & \xrightarrow{\underline{\varphi}_{\mathcal{P}}} & \mathcal{P}_3^{\otimes} \\ \downarrow{\scriptstyle \alpha_1\underline{\odot}\alpha_2} & & \downarrow & & \downarrow{\scriptstyle \alpha_3} \\ \mathcal{E}_1^{\otimes}\underline{\odot}\mathcal{E}_2^{\otimes} & \xrightarrow{\underline{M}} & \mathcal{E}_1^{\otimes}\underline{\wr}\mathcal{E}_1^{\otimes} & \xrightarrow{\underline{\varphi}_{\mathcal{E}}} & \mathcal{E}_3^{\otimes} \end{array}$$

and in sight of Proposition 1.69 and two-out-of-three, we're tasked with proving that $\underline{\varphi_{\mathcal{E}}} \circ \underline{M}$ is a Morita equivalence.

By Proposition 1.51 and Lemma 1.70 we know that α_3 and $\alpha_1\underline{\odot}\alpha_2$ are Γ-weak approximations, by Theorem 1.59 that each arrow marked M is a Morita equivalence, and by Corollary 1.50 that $\underline{\varphi_{\mathcal{P}}}$ is a Morita equivalence. By two-out-of-three for Morita equivalences, $\underline{\varphi_{\mathcal{P}}} \circ \underline{M}$ is a Morita equivalence, so by Corollary 1.50, applying $\mathrm{Seg}_{(-)}$ to the outer diagram yields a diagram of fully faithful inclusions

$$\begin{array}{ccc} \mathrm{Seg}_{\mathrm{Tot}_{\mathcal{T}}\mathcal{P}_1}\underline{\mathrm{Seg}}_{\mathrm{Tot}_{\mathcal{T}}\mathcal{P}_2}(\mathcal{S}) & \simeq & \mathrm{Seg}_{\mathrm{Tot}_{\mathcal{T}}\mathcal{P}_3}(\mathcal{S}) \\ \uparrow & & \uparrow \\ \mathrm{Seg}_{\mathrm{Tot}_{\mathcal{T}}\mathcal{E}_1}\underline{\mathrm{Seg}}_{\mathrm{Tot}_{\mathcal{T}}\mathcal{E}_2}(\mathcal{S}) & \hookrightarrow & \mathrm{Seg}_{\mathrm{Tot}_{\mathcal{T}}\mathcal{E}_3}(\mathcal{S}); \end{array}$$

It suffices to verify that the images of $(\alpha_1 \times \alpha_2)^*$ and $\varphi_{\mathcal{O}}^*\alpha_3^*$ agree. In fact, by Corollary 1.50 they correspond with $\mathcal{O}_1\underline{\odot}\mathcal{O}_2$-monoid $\mathcal{T}$-spaces attaining solutions to the inner and outer lifting problem, respectively:

$$\begin{array}{ccc} U\mathcal{O}_1 \times U\mathcal{O}_2 & \longrightarrow & \underline{\mathcal{S}}_{\mathcal{T}} \\ \downarrow & \nearrow & \\ U\mathcal{P}_1 \times U\mathcal{P}_2 & \xrightarrow[\sim]{U\varphi_{\mathcal{P}}} & U\mathcal{P}_3 \end{array}$$

Indeed, $U\varphi_{\mathcal{P}}$ is an equivalence by assumption, so the images agree. □

2. Embeddings of orthogonal representations

In this section, we collect the various results of equivariant elementary topology which we will need for the rest of the paper. The author does not claim this to be particularly original; it largely comes as a straightforward extension of the relevant parts of the existing literature on factorization algebras and factorization homology.

2.1. **Rectilinear little V-polydisks.** Our geometry admits the following as its local types.

Definition 2.1. An *orthogonal G-polyrepresentation* $V_\bullet = (V_1, \dots, V_n)$ is a finite list of orthogonal representations of G. The *polydisk in* $V_\bullet$ is the open sub-G-manifold

$$P^{V_\bullet} := \prod_{i=1}^n D^{V_i} \quad \subset \quad \prod_{i=1}^n V_i \simeq \bigoplus_{i=1}^n V_i.$$

When it will not cause confusion, we will write $V := \bigoplus_{i=1}^n V_i$, and we will omit the bullet and write $P^V := P^{V_\bullet}$. ◁

Note that the inclusion $P^V \subset V$ is an isotopy equivalence; indeed, this follows by taking products of the usual isotopies witnessing that $D^{V_i} \subset V_i$ is an isotopy equivalence. Crucially, little V-polydisks will provide a *homotopy-equivalent* theory to little V-disks. For additivity, we need the following.

Construction 2.2. Given two polyrepresentations $V_\bullet$ and $W_\bullet$, we write $V_\bullet \oplus W_\bullet$ for the orthogonal G-polyrepresentation given by the concatenation $(V_1, \dots, V_n, W_1, \dots, W_m)$. Note that

$$P^{V\oplus W} = P^V \times P^W \quad \subset \quad V \times W \simeq V \oplus W.$$ ◁

We henceforth fix $(V_\bullet, W_\bullet)$ a pair of orthogonal G-polyrepresentations.

Construction 2.3. Given $H \subset G$ a compact closed subgroup and $S \in \mathbb{F}_H$ a finite H-set, we let

$$S \cdot P^V := \coprod_{[H/K]\subset S} \mathrm{Ind}_K^H P^{\mathrm{Res}_K^G V}$$

denote the S-*indexed copower* of P^V; we refer to H-manifolds of this form as $V_\bullet$-*multi-polydisks.* ◁

2.1.1. *Rectilinear embedding spaces.*

Definition 2.4. We refer to embeddings $D^{V_i} \hookrightarrow D^{V_i}$ of the form $v \mapsto \lambda v + v_0$ for some $\lambda \in \mathbb{R}_{>0}$ and $v_0 \in D^V$ as *equidiameter embeddings* and more generally, we refer to embeddings $S \cdot P^V \hookrightarrow P^V$ given componentwise by products of equidiameter embeddings as *rectilinear embeddings.*

Given a map of fintie G-sets $f\colon T \to S$, we refer to an embedding $\iota\colon T \cdot P^V \to S \cdot P^V$ as an f-*indexed little* V-*polydisk multiembedding* if:

(a) the induced map $\underline{\pi}_0\iota\colon \underline{\pi}_0\left(T \cdot P^V\right) = T \to S = \underline{\pi}_0\left(S \cdot P^W\right)$ is f,
(b) ι is an embedding of H-manifolds, and
(c) for each orbit $[H/K] \subset T$, the mate $P^{\mathrm{Res}^G_K V} \to P^{\mathrm{Res}^G_K W}$ to the restricted map $\iota_{[H/K]}$ of

$$\begin{array}{ccc} \mathrm{Ind}^H_K P^{\mathrm{Res}^G_K V} & \xrightarrow{\iota_{[H/K]}} & P^{\mathrm{Res}^G_H V} \\ \downarrow & & \downarrow \\ T \cdot P^V & \xrightarrow{\iota} & S \cdot P^V \end{array}$$

is a rectilinear embedding of $\mathrm{Res}^G_K V$.

We refer to the topological subspace of H-equivariant f-indexed little V-polydisk multiembeddings as

$$\mathrm{Emb}^{H,\mathrm{Rect}}_f\left(T \cdot P^V, S \cdot P^V\right) \subset C^\infty\left(T \cdot P^V, S \cdot P^V\right),$$

the latter having the (strong) C^∞-topology. In the case $S = *_H$ is the terminal H-set, we simply write

$$\mathrm{Emb}^{H,\mathrm{Rect}}\left(T \cdot P^V, P^V\right) \coloneqq \mathrm{Emb}^{H,\mathrm{Rect}}_!\left(T \cdot P^V, *_H \cdot P^V\right).$$ ◁

Remark 2.5. Note that tracking diameters and centers under the diffeomorphism $V \simeq P^V$ yields an injective function

$$(\mathfrak{c}, \mathfrak{t})\colon \mathrm{Emb}^{H,\mathrm{Rect}}_f\left(T \cdot P^V, S \cdot P^V\right) \hookrightarrow \left(\mathbb{R}^{|V|} \times \mathbb{R}^n\right)^{|T|},$$

Now, the codomain admits a structure as a real subspace

$$(\mathbb{R}^n \times \mathbb{R})^{|T|} \subset C^\infty(T \cdot V, S \cdot V) \simeq C^\infty\left(T \cdot P^V, S \cdot P^V\right),$$

and its subspace topology endows it with the structure of a finite dimensional Hausdorff real topological vector space, under which $(\mathfrak{c}, \mathfrak{t})$ is a topological subspace embedding. But such a structure exists on a finite dimensional real vector space uniquely up to linear homeomorphism; hence $(\mathfrak{c}, \mathfrak{t})$ is also a topological subspace embedding for the (homemorphic) standard topological structure. ◁

Warning 2.6. In the case of the one-term polyrepresentation $V = V_1$, these are called *affine* in [BH15; Bon19; GM17; Hil22; Hor19]; to avoid confusion with the more common definition of *affine transformations*, we will avoid this term. ◁

Remark 2.7. In the case of an n-step polyrepresentation of the trivial group, $\mathbb{R}^{\oplus n} = \mathbb{R} \oplus \cdots \oplus \mathbb{R}$, this recovers May's open little disk embeddings [May72, Def. 4.1]. In the nonequivariant case of a one-term polyrepresentation $\mathbb{R}^n = \mathbb{R}^n$, these are *equidiameter embeddings* in the sense of [May72, Thm. 4.8]. ◁

Now, first off, note that these behave well with respect to restriction.

Observation 2.8. Given $K \subset H$ a subgroup inclusion, we acquire a *restriction* factorization inclusion

$$\mathrm{Res}^H_K\colon \mathrm{Emb}^{H,\mathrm{Rect}}_f\left(T \cdot P^V, S \cdot P^V\right) \hookrightarrow \mathrm{Emb}^{K,\mathrm{Rect}}_{\mathrm{Res}^H_K f}\left(T \cdot P^V, S \cdot P^V\right) \hookrightarrow C^\infty\left(T \cdot P^V, S \cdot P^V\right).$$

In particular, writing $C^{\infty,H}_f\left(T \cdot P^V, S \cdot P^V\right)$, for the maps satisfying Condition (a) of Definition 2.4, we find another restriction map

$$C^{\infty,xH}_f\left(T \cdot P^V, S \cdot P^V\right) \hookrightarrow C^{\infty,K}_{\mathrm{Res}^H_K f}\left(T \cdot P^V, S \cdot P^V\right)$$

furnishing the above map. ◁

Additionally, we observe that rectilinear embeddings satisfy a Segal condition.

Observation 2.9. By first restricting and then pulling back along the inclusion of the identity coset

$$*_K = [K/eKe^{-1}] \subset \coprod_{g\in[K\backslash H/K]} [K/gKg^{-1}] \simeq \mathrm{Res}^H_K[H/K].$$

we acquire a chain of homeomorphisms

$$\begin{aligned} C^{\infty,H}_f\left(T\cdot P^V, S\cdot P^V\right) &\simeq \prod_{[H/K]\subset S} C^{\infty,H}_{\mathrm{Ind}^H_K T_K\to[H/K]}\left(\mathrm{Ind}^H_K T_K\cdot P^V, [H/K]\cdot P^V\right) \\ &\simeq \prod_{[H/K]\subset S} C^{\infty,K}\left(T_K\cdot P^V, P^V\right), \end{aligned}$$

where we write $\mathrm{Ind}^H_K T_K$ for the canonical induction-expression for $T\times_S[H/K]$, compatibly with affineness. Moreover, note that the composite homeomorphism preserves and reflects Conditions (b) and (c) of Definition 2.4, restricting to another homeomorphism

$$\mathrm{Emb}^H_f\left(T\cdot P^V, S\cdot P^V\right) \simeq \prod_{[H/K]\subset S} \mathrm{Emb}^K\left(T_K\cdot P^V, P^V\right). \qquad \triangleleft$$

2.1.2. *Composition.* An essential structure on rectilinear embedding spaces is *composition*.

Lemma 2.10. *Composition of continuous maps restricts uniquely to operations*

$$\gamma\colon \mathrm{Emb}^{H,\mathrm{Rect}}_g\left(S\cdot P^V, R\cdot P^V\right)\times\mathrm{Emb}^{H,\mathrm{Rect}}_f\left(T\cdot P^V, S\cdot P^V\right)\longrightarrow \mathrm{Emb}^{H,\mathrm{Rect}}_{g\circ f}\left(T\cdot P^V, R\cdot P^V\right),$$

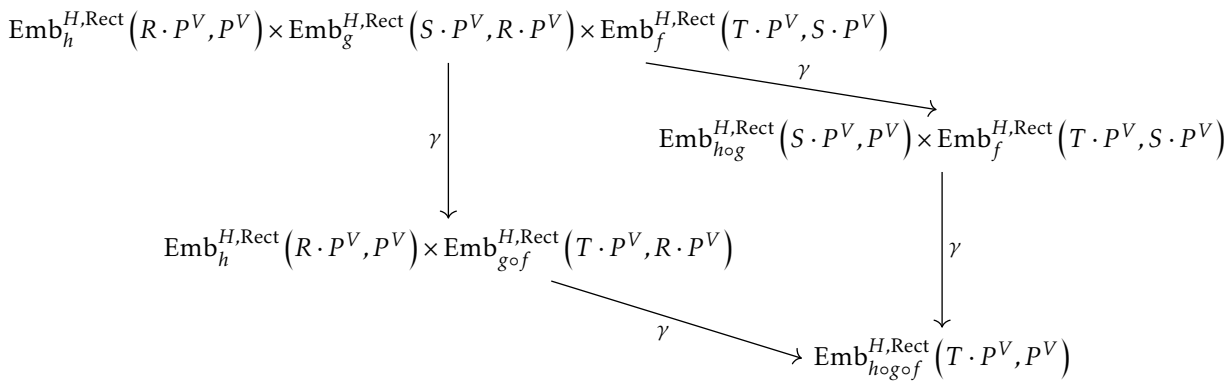
compatible with inclusions of structures. Moreover, γ is compatible with restriction, i.e. the following diagram commutes.

$$\begin{array}{ccc} \mathrm{Emb}^{H,\mathrm{Rect}}_g\left(S\cdot P^V, R\cdot P^V\right)\times\mathrm{Emb}^{H,\mathrm{Rect}}_f\left(T\cdot P^V, S\cdot P^V\right) & \xrightarrow{\gamma} & \mathrm{Emb}^{H,\mathrm{Rect}}_{g\circ f}\left(T\cdot P^V, R\cdot P^V\right) \\ \downarrow{\scriptstyle \mathrm{Res}^H_K} & & \downarrow{\scriptstyle \mathrm{Res}^H_K} \\ \mathrm{Emb}^{K,\mathrm{Rect}}_g\left(S\cdot P^V, R\cdot P^V\right)\times\mathrm{Emb}^{K,\mathrm{Rect}}_f\left(T\cdot P^V, S\cdot P^V\right) & \xrightarrow{\gamma} & \mathrm{Emb}^{K,\mathrm{Rect}}_{g\circ f}\left(T\cdot P^V, R\cdot P^V\right). \end{array}$$

Proof. Since $\underline{\pi}_0$ is functorial, we're tasked with verifying that rectilinear embeddings are closed under composition; unwinding definitions, this follows by taking products of the fact that equidiameter maps are closed under composition. Restriction compatibility follows from restriction compatibility of composition of smooth equivariant maps. □

We will use this as a G-operadic composition map, so we must observe associativity.

Observation 2.11. Given V an orthogonal G-representation, the following diagram commutes.



Indeed, this admits a natural monomorphism of topological spaces to the same square for C^∞ mapping spaces, which commutes by the fact that smooth manifolds form a topological category, so the property follows by unique factorization through monomorphisms. $\triangleleft$

2.1.3. *External products.* We now define a staple of the *additivity phenomenon.*

Construction 2.12. Given X, X', Y, Y' a quadruple of connected H-manifolds and composable morphisms $T \xrightarrow{f} S \xrightarrow{g} R$ we define the *external product map*

$$\varphi \colon C_g^{\infty,H}(S \cdot X, R \cdot Y) \times C_f^{\infty,H}(T \cdot X', S \cdot Y') \to C_{g\circ f}^{\infty,H}(T \cdot (X \times X'), R \cdot (Y \times Y'))$$

elementwise by the composition

$$\begin{array}{ccccc} T \cdot (X \times X') & & \xrightarrow{\qquad\qquad\varphi(\tilde{g};\tilde{f})\qquad\qquad} & & R \cdot (Y \times Y') \\ \wr\downarrow & & & & \wr\downarrow \\ X \times (T \cdot X') & \xrightarrow{(\mathrm{id},\tilde{g})} & X \times (S \cdot Y') \simeq (S \cdot X) \times Y' & \xrightarrow{(\tilde{f},\mathrm{id})} & (R \cdot Y) \times Y' \end{array} \tag{7}$$

Now, note that this restricts uniquely to a map

$$\begin{array}{ccc} \mathrm{Emb}_g^{H,\mathrm{Rect}}\left(S \cdot P^V, R \cdot P^V\right) \times \mathrm{Emb}_f^{H,\mathrm{Rect}}\left(T \cdot P^W, S \cdot P^W\right) & \dashrightarrow^{\varphi} & \mathrm{Emb}_{g\circ f}^{H,\mathrm{Rect}}\left(T \cdot P^{V\oplus W}, R \cdot P^{V\oplus W}\right) \\ \downarrow & & \downarrow \\ C_g^{\infty,H}\left(S \cdot P^V, R \cdot P^V\right) \times C_f^{\infty,H}\left(T \cdot P^W, S \cdot P^W\right) & \xrightarrow{\qquad\varphi\qquad} & C_{g\circ f}^{\infty,H}\left(T \cdot P^{V\oplus W}, R \cdot P^{V\oplus W}\right) \end{array}$$

◂

We see that φ behaves well with respect to restriction and composition.

Observation 2.13. Eq. (7) is manifestly restriction-stable, i.e. the following diagram commutes

$$\begin{array}{ccc} \mathrm{Emb}_g^{H,\mathrm{Rect}}\left(S \cdot P^V, R \cdot P^V\right) \times \mathrm{Emb}_f^{H,\mathrm{Rect}}\left(T \cdot P^W, S \cdot P^W\right) & \xrightarrow{\varphi} & \mathrm{Emb}_{g\circ f}^{H,\mathrm{Aff}}\left(T \cdot P^{V\oplus W}, R \cdot P^{V\oplus W}\right). \\ \downarrow \mathrm{Res}_K^H & & \downarrow \mathrm{Res}_K^H \\ \mathrm{Emb}_{\mathrm{Res}_K^H g}^{K,\mathrm{Rect}}\left(S \cdot P^V, R \cdot P^V\right) \times \mathrm{Emb}_{\mathrm{Res}_K^H f}^{K,\mathrm{Rect}}\left(T \cdot P^W, S \cdot P^W\right) & \xrightarrow[\varphi]{} & \mathrm{Emb}_{\mathrm{Res}_K^H g\circ f}^{K,\mathrm{Rect}}\left(T \cdot P^{V\oplus W}, R \cdot P^{V\oplus W}\right). \end{array}$$

Moreover, it is compatible with composition, i.e. the top horizontal composite

$$\begin{array}{ccccccc} P \cdot P^V & \xrightarrow{\varphi(\tilde{h};\mathrm{id})} & T \cdot P^{V\oplus W} & \xrightarrow{\qquad\varphi(\tilde{g};\tilde{f})\qquad} & R \cdot P^{V\oplus W} & \xrightarrow{\varphi(\mathrm{id};\tilde{j})} & Q \cdot P^{V\oplus W} \\ \wr\downarrow & & \wr\downarrow & & \wr\downarrow & & \wr\downarrow \\ P^V \times P \cdot P^W & \xrightarrow{(\mathrm{id},\tilde{h})} & P^V \times T \cdot P^W & \xrightarrow{(\mathrm{id},\tilde{g})} & P^V \times S \cdot P^W & & \\ & & & & \wr\downarrow & & \\ & & & & S \cdot P^V \times P^W & \xrightarrow{(\tilde{f},\mathrm{id})} R \cdot P^V \times P^W & \xrightarrow{(\tilde{j},\mathrm{id})} Q \cdot P^V \times P^W \end{array}$$

clearly agrees with $\varphi\left(\tilde{g} \circ \tilde{h}; \tilde{j} \circ \tilde{f}\right)$. Here, for visual clarity, we've denoted equivalences as squiggly headless double arrows. Moreover, Eq. (7) is associative, in the sense that we may define an element $\varphi(f;g;h)$ making following diagram commute

$$\begin{array}{ccccc} T \cdot P^{U\oplus V\oplus W} & & \xrightarrow{\qquad\qquad\varphi(f;g;h)\qquad\qquad} & & P \cdot P^{U\oplus V\oplus W} \\ \wr\downarrow & & & & \wr\downarrow \\ P^U \times T \cdot P^{V\oplus W} & & \xrightarrow{\qquad(\mathrm{id},\varphi(g;h))\qquad} & & P^U \times R \cdot P^{V\oplus W} \\ \wr\downarrow & & & & \wr\downarrow \\ & & P^U \times S \cdot P^V \times P^W & \xrightarrow[(\mathrm{id},g,\mathrm{id}))]{} & P^U \times R \cdot P^V \times P^W \\ & & \wr\downarrow & & \wr\downarrow \\ P^U \times P^V \times T \cdot P^W & \xrightarrow{(\mathrm{id},\mathrm{id},h)} & P^U \times P^V \times S \cdot P^W & & R \cdot P^U \times P^V \times P^W \\ & & \wr\downarrow & & \\ & & S \cdot P^{V\oplus W} \times P^W & \xrightarrow[(\varphi(f;g),\mathrm{id})]{\qquad\qquad} & P \cdot P^{V\oplus U} \times P^W \end{array}$$

We moreover see that "products with the identity self-embedding" yields a map

$$\iota \colon \mathrm{Emb}_f^{H,\mathrm{Rect}}\left(S \cdot P^V, R \cdot P^V\right) \longrightarrow \mathrm{Emb}_f^{H,\mathrm{Rect}}\left(S \cdot P^{U\oplus V}, R \cdot P^{U\oplus V}\right),$$

which makes the following diagram commute

$$\begin{array}{ccc} \mathrm{Emb}_g^{H,\mathrm{Rect}}\left(S\cdot P^V, R\cdot P^V\right)\times \mathrm{Emb}_f^{H,\mathrm{Rect}}\left(T\cdot P^W, S\cdot P^W\right) & \xrightarrow{\varphi} & \mathrm{Emb}_{g\circ f}^{H,\mathrm{Rect}}\left(T\cdot P^{V\oplus W}, R\cdot P^{V\oplus W}\right) \\ \downarrow{\scriptstyle (\iota,\mathrm{id})} & & \downarrow{\scriptstyle \iota} \\ \mathrm{Emb}_g^{H,\mathrm{Rect}}\left(S\cdot P^{U\oplus V}, R\cdot P^{U\oplus V}\right)\times \mathrm{Emb}_f^{H,\mathrm{Rect}}\left(T\cdot P^W, S\cdot P^W\right) & \xrightarrow{\varphi} & \mathrm{Emb}_{g\circ f}^{H,\mathrm{Rect}}\left(T\cdot P^{U\oplus V\oplus W}, R\cdot P^{U\oplus V\oplus W}\right) \end{array}$$

Moreover, ι intertwines with restriction and composition in the obvious way. ◁

2.2. **Weiss covers and factorization.**

2.2.1. *Configuration spaces and Weiss covers.* For now, fix X a topological space with G-action. We will study configurations in X in the following sense.

Definition 2.14. We write $\mathrm{Conf}_S^H(X)\subset \mathrm{Map}(S,X)$ for the topological subspace of H-equivariant embeddings of S into X.[4] ◁

These satisfy a type of *codescent*, for which we need the following notions.

Definition 2.15. Let $\mathrm{Open}(X)\subset \mathscr{P}(X)$ denote the poset of open subspaces of X. The poset of *H-invariant open precovers of X* is the fixed points

$$\mathrm{Open}^H(X) \coloneqq \mathrm{Open}(X)^H,$$

i.e. the collection of open subspaces $U\subset X$ such that $gu\in U$ for all $u\in U$ and $g\in G$. These naturally assemble into a G-poset $\underline{\mathrm{Open}}(X)$ such that $\underline{\mathrm{Open}}(X)^H\simeq \mathrm{Open}^H(X)$. An *open G-precover of X* is a G-subposet $\mathcal{U}\subset \underline{\mathrm{Open}}(X)$.

We additionally make the following definitions.

- A *G-invariant open cover of X* is a subposet $\mathcal{U}\subset \mathrm{Open}^G(X)\subset \mathrm{Open}(X)$ which is an open cover. An *open G-cover of X* is a G-subposet $\mathcal{U}\subset \underline{\mathrm{Open}}(X)$ such that $\mathcal{U}^H\subset \mathrm{Open}(X)$ is an H-invariant open cover for all $H\subset G$.
- A subposet $\mathcal{U}\subset \mathrm{Open}(X)$ is a *Weiss cover* if, for all finite configurations $S\subset X$, there exists a single open $U\in\mathcal{U}$ such that $S\subset U$; G-invariant Weiss covers and G-Weiss covers are defined similarly.
- A subposet $\mathcal{U}\subset \mathrm{Open}(X)$ is a *complete Weiss cover* if, for all finite subsets $\{U_i \mid i\in I\}\subset\mathcal{U}$,

$$\mathcal{U}_{\bigcap_{i\in I}U_i} \coloneqq \mathrm{Open}\left(\bigcap_{i\in I}U_i\right)\cap\mathcal{U}\subset \mathrm{Open}\left(\bigcap_{i\in I}U_i\right)$$

  is a Weiss cover. G-invariant complete Weiss covers and complete G-Weiss covers are defined similarly.
- A subposet $\mathcal{U}\subset \mathrm{Open}(X)$ is a *complete separating cover* if, for all finite subsets $\{U_i \mid i\in I\}\subset\mathcal{U}$, and all configurations $c\colon [n]\hookrightarrow \bigcap_{i\in I}U_i$ the poset of opens $\mathcal{U}_{\bigcap_{i\in I}U_i}$ contains a neighborhood $\mathrm{im}(c)\subset U$ such that the induced map $[n]\to \pi_0 U$ is a bijection.

◁

The following result of [Ste25c] is will 2be crucial to our study of factorization.

Proposition 2.16. *Given a complete G-Weiss cover $\mathcal{U}\subset\mathrm{Open}(X)$, the associated assembly map*

$$\mathrm{colim}_{U\in\mathcal{U}}\mathrm{Conf}_S^{\mathrm{stab}(U)}(U)\longrightarrow \mathrm{Conf}_S(V)$$

is an equivalence.

Construction 2.17. Given a configuration $c\colon S\to P^V$ and a separating neighborhood $c\subset U$, we let

$$\mathrm{Conf}_{S,U}^H(V)\subset \mathrm{Conf}_S^H(V)$$

be the (contractible) subspace of configurations $c'\colon S\to U\to P^V$ subordinate to U such that, for all $[G/H]\subset S$, the points $c([G/H])$ and $c'([G/H])$ lie in the same component of U.

This is compatible with inclusions, and thus defines a strict diagram

$$\mathcal{U}_S^H(V)^{\triangleright}\to \mathrm{Top}$$

[4] Given a smooth structure on X, it follows by triviality of tangent spaces on a 0-manifold that this agrees with the strong C^∞-topology. We will implicitly use this later.

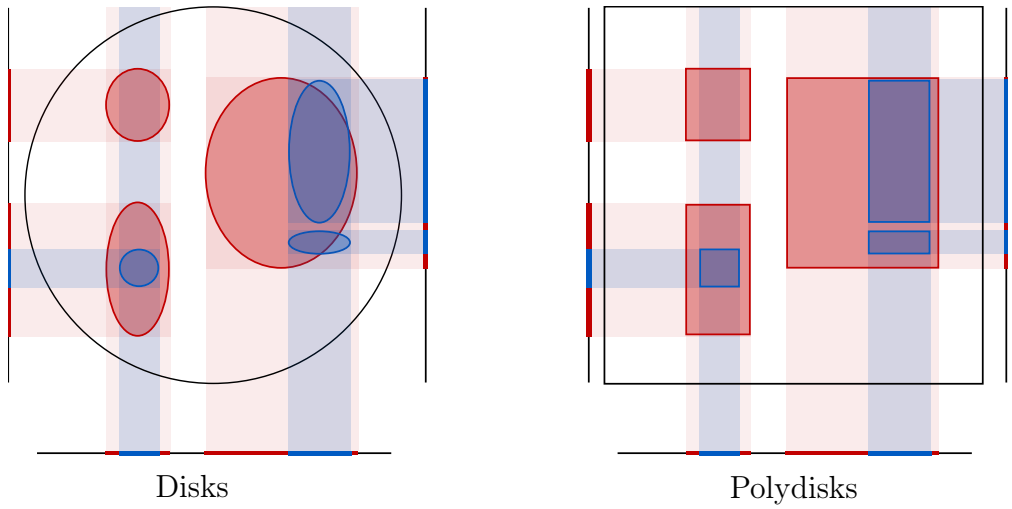


Figure 1. Factorizations of decomposable rectilinear $V \oplus W$-disks do not decompose; luckily, factorization of decomposable rectilinear $V \oplus W$-polydisks do decompose.

with cone point $\mathrm{Conf}_S^H(V)$ for any V-disk G-precover $\mathcal{U}$. In particular, taking coherent nerves, the universal property for colimits then yields a map

$$\alpha_{\mathcal{U},\mathrm{Conf}} \colon B\mathcal{U}_S^H \simeq \mathrm{colim}_{U \in \mathcal{U}} \mathrm{Conf}_{S,U}^H(V) \longrightarrow \mathrm{Conf}_S^H(V). \qquad \triangleleft$$

We also proved the following in [Ste25c].

Lemma 2.18. *If $\mathcal{U}$ is a separating G-cover, then $\alpha_{\mathcal{U},\mathrm{Conf}}$ is an equivalence.*

2.2.2. *V-polydisk G-precovers and (de)composition.* Our G-precovers will generally come as follows.

Construction 2.19. Let $\underline{\mathrm{RPDisk}}_S(V) \subset \underline{\mathrm{Open}}\left(P^V\right)$ denote the images of little polydisk embeddings $S \cdot P^V \hookrightarrow P^V$, and define the G-precover

$$\underline{\mathrm{RPDisk}}(V) \coloneqq \bigcup_{S \in \mathbb{F}_G} \underline{\mathrm{RPDisk}}_S(V).$$

We refer to sub-G-precovers of $\underline{\mathrm{RPDisk}}(V)$ as *V-polydisk G-precovers.* $\triangleleft$

The following proposition is our entire reason for considering polydisks; see Fig. 1.

Proposition 2.20 (Factorization decomposability)**.** *φ underlies an embedding of posets*

$$\varphi_{\underline{\mathrm{RPDisk}}} \colon \underline{\mathrm{RPDisk}}(V) \times_{\pi_0, \mathrm{Ob}\,\mathbb{F}_G} \coprod_{T \in \mathrm{Ob}\,\mathbb{F}_G} \underline{\mathrm{RPDisk}}(W)^T \longrightarrow \underline{\mathrm{RPDisk}}(V \oplus W);$$

that is, factorizations of decomposable rectilinear little $V \oplus W$-polydisk embeddings factor if and only if their decomposed V- and W-polydisk embeddings factor.

Proof. A rectilinear embedding $\iota \colon S \cdot P^V \to P^V$, by definition, consists of a configuration $\mathfrak{c}(\iota) \colon T \hookrightarrow P^{V \oplus W}$ together with a function $d \colon S \times n_V \to \mathbb{R}_{>0}$ subject to an injectivity condition. Moreover, we may explicitly describe factorization of another embedding $\iota' \colon P^V \hookrightarrow P^V$ through ι on a per-representation basis: ι' factors through ι if and only if, for all $i \in [n_V]$ and $[G/H] \subset S$,

(a) $\pi_{V_i} \mathfrak{c}_\iota([G/H]) \in \mathrm{im}(\pi_{V_i} \iota)$, and
(b) $d_{\iota'}([G/H], i) \leq w_{\pi_{V_i}\iota(V_i)}\left(\pi_{V_i}\iota'([G/H])\right)$.

The result follows by noting that, given a *decomposable* polydisk embedding, this precisely reduces to the same conditions on decomposed pieces. $\square$

We refer to the G-poset essential image of $\mathcal{U}_V \times_{\pi_0 \mathrm{Ob}\,\mathbb{F}_G} \coprod_{T \in \mathrm{Ob}\,\mathbb{F}_G} \mathcal{U}_W$ under $\varphi_{\underline{\mathrm{RPDisk}}}$ as $\mathcal{U}_V \mid \mathcal{U}_W$; in particular, we will write

$$\underline{\mathrm{RPDisk}}(V \mid W) \coloneqq \underline{\mathrm{RPDisk}}(V) \mid \underline{\mathrm{RPDisk}}(W).$$

For the purpose of *deformation*, we also note a weak convexity property of $\underline{\mathrm{RPDisk}}(V \mid W)$.

Proposition 2.21. *Suppose $\iota, \iota' \colon S \cdot P^{V\oplus W} \hookrightarrow P^{V\oplus W}$ are decomposable little disk embeddings with the same center. Then, for all $t \in [0,1]$, the linear combination*

$$h_t(\iota,\iota')(v) = t\cdot \iota(v) + (1-t)\iota'(v)$$

is a decomposable little disk embedding.

This follows immediately from the following lemma, which itself is obvious.

Lemma 2.22. *An S-indexed general little $V\oplus W$-polydisk embedding ι is decomposable precisely when the following conditions hold for all distinct orbits $[G/H] \neq [G/K] \subset S$:*

(a) If $\mathfrak{c}\pi_V \iota_{[G/H]} = \mathfrak{c}\pi_V \iota_{[G/K]}$, then $\mathfrak{t}\pi_V \iota_{[G/H]} = \mathfrak{t}\pi_V \iota_{[G/K]}$, and
(b) if $\pi_V \mathfrak{c}\iota_{[G/H]} \neq \pi_V \mathfrak{c}\iota_{[G/K]}$, then for all $i \in [n_V]$, we have

$$d\left(\pi_{V_n}\mathfrak{c}\iota_{[G/H]}, \pi_{V_n}\mathfrak{c}\iota_{[G/H]}\right) \geq \mathfrak{t}\pi_{V_n}\iota_{[G/H]} + \mathfrak{t}\pi_{V_n}\iota_{[G/K]}.$$

2.2.3. *Factorization and topology.* We now somewhat systematize the strategy of [May72, Thm. 4.8].

Definition 2.23. Given a configuration $i \colon [n] \hookrightarrow V^e$ and a sufficiently small number $\varepsilon \in \mathbb{R}_{>0}$, we define the *equidiameter embedding*

$$e_{i,\epsilon}(v) \coloneqq \iota + \epsilon \cdot v.$$

This assembles to an $\mathrm{Aut}^H(S)$-equivariant continuous map

$$e \colon \mathrm{Conf}_S^H(V) \times \mathbb{R}_{>0} \longrightarrow C^{\infty,H}\left(S\cdot P^V, P^V\right)$$

landing on maps which are componentwise-rectilinear, but which may not be embeddings. Given a V-disk G-precover $\mathcal{U}$, an *equidiameter datum for* $\mathcal{U}$ is a one parameter family of $G\times\Sigma_n$-equivariant continuous maps,

$$\varepsilon_{(-)}(-) \colon (0,1) \to \mathrm{Map}\left(\mathrm{Conf}_n(V), \mathbb{R}_{>0}\right),$$

(with righthand side having the compact-open topology) such that

(1) (infinitesimal values) $\lim_{s\to 0}\varepsilon_s = 0$, and
(2) (admissibility) for all $s \in (0,1)$, all $t\in[0,1]$, and all $\mathcal{U}$-admissible ι, the linearly interpolated map

$$h_{s,t}\iota(v) \coloneqq t\cdot e_{c(\iota),\varepsilon_s(c(\iota))} + (1-t)\iota(v)$$

is a $\mathcal{U}$-admissible embedding.

If $\mathcal{U}$ admits such ε for all n, we say that it is an *equidiameterizable V-disk G-cover.* ◁

Observation 2.24. Equidiameter G-covers are automatically complete separating G-covers. ◁

Now, we may set up a comparisons between embedding spaces, factorization, and configuration spaces.

Construction 2.25. Given an S-indexed little V-polydisk embedding $\iota \colon S\cdot P^V \hookrightarrow P^V$ with image U, there is a natural inclusion of *ι-subordinate embeddings* and *ι-subordinate configurations* making the following commute

$$\begin{array}{ccc} \mathrm{Emb}^{H,\mathrm{Rect}}_{\mathrm{id}_S}\left(S\cdot P^V, U\right) & \longrightarrow & \mathrm{Emb}^{H,\mathrm{Rect}}\left(S\cdot P^V, P^V\right) \\ \downarrow & & \downarrow \\ \mathrm{Conf}^H_{S,U}(V) & \longrightarrow & \mathrm{Conf}^H_S(V) \end{array}$$

It is easy to see that $\mathrm{Conf}^H_{S,U}(V)$ is contractible, so this yields a diagram

$$\begin{array}{ccccc} & & \mathrm{colim}_{U\in \mathrm{RPDisk}_S(V)} \mathrm{Emb}^{H,\mathrm{Rect}}_{\mathrm{id}_S}\left(S\cdot P^V, U\right) & \xrightarrow{\alpha} & \mathrm{Emb}^{H,\mathrm{Rect}}\left(S\cdot P^V, P^V\right) \\ & & \mathfrak{c}\downarrow & & \mathfrak{c}\downarrow \\ B\mathrm{RPDisk}_S(V) & \xrightarrow{\sim} & \mathrm{colim}_{U\in\mathrm{RPDisk}_S(V)} \mathrm{Conf}^H_{S,U}(V) & \xrightarrow{\alpha_{\mathrm{Conf}}} & \mathrm{Conf}^H_S(V) \end{array} \tag{8}$$

Here, α_{Conf} is the map of Lemma 2.18 in the case $\mathcal{U} = \underline{\mathrm{RPDisk}}(V)$. ◁

We find that, in good cases, these comparison maps are actually equivalences.

Proposition 2.26 (c.f. [May72, Thm. 4.8])**.** *Every map in Eq. (8) is an equivalence. Moreover, if $\mathcal{U}$ is an equidiameterizable V-disk G-cover, then taking centers yields a commutative diagram of weak equivalences*

$$\begin{array}{ccccc} B\mathcal{U}_S^H & \xrightarrow{\alpha_{\mathcal{U}}} & \mathrm{Emb}^{H,\mathrm{Rect},\mathcal{U}}(S\cdot D(V), D(V)) & \xrightarrow{\mathfrak{c}} & \mathrm{Conf}_S^H(V) \\ \downarrow & & \downarrow & & \| \\ B\mathrm{RPDisk}_S^H(V) & \xrightarrow{\alpha} & \mathrm{Emb}^{H,\mathrm{Rect},}(S\cdot D(V), D(V)) & \xrightarrow{\mathfrak{c}} & \mathrm{Conf}_S^H(V) \end{array}$$

Now, for the first statement to follow from the second, it suffices to prove the following.

Lemma 2.27. $\underline{\mathrm{RPDisk}}(V)$ *is equidiameterizable.*

Proof. Take the equidiameter datum $\varepsilon_s := s\cdot\varepsilon^{\max}$, defined as follows.

$$\varepsilon^{\max}\{x_1,\dots,x_n\} := \min_{i\neq j}\frac{d(x_i,x_j)}{2}.$$

This is clearly continuous, as it's a polynomial, so to verify that this is an equidiameter datum, we're tasked with verifying that $h_t(\iota, \lambda\cdot\varepsilon^{\max})$ is an embedding; as it's affine, it suffices to check that it's injective. But this the defining property of $\varepsilon^{\max}$. □

Proof of Proposition 2.26. Choose an equidiameter datum ε for $\mathcal{U}$. We begin with each map $\mathfrak{c}$. The (essentially unique) lifted $\mathrm{Aut}^H(S)$-equivariant map s of

$$\begin{array}{ccc} \mathrm{Conf}_S^H(V) & \overset{s}{\dashrightarrow} & \mathrm{Emb}^{H,\mathrm{Rect},\mathcal{U}}\left(S\cdot P^V, P^V\right) \\ {\scriptstyle (\mathrm{id},\varepsilon)}\downarrow & & \downarrow \\ \mathrm{Conf}_S^H(V)\times\mathbb{R}_{>0} & \xrightarrow{e} & C^\infty(S\cdot V, V) \end{array}$$

is a section of $\mathfrak{c}$, and h_t is a deformation retract of $\mathrm{Conf}_S^H(V)$ onto $\mathrm{im}(s)$, so $\mathfrak{c}$ is an equivalence.

This and Observation 2.9 construct a composite equivalence

$$\mathrm{Emb}_{\mathrm{id}_S}^{H,\mathrm{Rect}}\left(S\cdot P^V, S\cdot P^V\right) \simeq \prod_{[H/K]\subset S}\mathrm{Conf}_{*_K}^K(V) \simeq \prod_{[H/K]\subset S} V^K \simeq *.$$

In particular, the map $\mathrm{Emb}_{\mathrm{id}_S}^{H,\mathrm{Rect}}\left(S\cdot P^V, U\right)\to\mathrm{Conf}_{S,U}^H(V)$ is an equivalence, which allows us to construct the commutative diagram and note that $\mathfrak{c}\circ\alpha_{\mathcal{U}}$ is homotopic to $\alpha_{\mathcal{U},\mathrm{Conf}}$; Lemma 2.18 shows that this is an equivalence, so $\alpha_{\mathcal{U}}$ is as well. The remaining equivalences follow from Lemma 2.27 and two-out-of-three. □

Now, our main example to which we'll apply Proposition 2.26 is the following.

Proposition 2.28. $\underline{\mathrm{RPDisk}}(V\mid W)$ *is equidiameterizable; hence we have an equivalence*

$$\alpha\colon B\underline{\mathrm{RPDisk}}_S^H(V\mid W)\xrightarrow{\sim}\mathrm{Emb}^{H,\mathrm{Rect}}\left(S\cdot P^{V\oplus W}, P^{V\oplus W}\right).$$

By Proposition 2.21 it suffices to prove the following lemma and use equidiameter datum $\varepsilon_s = s\cdot\varepsilon^{V|W}$.

Lemma 2.29. *Define the map $\varepsilon^{V|W}\colon\mathrm{Conf}_n(V)\to\mathbb{R}_{>0}$ by*

$$\varepsilon^{V|W}(x_1,\dots,x_n) := \min\left\{\epsilon^{\max}\{x_1,\dots,x_n\}, \min_{\pi_V(x_i)\neq\pi_V(x_j)}\frac{d(\pi_V(x_i),\pi_V(x_j))}{2}\right\}.$$

Then, the equidiameter embedding $e_{\varepsilon,c}$ is decomposable if and only if $\varepsilon\le\varepsilon^{V|W}(c)$.

Proof. This is a straightforward application of Lemma 2.22. □

2.3. **Germs and tangential structures.** We now connect our work to equivariant differential topology.

2.3.1. *Smooth self-embeddings of representations.* Let $\mathrm{Diff}^G(V,0)$ denote the topological group of smooth origin-preserving diffeomorphisms of V under the strong C^∞-topology, $\mathrm{Emb}_0^G(V,V)$ the self-embedding space under the strong C^∞-topology. Let $O^G(V) \subset O(\dim V)$ and $\mathrm{GL}^G(V) \subset \mathrm{GL}(\dim V)$ be the topological subgroups of equivariant linear isometries and isomorphisms, respectively. Note that, by the same argument as Remark 2.5, these agree with the strong C^∞ topologies, and in particular, we acquire a diagram of topological submonoid inclusions

(9)
$$\begin{array}{ccccc}
\mathrm{Emb}_0^G(V,V) & \xrightarrow{a_{\mathrm{Emb}}} & \mathrm{Emb}^G(V,V) & & \\
\uparrow{\scriptstyle \iota_{\mathrm{Emb}}} & \nearrow & & & \\
\mathrm{Diff}^G(V,0) & \xrightarrow{a_{\mathrm{Diff}}} & \mathrm{Diff}^G(V) & & \\
\uparrow{\scriptstyle \iota_{\mathrm{Diff}}} & \nearrow & & & \\
O^G(V) & \xrightarrow{s} & \mathrm{GL}^G(V) & \xrightarrow{a_{\mathrm{Aff}}} & \mathrm{Emb}^{G,\mathrm{Aff}}(V,V)
\end{array}$$

Proposition 2.30. *All of the maps in Eq. (9) are equivalences.*

Proof. It suffices to prove that the solid arrows are equivalences, after which the dashed arrows become equivalences by two out of three. s is an equivalence by the Gram-Schmidt procedure. a_{Emb} admits a deformation retract by $h_t(\iota) = \iota - t \cdot \iota(0)$; this restricts to deformation retracts of a_{Diff} and a_{Aff}. To finish, we simply note that the deformation retract constructed in [Kup17, Thm. 2.4] restricts to a deformation retract for ι_{Emb} and ι_{Diff}. □

2.3.2. *Germs of embeddings.* It is a classical fact that, up to homotopy, germs of embeddings $k \cdot \mathbb{R}^n \to M$ differ from configurations of k points in $\mathbb{R}^n$ by a power of $O(n)$. We will equivariantize this, for which we need the following definition.

Definition 2.31. An *S-indexed multidisk* is a smooth G-manifold D admitting a sequence of maps

$$D \hookrightarrow E \twoheadrightarrow S,$$

where $E \twoheadrightarrow S$ is an equivariant vector bundle and $D \subset R$ is the open unit disk. Given $[G/H] \subset S$ an orbit, we write $D_{[G/H]}$ for the corresponding path orbit lying over $[G/H]$, and we write D_H for the corresponding H-representation over the identity coset, so that

$$D \simeq \coprod_{[G/H] \subset S} D_{[G/H]} \simeq \coprod_{[G/H] \subset S} G \times_H D_H. \qquad \triangleleft$$

These are the local type about a configuration. Indeed, we will find the following.

Proposition 2.32. *Given a G-manifold M, the point-set sequential colimit*

$$\mathrm{Germ}_D^G(M) \coloneqq \mathrm{colim}\left(\mathrm{Emb}^G(D,M) \xrightarrow{e_{0,\frac{1}{2}}^*} \mathrm{Emb}^G(D,M) \xrightarrow{e_{0,\frac{1}{2}}^*} \cdots \right) \xrightarrow{\mathfrak{c}} \mathrm{Conf}_S^G(M)$$

admits the structure of a fiber bundle over its image with fiber $\prod_{[H/K] \subset S} O^K(D_K)$ and

We prove local triviality by an equivariant lift of the usual argument.

Lemma 2.33 (Isotopy extension)**.** *If N is compact, then every continuous $\mathrm{Diff}^G(M)$-equivariant map $X \to \mathrm{Emb}^G(N,M)$ is a fiber bundle over its image.*

Proof. By [Pal60, Thm. A], we must check that all $\iota \in \mathrm{Emb}^G(N,M)$ admit a neighborhood $\iota \in U$ and map $\chi \colon U \to \mathrm{Diff}^G(M)$ such that $\iota' = \chi(\iota') \cdot \iota$. The proof of [Lim64] straightforwardly shows this, by choosing equivariant tubular neighborhoods using [Bre72].[5] □

[5] The author will explain this in more explicit detail in forthcoming work with Branko Juran.

To see this as a form of isotopy extension, consider the case of $\iota^*\colon N \hookrightarrow N'$ a compact submanifold, in which case we find that $\iota\colon \mathrm{Emb}^G(N',M) \to \mathrm{Emb}^G(N,M)$ is onto its image; since $C^\infty(N,M)$ is metrizable when N is compact, its subspaces are paracompact, and hence ι^* is a Serre fibration, and one may extend isotopies using the associated lifting property.

Now, to identify germs with embeddings, we note the following.

Lemma 2.34 (Isototopy to sub-disk)**.** *Given any $\epsilon \in (0,1]$, the precomposition embedding of topological spaces*

$$e^*_{0,\epsilon}\colon \mathrm{Emb}^G(D,M) \longrightarrow \mathrm{Emb}^G(D,M)$$

admits a deformation retract; in particular, it is an acyclic Serre cofibration.

Proof. It suffices to construct an equivariant isotopy $e_{0,\varepsilon} \sim \mathrm{id}_{P^V}$; we choose $h_{t,\varepsilon} = e_{0,t+(1-t)\cdot\varepsilon}$. □

Proof of Proposition 2.32. Choosing $N = S$ in Lemma 2.33 shows that $\mathfrak{c}$ is a fiber bundle over its image and Lemma 2.33 shows that it is the homotopy colimit in $\mathcal{S}_{/\mathrm{Conf}^G_S(V)}$ such that $\mathrm{Germ}^G_D(M) \to \mathrm{Emb}^G(D,M)$ is an equivalence.

We're left with computing the point-set fiber F_c over a configuration c, showing that the derivative map

$$d\colon F_c \longrightarrow \prod_{[H/K]\subset S} O^K(D_K).$$

admits an inverse given by precomposing a fixed embedding with well-chosen elements of $\prod_{[H/K]\subset S} \mathrm{GL}^K(D_K)$. This argument is routine, so we omit it. □

Remark 2.35. Let $\mathrm{Conf}^G_D(M) \subset \mathrm{Conf}^G_S(M)$ be the image of $\mathrm{Germ}^G_D(M)$ under $\mathfrak{c}$. We claim three things:

(i) $\mathrm{Conf}^G_D(M) \subset \mathrm{Conf}^G_S(M)$ is a disjoint union of path-components.
(ii) A configuration $c\colon S \hookrightarrow M$ lies in $\mathrm{Conf}^G_D(M)$ if and only if, for each orbit $[G/H] \subset S$, there exists an equivalence of H-representations $T_{c(G/H)}M \simeq D_H$.
(iii) In the case that M admits a V-framing (such as $M = V$), we have

$$\mathrm{Conf}^G_D(M) = \begin{cases} \mathrm{Conf}^G_S(M) & D \simeq S \cdot D^V, \\ \varnothing & \text{otherwise.} \end{cases}$$

Claim (iii) follows straightforwardly from Claim (ii), and Claim (ii) follows by noting that the derivative of an embedding $S \cdot D(V) \hookrightarrow M$ at $[G/H]$ is such an equivalence, and conversely by taking equivariant tubular neighborhoods using [Bre72].

For Claim (i), note that we may express configurations spaces homeomorphically as products

$$\begin{aligned} \mathrm{Conf}^G_S(M) &\simeq \prod_{(H)\subset G} \mathrm{Conf}^G_{S_{(H)}}(M) \\ &\simeq \prod_{(H)\subset G} \mathrm{Conf}^G_{S_{(H)}}(M_{(H)}), \end{aligned}$$

where $M_{(H)} \subset M$ denotes the topological subspace of points x such that $(\mathsf{stab}(x)) = (H)$. As such, it suffices to prove this in the case that S and M have fixed isotropy (H).

Now, in this case, note that the representation T_xM is *locally constant* in x; Claim (ii) then implies that all paths in $\mathrm{Conf}^G_S(M)$ beginning in $\mathrm{Conf}^G_D(M)$ must end in $\mathrm{Conf}^G_D(M)$, which is what we wanted to show. ◂

Variant 2.36. By Remark 2.35, the assembly equivalence of Proposition 2.16 restricts to an equivalence

$$\mathrm{colim}_{U\in\mathcal{U}} \mathrm{Conf}^G_D(U) \xrightarrow{\sim} \mathrm{Conf}^G_D(M)$$

for any complete Weiss G-cover $\mathcal{U}$. ◂

Before moving on, we acquire a useful technicality about embedding spaces.

Corollary 2.37. *The following diagram consists of homotopy-cartesian squares:*

$$\begin{array}{ccccc} \mathrm{Emb}^G(D,M) & \simeq & \mathrm{Germ}^G_D(M) & \xrightarrow{\mathfrak{c}} & \mathrm{Conf}^G_D(M) \\ \downarrow & & \downarrow & & \downarrow \\ \prod_{[G/H]\subset S} \mathrm{Emb}^H(D_H,M) & \simeq & \prod_{[G/H]\subset S} \mathrm{Germ}^H_{D_H}(M) & \xrightarrow{\mathfrak{c}} & \prod_{[G/H]\subset S} M^H \end{array}$$

Proof. By [HTT, Lem. 4.4.2.1] , the lefthand square is homotopy-cartesian, so it suffices to prove the same for the righthand square. Since M^H and $\mathrm{Conf}^G_D(M)$ are smooth manifolds, they are paracompact, and so each map $\mathfrak{c}$ is a Serre fibration. Hence it suffices to prove that this is a point-set cartesian square; equivalently, it suffices to prove that it induces a homeomorphism on point-set fibers. Indeed, each of the vertical maps are injections, and the bottom horizontal map takes image bijectively onto image. □

2.3.3. *Embeddings with tangential structures.*

Definition 2.38. Given $\Theta\colon B_\Theta \to B_G O(n)$ a map of G-spaces, a Θ-*framed G-manifold* is a pair of M a G-manifold and $T^\Theta M$ a lift

$$\begin{array}{ccc} & & B_\Theta \\ & {\scriptstyle T^\Theta M}\nearrow & \downarrow{\scriptstyle \Theta} \\ M & \xrightarrow[TM]{} & B_G O(n) \end{array}$$

We will usually abusively refer to the pair $(M, T^\Theta M)$ simply as M. Given M, N a pair of Θ-framed G-manifolds, we define the *space of H-equivariant Θ-framed embeddings of M into N* as the (homotopy) pullback

$$\begin{array}{ccc} \mathrm{Emb}^{\Theta,H}(M,N) & \longrightarrow & \mathrm{Emb}^H(M,N) \\ \downarrow & \lrcorner & \downarrow{\scriptstyle d} \\ \mathrm{Map}^H_{/Res^G_H B_\Theta}(M,N) & \xrightarrow[\mathrm{Res}^G_H \Theta_!]{} & \mathrm{Map}^H_{/B_H O(n)}(M,N) \end{array} \tag{10}$$

In the case that $B_\Theta = *_G$ and the map $*_G \to B_G O(n)$ classifies the n-dimensional orthogonal representation V, we call these (embeddings of) *V-framed G-manifolds*.

We refer to a Θ-framed G-manifold D whose underlying G-manifold is an S-indexed multidisk as a *Θ-framed S-indexed multidisk*. ◁

Given D a Θ-framed S-indexed multidisk, we have a composite map

$$\mathfrak{c}_M\colon \mathrm{Emb}^{H,\Theta}(D,M) \longrightarrow \mathrm{Emb}^H(D,M) \longrightarrow \mathrm{Conf}^H_D(M).$$

We may bootstrap a result of Horev to compute its homotopy fibers.

Proposition 2.39. *The homotopy fibers of $\mathfrak{c}$ are computed by*

$$\mathfrak{c}_M^{-1}(\iota) \simeq \prod_{G/H\subset S} \Omega_{D_{G/H}} B^H_\Theta.$$

In particular, any Θ-framed embedding $f\colon M \hookrightarrow N$ induces an equivalence $\mathfrak{c}_M^{-1}(\iota) \xrightarrow{\sim} \mathfrak{c}_N^{-1}(f\iota)$.

Proof of Proposition 2.39. Consider the diagram

$$\begin{array}{ccccc} \prod_{G/H\in S} \Omega_{\Theta\iota(G/H)} B^H_\Theta & \longrightarrow & \mathrm{Emb}^{\Theta,G}_M(S,M) & \longrightarrow & \prod_{G/H\subset S} \mathrm{Emb}^{\Theta,G}_M(G/H,M) \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ \{\iota\} & \longrightarrow & \mathrm{Conf}^G_D(M) & \longrightarrow & \prod_{G/H\subset S} M^H \end{array}$$

The righthand rectangle is cartesian by Corollary 2.37, and the outer diagram is cartesian by taking products of [Hor19]. Hence the lefthand diagram is cartesian by [HTT, Lem. 4.4.2.1] . □

Lastly, we will use the following to assemble (equivariant) smooth invariants from rectilinear ones.

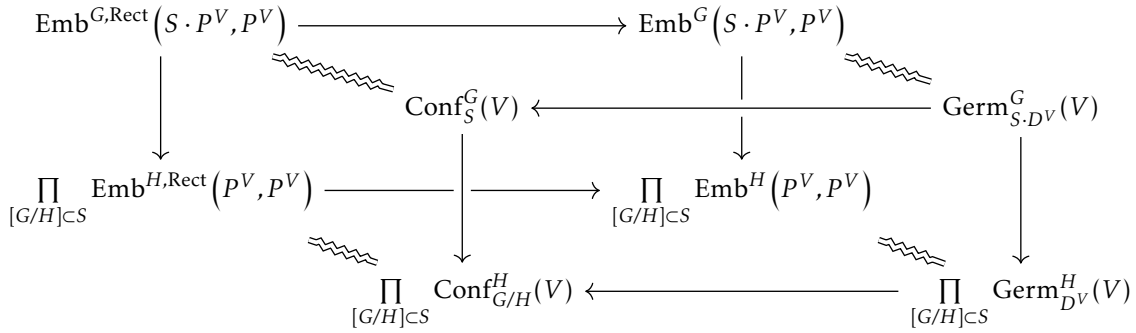
Corollary 2.40. *The following diagram consists of homotopy-cartesian squares:*

$$\begin{array}{ccccc}
\mathrm{Emb}^{G,\mathrm{Rect}}\left(S\cdot P^V, P^V\right) & \longrightarrow & \prod_{[G/H]\subset S}\mathrm{Emb}^{H,\mathrm{Rect}}\left(S\cdot P^V, P^V\right) & \xrightarrow{t} & * \\
\downarrow{\scriptstyle \iota} & \lrcorner & \downarrow{\scriptstyle \iota} & \lrcorner & \downarrow \\
\mathrm{Emb}^{G}\left(S\cdot P^V, P^V\right) & \longrightarrow & \prod_{[G/H]\subset S}\mathrm{Emb}^{H}\left(P^V, P^V\right) & \xrightarrow{t} & \prod_{[G/H]\subset S} O^H(V)
\end{array}$$

Proof. Both arrows marked t are equivalences by Propositions 2.26 and 2.30, so the righthand square is homotopy-cartesian by [HTT, Lem. 4.4.2.1] ; by another application of [HTT, Lem. 4.4.2.1] , it suffices to prove that the lefthand square is homotopy-cartesian. Now, note that the sequence

$$\mathrm{Emb}^{G,\mathrm{Rect}}\left(S\cdot P^V, P^V\right)\longrightarrow \mathrm{Emb}^{G}\left(S\cdot D^V, D^V\right)\longrightarrow \mathrm{Germ}^G_{S\cdot D^V}(V)\longrightarrow \mathrm{Conf}^G_S(V)$$

is natural in S and has composite and middle arrow an equivalence; hence it has a coassembly natural transformation



whose back face is exactly the square in question. Here, the squiggly double headless arrows are equivalences by Propositions 2.26 and 2.32, so the left and right faces are homotopy-cartesian by [HTT, Lem. 4.4.2.1] .

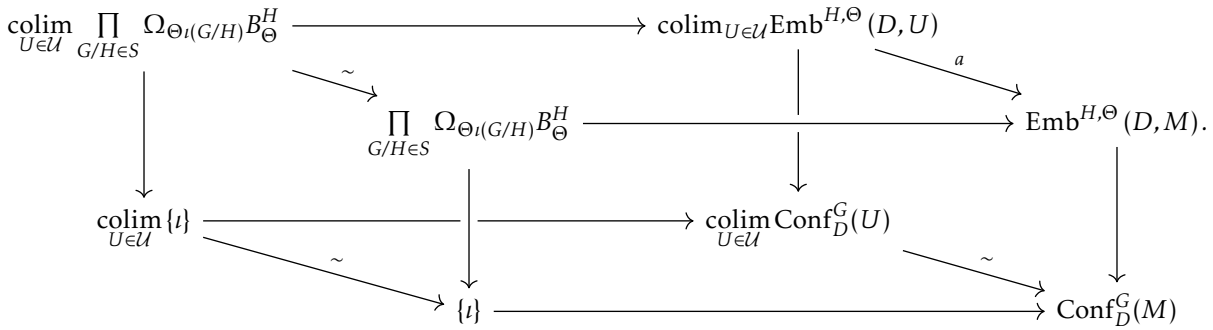
By repeatedly applying [HTT, Lem. 4.4.2.1] , it suffices to prove that the front face is homotopy-cartesian; that is, our square admits a retract (as an arrow in the arrow category), so we may check that the retraction square is homotopy-cartesian. But this is Corollary 2.37. □

2.3.4. *Hypercodescent.* We acquire an important hypercodescent phenomenon for little disk spaces.

Corollary 2.41 (Filtered Weiss hypercodescent of embedding spaces)**.** *Given a Θ-framed G-manifold M, a Θ-framed multidisk D, and a complete G-Weiss cover $\mathcal{U}\subset \underline{\mathrm{Open}}(M)$ such that $\mathcal{U}^H$ is filtered for all $H\subset G$, assembly yields an equivalence*

$$a\colon \mathrm{colim}_{U\in\mathcal{U}}\mathrm{Emb}^{H,\Theta}(D,U)\longrightarrow \mathrm{Emb}^{H,\Theta}(D,M).$$

Proof. Since filtered colimits commute with finite limits in spaces, we have an assembly transformation



of (homotpy) cartesian squares. The upper left assembly arrow is an equivalence by Proposition 2.39, the bottom left by weak contractibility of filtered diagrams, and the bottom right by Variant 2.36. Hence the top right assembly arrow is an equivalence, as desired. □

Example 2.42. We refer to the images of elements of $\mathrm{Emb}^{H,\Theta}(D,M)$ as $\underline{\mathrm{Disk}}^{\Theta}_{D}(M)$, and we write

$$\underline{\mathrm{Disk}}^{\Theta}(M) \coloneqq \bigcup_{\substack{D\to S\\ \Theta\text{-framed}}} \underline{\mathrm{Disk}}^{\Theta}_{D}(M).$$

We write $\underline{\mathrm{Disk}}^{\Theta}(M)\times\underline{\mathrm{Disk}}^{\Psi}(N)\subset\underline{\mathrm{Disk}}^{\Theta\oplus\Psi}(M\times N)$ for the sub-G-precover of disjoint unions of products of Θ-framed little disks.

Each of these precovers is closed under disjoint unions; moreover, taking tubular neighborhoods demonstrates that each is a complete G-Weiss cover. Hence Corollary 2.41 assembles $\mathrm{Emb}^{H,\Theta}(D,M)$ from Θ-framed embeddings in (disjoint unions of Θ-framed) *Euclidean* G-manifolds. ◁

3. Little disk and prefactorization algebras and the additivity theorems

We finish by defining the objects appearing in our main theorems and proving the theorems. In particular, Theorem A corresponds with Section 3.1, Corollary B with Section 3.2, Corollary C with Section 3.3, and Theorem D with Section 3.4.

3.1. Additivity of little disk G-operads.

3.1.1. *A few models for little disk algebras.* We first define the *topological little V-polydisk G-operad.*

Construction 3.1. We define the topological G-symmetric sequence

$$\mathbb{E}^{t}_{V_\bullet}(S) \coloneqq \mathrm{Emb}^{H,\mathrm{Rect}}\left(S\cdot P^{V_\bullet}, P^{V_\bullet}\right).$$

with $\mathrm{Aut}_H S$ acting by precomposition and restriction as in Observation 2.8; note that restriction is Borel $\mathrm{Aut}_H S$-equivariant by Lemma 2.10. ◁

Note that the Segal condition of Observation 2.9 characterizes $\mathbb{E}^{t}_{V}$-multioperations:

$$\mathrm{Mul}^{f}_{\mathbb{E}^{t}_{V_\bullet}}(T;S) \simeq \mathrm{Emb}^{H,\mathrm{Rect}}_{f}\left(T\cdot P^{V}, S\cdot P^{V}\right).$$

In particular, the composition map γ yields a composition operation

$$\gamma\colon \mathrm{Mul}^{g}_{\mathbb{E}^{t}_{V_\bullet}}(S;R)\times\mathrm{Mul}^{f}_{\mathbb{E}^{t}_{V_\bullet}}(T;S)\longrightarrow \mathrm{Mul}^{g\circ f}_{\mathbb{E}^{t}_{V_\bullet}}(T;R).$$

Proposition 3.2. *$(\mathbb{E}^{t}_{V_\bullet}, (\mathrm{id}_{P^V}), \gamma)$ is a genuine topological G-operad.*

Proof. We're tasked with verifying Conditions (OP-a) to (OP-e). To start, note that Conditions (OP-a) and (OP-c) (i.e. unitality and restriction-stability of units) are obvious. Moreover, Condition (OP-b) is simply Observation 2.11 and Condition (OP-d) is verified in the second statement of Lemma 2.10. Lastly, equivariance of γ follows from the same property for composition of equivariant maps. □

We write $\mathbb{E}^{\otimes}_{V}\coloneqq N^{\otimes}\mathbb{E}^{t}_{V_\bullet}$. We'd like to discuss change of decomposition, for which we need the following.

Variant 3.3. Given $\mathcal{U}$ a V-polydisk G-precover satisfying the property that $\mathcal{U}$-admissible embeddings are closed under composition, we acquire a topological suboperad $\mathbb{E}^{t}_{V_\bullet,\mathcal{U}}\subset\mathbb{E}^{t}_{V_\bullet}$ which, on structure spaces, yields the inclusion

$$\mathrm{Emb}^{H,\mathrm{Rect},\ \mathcal{U}}_{f}\left(T\cdot P^{V}, S\cdot P^{V}\right)\subset\mathrm{Emb}^{H,\mathrm{Rect}}_{f}\left(T\cdot P^{V}, S\cdot P^{V}\right).$$ ◁

Proposition 3.4. *If $\mathcal{U}$ is closed under composition and equidiameterizable, then the inclusion $\mathbb{E}^{t}_{V,\mathcal{U}}\to\mathbb{E}^{t}_{V}$ is an equivalence of genuine topological G-operads.*

Proof. This follows immediately from Proposition 2.26. □

Using this, we are free to choose convenient decompositions:

Corollary 3.5. *If V and V' are polyrepresentations with the same direct sum, then there exists a zigzag of equivalences connecting $\mathbb{E}^{t}_{V}$ and $\mathbb{E}^{t}_{V'}$.*

Proof. After Proposition 3.4, it suffices to construct equidiameterizable V- and V'-polydisk G-covers $\mathcal{U}$ and $\mathcal{U}'$ and an equivalence $\mathbb{E}^t_{V,\mathcal{U}} \xrightarrow{\sim} \mathbb{E}^t_{V',\mathcal{U}'}$. Indeed, we let $\mathcal{U}$ be the equidiamter V-polydisk embeddings whose corresponding equidiameter V'-polydisk map (with the same diameter and center) is an embedding, and we define $\mathcal{U}'$ to be the corresponding V'-polydisk embeddings.

It is obvious that this correspondence yields an equivalence $\mathbb{E}^t_{V,\mathcal{U}} \xrightarrow{\sim} \mathbb{E}^t_{V',\mathcal{U}'}$, and $\mathcal{U}$ and $\mathcal{U}'$ are tautologically equidiameterizable. □

The reader is free to henceforth always assume the symbols V and W correspond with one-step polyrepresentations.

Observation 3.6. Observation 2.13 constructs a map $\mathbb{E}^t_V \to \mathbb{E}^t_{V\oplus W}$; given a sub-representation inclusion $U \subset V$, choosing an orthogonal complement to U and taking nerves yields a map $\mathbb{E}^\otimes_U \to \mathbb{E}^\otimes_V$. ◁

Now, the following constructs a *Dunn map.* $\varphi_{\mathbb{E}} \colon \mathbb{E}^\otimes_V \wr \mathbb{E}^\otimes_W \to \mathbb{E}^\otimes_{V\oplus W}$.

Proposition 3.7. *The maps* $\varphi \colon \mathbb{E}^t_V(S) \times \mathrm{Mul}^{T\to S}_{\mathbb{E}^t_W}(T;S) \to \mathbb{E}^t_{V\oplus W,\omega(V,W)}(T)$ *of Construction 2.12 satisfy Proposition 1.63.*

Proof. The first and second diagram of Observation 2.13 correspond with Eq. (2) and Eq. (3), respectively. □

Of course, composition yields a diagram of G-preoperads

$$\mathbb{E}^\otimes_V \odot \mathbb{E}^\otimes_W \to \mathbb{E}^\otimes_V \wr \mathbb{E}^\otimes_W \to \mathbb{E}^\otimes_{V\oplus W},$$

which corresponds canonically with a map of G-operads $\varphi \colon \mathbb{E}^\otimes_V \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_W \to \mathbb{E}^\otimes_{V\oplus W}$. The precise form of Theorem A is that φ is an equivalence together with the following observation.

Observation 3.8. Observation 2.13 demonstrates that the right square of the following commutes given $U \subset V$:

$$\begin{array}{ccccccc}
\mathbb{E}^\otimes_U \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_W & \longleftarrow & \mathbb{E}^\otimes_U \odot \mathbb{E}^\otimes_W & \longrightarrow & \mathbb{E}^\otimes_U \wr \mathbb{E}^\otimes_W & \longrightarrow & \mathbb{E}^\otimes_{U\oplus W} \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
\mathbb{E}^\otimes_V \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_W & \longleftarrow & \mathbb{E}^\otimes_V \odot \mathbb{E}^\otimes_W & \longrightarrow & \mathbb{E}^\otimes_V \wr \mathbb{E}^\otimes_W & \longrightarrow & \mathbb{E}^\otimes_{V\oplus W}
\end{array}$$

We've already verified that the rest of the inner diagram commutes, so in particular, the outer square commutes; that is, the Dunn map is natural with respect to summand inclusions in each factor. In particular, taking $U = 0$ implies that the composite map

$$\mathbb{E}^\otimes_W \simeq \mathbb{E}^\otimes_0 \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_W \xrightarrow{(\iota,\mathrm{id})} \mathbb{E}^\otimes_V \overset{\mathrm{BV}}{\otimes} \mathbb{E}^\otimes_W \xrightarrow{\varphi} \mathbb{E}^\otimes_{V\oplus W}$$

is ι; we interpret this as showing that "φ extends ι." ◁

We will define a variant of this using the smooth category, which we now recall.

Recollection 3.9 ([Hor19])**.** There exists a topological G-symmetric monoidal category $\mathbf{Mfld}^{\sqcup}_G$ such that

(1) The H-value topological category $\mathbf{Mfld}_H$ has objects the smooth H-manifolds and mapping topological spaces $\mathrm{Emb}^H(M,N)$ under the strong C^∞ topology.
(2) The restriction functor $\mathbf{Mfld}_H \to \mathbf{Mfld}_K$ takes $H \to \mathrm{Diff}(M)$ to the restriction $K \to H \to \mathrm{Diff}(M)$.
(3) The tensor products in $\mathbf{Mfld}_H$ are disjoint unions $(M,N) \mapsto M \sqcup N$.
(4) The norm functor $\mathbf{Mfld}_K \to \mathbf{Mfld}_H$ is topological induction $M \mapsto H \times_K M$. ◁

Variant 3.10. We define a "fattened" version of $\mathbb{E}^t_V$. Consider the evident inclusion $\underline{\mathrm{RPDisk}}_*(V) \hookrightarrow \mathbf{Mfld}_G$. Let $\mathcal{C}^t \subset \mathbf{Mfld}_G$ be the full topological G-suboperad spanned by its essential image. We define the wide topological G-suboperad $\widetilde{\mathbb{E}}^t_V \subset \mathcal{C}^t$ to have operations

$$\widetilde{\mathbb{E}}^t_V(U';U) \simeq \mathrm{Emb}^{H,\mathrm{Rect}}(U',U),$$

where here we write $\mathrm{Emb}^{H,\mathrm{Rect}}(U',U)$ for those embeddings which become rectilinear under the pre-postcomposition homeomorphism $\mathrm{Emb}^H(U',U) \simeq \mathrm{Emb}^H\left(S \cdot P^V, P^V\right)$. Of course, this a topological full

G-suboperad embedding $\phi\colon \mathbb{E}^t_V \to \widetilde{\mathbb{E}}^t_V$ by including in P^V; moreover, the same recipe as Proposition 3.7 constructs an extension $\widetilde{\varphi}_{\mathbb{E}}$ making the following commute:

$$\begin{array}{ccc} \mathbb{E}^t_{V_\bullet} \wr \mathbb{E}^t_{W_\bullet} & \xrightarrow{\phi \wr \phi} & \widetilde{\mathbb{E}}^t_{V_\bullet} \wr \widetilde{\mathbb{E}}^t_{W_\bullet} \\ \downarrow{\scriptstyle \varphi_{\mathbb{E}}} & & \downarrow{\scriptstyle \widetilde{\varphi}_{\mathbb{E}}} \\ \mathbb{E}^t_{V_\bullet \oplus W_\bullet} & \xrightarrow{\phi} & \widetilde{\mathbb{E}}^t_{V_\bullet \oplus W_\bullet} \end{array}$$

◂

Proposition 3.11. *The map $\phi\colon \mathbb{E}^t_V \to \widetilde{\mathbb{E}}^t_V$ is an equivalence.*

Proof. First note that $U\widetilde{\mathbb{E}}^t_V$ is a connected G-space; moreover, its loop space is $\mathrm{Emb}^{H,\mathrm{Rect}}\left(P^V, P^V\right) \simeq *$, so $\widetilde{\mathbb{E}}^t_V$ has one color. Hence it suffices to prove that ϕ induces equivalences on operation spaces, i.e. the map

$$\mathrm{Emb}^{H,\mathrm{Rect}}(U', U) \to \mathrm{Emb}^{H,\mathrm{Rect}}\left(S \cdot P^V, P^V\right)$$

is an equivalence; this follows from the fact that $S \cdot P^V \to U'$ and $U \to P^V$ are diffeomorphisms. □

3.1.2. *Prefactorization algebras and self-similarity prefactorization algebras.* Let $\underline{\mathrm{Open}}_*(X) \subset \underline{\mathrm{Open}}(X)$ denote the G-subposet of connected open subsets.

Construction 3.12. Given a topological space X with G-action, we define the *factorization* $\underline{\pi}_0\underline{\mathrm{Open}}_*(X)$-*symmetric sequence*

$$\underline{\mathrm{Open}}(X)((Y_K)_S; Z) = \begin{cases} * & \exists \text{ disjoint factorization} \coprod_K^S Y_K \hookrightarrow Z \hookrightarrow X; \\ \varnothing & \text{otherwise.} \end{cases}$$

More generally, we define $\mathcal{U}(-;-)$ by restriction along $\underline{\pi}_0\mathcal{U}_* \hookrightarrow \underline{\pi}_0\underline{\mathrm{Open}}_*(X)$. By "disjoint factorization," we mean that $\coprod_K^S Y_K \hookrightarrow Z$ itself is an embedding. Note that the factorization is unique, if it exists. ◂

Proposition 3.13. *There exists an essentially unique $\underline{\pi}_0\mathcal{U}$-colored G-0-preoperad $\mathcal{U}^\otimes$ with $\underline{\pi}_0\mathcal{U}$-symmetric sequence $\mathcal{U}(-;-)$.*

Proof. Since $\mathcal{U}(-;-)$ has 0-truncated structure spaces, there is at most one choice of maps 1_C and of γ, and if they exist, the associativity, unitality, equivariance, and restriction-stability diagrams all commute.

For existence of 1_C, it suffices to note that $\mathcal{U}(U;U)$ is nonempty. For γ, we're tasked with verifying that, whenever $\mathcal{U}(U';U'') \times \prod_{[H/K] \subset \underline{\pi}_0 U'} \mathcal{U}\left(U_K; U'_K\right)$ is nonempty, so is $\mathcal{U}(U;U'')$ where $U = \bigcup_K U_K$. The former condition says that each arrow in $U \to U' \to U''$ is injective and implies that $U \to P^V$ is in $\mathcal{U}$. By composing injections, this implies that $\mathcal{U}(U;U'')$ is nonempty, as desired. □

$\mathcal{U}$ encodes a *genuine equivariant* version of prefactorization algebras. We would like to use this to exercise fine control over different versions of $\mathbb{E}_V$, as *locally constant $\mathcal{U}$-admissible rectilinear embeddings*, but to do so, we this must be realized as *all* spaces of active arrows in our corepresenting object, unlike the case of $\mathcal{U}$, where these spaces of active arrows may vary. As such, we introduce a variant.

Construction 3.14. Given a V-polydisk G-precover $\mathcal{U} \subset \underline{\mathrm{RPDisk}}(V)$, we define the replete G-subpreoperad

$$\mathbb{P}^\otimes_{\mathcal{U}} \subset \underline{\mathrm{RPDisk}}(V)^\otimes$$

with colors the G-subposet $\mathcal{U} \subset \underline{\mathrm{RPDisk}}_*(V)$ and operations

$$\mathbb{P}_{\mathcal{U}}(U';U) \simeq \begin{cases} * & U' \in i^*\mathcal{U} \subset \underline{\mathrm{RPDisk}}(U) \subset \underline{\mathrm{Open}}(U) \subset \underline{\mathrm{Open}}(X), \\ \varnothing & \text{otherwise;} \end{cases}$$

That is, the operations with target U correspond with tuples of operations with terminal target in $\mathrm{Res}^G_{\mathrm{stab}(U_i)}\mathcal{U}^\otimes$ under the diffeomorphism $U_i \simeq \mathrm{Res}^G_{\mathrm{stab}(U_i)}$ for each path orbit $U_i \subset V$. ◂

Indeed, the following proposition is immediate and establishes our desired property.

Proposition 3.15. *Given $\mathcal{U} \subset \mathcal{U}'$ a related pair of V-polydisk G-precovers, the inclusion $\mathcal{U} \subset \mathcal{U}'$ underlies an essentially-unique G-preoperad map $\mathbb{P}^{\otimes}_{\mathcal{U}} \to \mathbb{P}^{\otimes}_{\mathcal{U}'}$; given $U \subset V$ an $\mathcal{U}$-admissible S-indexed little embedded little V-multipolydisk, the induced map*

$$\mathcal{U}^S_{/P^V} \simeq \mathbb{P}^{\mathrm{act}}_{\mathcal{U},/U} \to \mathbb{P}^{\mathrm{act}}_{\mathcal{U}',/U} \simeq \mathcal{U}'^S_{/X}$$

is a sliced power of the inclusion $\mathcal{U} \subset \mathcal{U}'$.

Remark 3.16. If a V-polydisk G-precover $\mathcal{U} \subset \underline{\mathrm{RPDisk}}(V)$ satisfies the property that $U' \in i^*\mathcal{U}$ if and only if $U' \in \mathcal{U}$, then $\mathbb{P}^{\otimes}_{\mathcal{U}} = \mathcal{U}^{\otimes}$ as replete suboperads of $\underline{\mathrm{Open}}(X)$, and in particular, $\mathbb{P}^{\otimes}_{\mathcal{U}}$ is a G-operad. ◁

Example 3.17. The G-cover $\mathcal{U} = \underline{\mathrm{RPDisk}}(V)$ satisfies the property called for in Remark 3.16; we define the G-operad $\mathbb{P}^{\otimes}_V := \mathbb{P}^{\otimes}_{\underline{\mathrm{RPDisk}}(V)} \simeq \underline{\mathrm{RPDisk}}(V)^{\otimes}$. In the case $G = e$, these are called $\mathbb{D}_n$ in [Har, Def. 5.2.3]. We also write $\mathbb{P}^{\otimes}_{V|W} := \mathbb{P}^{\otimes}_{\underline{\mathrm{RPDisk}}(V|W)}$. ◁

3.1.3. *Locally constant prefactorization algebras.* We now need to define a map $\mathbb{P}^{\otimes}_V \to \mathbb{E}^{\otimes}_V$.

Construction 3.18. Given a rectilinear little V-polydisk embedding $\iota\colon U \to U'$, the tautological point $\iota \in \mathrm{Emb}^{H,\mathrm{Rect}}(U, U')$ is compatible with G-operadic composition in $\widetilde{\mathbb{E}}^t_V$, and so it constructs a wide topological G-suboperad

$$\widetilde{\alpha}\colon \mathbb{P}^t_V \subset \widetilde{\mathbb{E}}^t_V;$$

such that $N^{\otimes}\mathbb{P}^t_V \simeq \mathbb{P}^{\otimes}_V$. We refer to the associated composite G-operad map under Proposition 3.11 as $\alpha_{\mathcal{U}}\colon \mathbb{P}^{\otimes}_{\mathcal{U}} \to \mathbb{P}^{\otimes}_V \to \mathbb{E}^{\otimes}_V$. ◁

Remark 3.19. It is unclear to the author how to interpret the construction called for in [Har, Eqn. (5.8)] directly; to define a topological operad map $\mathbb{P}^t_V \to \mathbb{E}^t_V$ lying over the unique map $f\colon \mathrm{Disk}_*(V) \to *$ would require choices of points $\chi_U \in \mathrm{Emb}^{H,\mathrm{Rect}}(S \cdot P^V, P^V)$ for each S-indexed U, which must be strictly compatible with composition and *strictly agree* along rectilinear isotopy equivalences $U' \hookrightarrow U$. Instead, our construction appears closely related to Lurie's map ψ used in [HA, Rmk. 5.4.5.8]. ◁

Our central observation about $\alpha_{\mathcal{U}}$ is the following.

Observation 3.20. Suppose $P^V \in \mathcal{U}$. Under Proposition 3.15, α_U induces on active arrow categories the functor $\mathcal{U}^H_S \to \mathrm{Emb}^{H,\mathrm{Rect}}\left(S \cdot P^V, U\right)$ whose restriction to $U' \hookrightarrow U$ is equivalent to the inclusion of U'-subordinate $\mathcal{U}$-admissible embeddings:

$$\begin{array}{ccccc}
\mathbb{P}^{\mathrm{act}}_{\mathcal{U},S//U} & \simeq & \mathcal{U}_S & \hookleftarrow & \mathrm{Emb}^{H,\mathrm{Rect},\mathcal{U}}_{\mathrm{id}_S}(S \cdot P^V, U') \\
\downarrow & & \downarrow & \swarrow_{\iota} & \\
\widetilde{\mathbb{E}}^{\mathrm{act}}_{V,S//\alpha(U)} & \simeq & \mathrm{Emb}^{H,\mathrm{Rect}}\left(S \cdot P^V, U\right) & &
\end{array}$$

In particular, if $\mathcal{U}$ is equidiameterizable, then under Proposition 2.26 we find that $\alpha_{\mathcal{U}}$ induces an equivalence

$$\alpha\colon B\mathbb{P}^{\mathrm{act}}_{\mathcal{U},S//U} \xrightarrow{\sim} B\widetilde{\mathbb{E}}^{\mathrm{act}}_{V,S//\alpha(U)}.$$ ◁

We have already observed the difficult step of the following proposition.

Proposition 3.21. *Given an equidiameterizable V-polydisk G-cover $\mathcal{U} \subset \underline{\mathrm{RPDisk}}(V)$ such that $P^V \in \mathcal{U}^G$,*

$$\alpha_U\colon \mathbb{P}^{\otimes}_{\mathcal{U}} \to \mathbb{E}^{\otimes}_V$$

is a weak approximation. In particular, it yields natural equivalences

$$\mathrm{Alg}_{\mathbb{E}_V}(\mathcal{C}) \xrightarrow{\sim} \mathrm{Alg}^{\mathrm{lc}}_{\mathbb{P}_V}(\mathcal{C}) \xrightarrow{\sim} \mathrm{Alg}^{\mathrm{lc}}_{\mathbb{P}_{\mathcal{U}}}(\mathcal{C}).$$

Proof. The fact that $P^V \in \mathcal{U}$ implies that $U\mathbb{P}_{\mathcal{U}}$ has a terminal G-object, giving Condition (WA-b). Moreover, applying Proposition 1.56 for $\mathcal{C} = *_G$, we find that Observation 3.20 implies Condition (WA-a). □

Corollary 3.22. *The maps $\mathbb{P}^{\otimes}_{V|W} \to \mathbb{E}^{\otimes}_{V\oplus W}$ and $\mathbb{P}^{\otimes}_{V\oplus W} \to \mathbb{E}^{\otimes}_{V\oplus W}$ are weak approximations.*

Proof. These each satisfy the conditions of Proposition 3.21 by Proposition 2.28. □

3.1.4. *The proof of the additivity theorem.* We begin with fine prefactorization algebra additivity.

Lemma 3.23. *Suppose $\mathcal{U}_V$ and $\mathcal{U}_W$ are V- and W-polydisk G-precovers. There exists a strong approximation $\varphi_{\mathbb{P}}$ making the following diagram strictly commute.*

$$\begin{array}{ccccc}
\mathbb{P}^t_{\mathcal{U}_V} \wr \mathbb{P}^t_{\mathcal{U}_W} & \xrightarrow{\widetilde{\alpha}\wr\widetilde{\alpha}} & \widetilde{\mathbb{E}}^t_V \wr \widetilde{\mathbb{E}}^t_W & \xleftarrow{\phi\wr\phi} & \mathbb{E}^t_V \wr \mathbb{E}^t_W \\
\downarrow{\scriptstyle \varphi_{\mathbb{P}}} & & \downarrow{\scriptstyle \widetilde{\varphi}_{\mathbb{E}}} & & \downarrow{\scriptstyle \varphi_{\mathbb{E}}} \\
\mathbb{P}^t_{\mathcal{U}_V \mid \mathcal{U}_W} & \xrightarrow{\widetilde{\alpha}} & \widetilde{\mathbb{E}}^t_{V\oplus W,\omega(V\oplus W)} & \xleftarrow{\phi} & \mathbb{E}^t_{V\oplus W,\omega(V\oplus W)}
\end{array}$$

Proof. In general, we have

$$U\left(\mathbb{P}^{\otimes}_{\mathcal{U}_V} \wr \mathbb{P}^{\otimes}_{\mathcal{U}_W}\right) \simeq \mathcal{U}_V \times_{\mathrm{Ob}\,\mathbb{F}_g} \coprod_T \mathcal{U}_W^T$$

so we may choose the restriction of the map $\varphi_{\underline{\mathrm{RPDisk}}}$ of Proposition 2.20 to be $U\varphi_{\mathbb{P}}$.

To see that this underlies an (essentially unique) strong approximation $\varphi_{\mathbb{P}}$ under Proposition 1.62 we're left with checking the dashed equivalence of the following conditions

$$\begin{array}{ccc}
\mathrm{Mul}^g_{\mathbb{P}_{\mathcal{U}_V \mid \mathcal{U}_W}}\left((\mathbf{U}_V;\mathbf{U}_W);\left(\mathbf{U}'_V;U_W\right)\right) = * & \Longleftarrow\!\!=\!\!\Rightarrow & \mathrm{Mul}^g_{\mathbb{P}_{\mathcal{U}_V}}\left(\mathbf{U}_V;\mathbf{U}'_V\right) \times \mathbb{P}_{\mathcal{U}_W}\left(\mathbf{U}_W;U'_W\right) = * \\
\Updownarrow & & \Updownarrow \\
\varphi(\mathbf{U}_V;\mathbf{U}_W) \in \iota^*(\mathcal{U}_V \mid \mathcal{U}_W) & \Longleftrightarrow & \mathbf{U}_V \in \iota^*\mathcal{U}_V \text{ and } \mathbf{U}_W \in \iota^*\mathcal{U}_W
\end{array}$$

Now, the vertical equivalences follow simply by definition and the bottom horizontal by the fact that φ is an equivalence (see Proposition 2.20), so the desired implication follows by two out of three.

After Variant 3.10, to see that the diagram strictly commutes, we only need to check the lefthand square commutes. Now, the involved maps commute commute on U by fiat, so it suffices to check that the induced square on structure spaces commutes; this follows by construction, as $\varphi_{\underline{\mathrm{RPDisk}}}$ is the direct image under the map φ defining $\widetilde{\varphi}_{\mathbb{E}}$. □

We're now ready to conclude homotopical equivariant Dunn additivity.

Proof of Theorem A. As telegraphed in the introduction, it suffices to verify that the commutative square

$$\begin{array}{ccc}
\mathbb{P}^{\otimes}_V \wr \mathbb{P}^{\otimes}_W & \xrightarrow{\widetilde{\varphi}_{\mathbb{P}}} & \mathbb{P}^{\otimes}_{V\mid W} \\
{\scriptstyle \alpha_V\wr\alpha_W}\downarrow & & \downarrow{\scriptstyle \alpha_{V\mid W}} \\
\mathbb{E}^{\otimes}_V \wr \mathbb{E}^{\otimes}_W & \xrightarrow{\varphi_{\mathbb{E}}} & \mathbb{E}^{\otimes}_{V\oplus W}
\end{array}$$

satisfies the conditions of our bookkeeping theorem, Theorem 1.1. Indeed, Condition (BK-a) is Proposition 3.21, Condition (BK-b) is Lemma 3.23, and Condition (BK-c) follows simply by noting that all of the G-operads involved in this condition have one color. □

3.2. **Additivity with tangential structures.** Let $\underline{\mathrm{Mfld}}^{\sqcup}_G$ be the G-symmetric monoidal ∞-category presented by $\mathbf{Mfld}^{\sqcup}_G$, as recalled in Recollection 3.9. We refer to the full G-symmetric monoidal subcategory of n-dimensional G-manifolds as $\underline{\mathrm{Mfld}}^{n,\sqcup}_G \subset \underline{\mathrm{Mfld}}^{\sqcup}_G$ and the tangent classifier G-functor as $\tau\colon \underline{\mathrm{Mfld}}^n_G \to \underline{\mathcal{S}}_{G,/B_GO(n)}$.

Remark 3.24. [Hor19, § 3.5] is fraught with the issue that they appeared to claim two mapping-in properties for cocartesian G-symmetric monoidal categories–one for algebras and another for G-symmetric monoidal functors–years before any such results were shown. In particular, [Hor19, Lem. 3.5.1] was not proved there, and remained a gap in that paper and subsequent articles until [Ste25d].

We claim that Proposition 1.25 fills this gap, and in particular exhibits contractible spaces of G-symmetric monoidal structure on postcomposition between cocartesian slice G-symmetric monoidal ∞-categories and on the tangent classifier $\tau\colon \underline{\mathrm{Mfld}}^n_G \to \underline{\mathcal{S}}_{G,/B_GO(n)}$. ◁

Given a map $\Theta\colon B_\Theta \longrightarrow B_G O(n)$, we define the *$G$-symmetric monoidal ∞-category of Θ-framed G-manifolds* to be

$$\begin{array}{ccc} \underline{\mathrm{Mfld}}_G^{\Theta,\sqcup} & \longrightarrow & \underline{\mathrm{Mfld}}_G^{n,\sqcup} \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \tau} \\ \underline{\mathcal{S}}_{G,/B_\Theta}^{\sqcup} & \xrightarrow[\Theta_!]{} & \underline{\mathcal{S}}_{G,/B_G O(n)}^{\sqcup} \end{array}$$

3.2.1. *Θ-framed little disk algebras.* The following forms the basis for our convenient model of $\mathbb{E}_\Theta^\otimes$.

Definition 3.25. The *representation topological G-category* is the (nonfull) topological G-subcategory

$$\mathbf{Rep}_G \subset \mathbf{Mfld}_G$$

defined by H-objects the orthogonal H-representations and H-mapping spaces $\mathrm{Map}(V,W) \simeq \mathrm{Emb}^{H,\mathrm{Lin}}(V,W) \subset \mathrm{Emb}^H(V,W)$ the *linear* embeddings. Let $\mathbf{B}_G O(n) \subset \mathbf{Rep}_G$ denote the topological G-subgroupoid of representations of dimension n, and

$$i\colon \mathbf{B}_G O(n) \longrightarrow \mathbf{Mfld}_G$$

the composite inclusion. We write $\mathbf{Emb}_G^n$ for its topological essential image. ◁

Proposition 3.26. *i is homotopy fully-faithful; in particular, $\underline{\mathrm{Emb}}_G^n := N^{hc}\mathbf{Emb}_G^n$ is a classifying G-space for n-dimensional equivariant vector bundles.*

Proof. Note that $\mathbf{Emb}_G^{n,H}$ is a disjoint union of one-object topological categories $\mathbf{B}\mathrm{Emb}^H\left(D^V, D^V\right)$ with endomorphisms $\mathrm{Emb}^H\left(D^V, D^V\right)$. Moreover, the induced topological functor $i^H\colon \mathbf{B}_G O(n)^H \to \mathbf{Mfld}_G$ is a disjoint union of the functors

$$\mathbf{B}O^H(V) \longrightarrow \mathbf{B}\mathrm{Emb}^H\left(D^V, D^V\right)$$

classifying the inclusions $O^H(V) \hookrightarrow \mathrm{Emb}^H\left(D^V, D^V\right)$. By Proposition 2.30, this is classified by an equivalence, and the proposition follows. □

Definition 3.27. We define the topological wide G-suboperad

$$\mathbb{E}_{\mathrm{id}_n}^t \subset \left(\mathbf{Emb}_G^n\right)^\sqcup$$

as having operations the topological subspace of embeddings

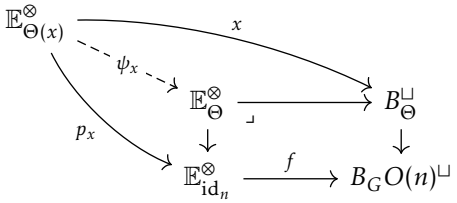
$$\mathrm{Emb}^H\left(S\cdot D^V, D^V\right) \subset \prod_{[H/K]\subset S} \mathrm{Emb}^K\left(D^V, D^V\right) \subset \prod_{[H/K]\subset S} C^{\infty,K}\left(D^V, D^V\right) = C^{\infty,H}\left(S\cdot D^V, D^V\right).$$

We write $\mathbb{E}_{\mathrm{id}_n}^\otimes := N^\otimes \mathbb{E}_{\mathrm{id}_n}^t$, and more generally, given a map $\Theta\colon B_\Theta \to B_G O(n)$, we define the pullback G-operad

$$\begin{array}{ccc} \mathbb{E}_\Theta^\otimes & \longrightarrow & B_\Theta^\sqcup \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{E}_{\mathrm{id}_n}^\otimes & \xrightarrow{f} & B_G O(n)^\sqcup \end{array}$$

where f classifies $U\mathbb{E}_{\mathrm{id}_n}^\otimes \simeq \underline{\mathrm{Emb}}_n^G \simeq B_G O(n)$. We refer to $\mathbb{E}_\Theta$ as the *Θ-framed little disk G-operad.* ◁

Construction 3.28. Given an orbit $[x] \in B_\Theta$, we have a diagram



in which p_x is modelled by the inclusion $\widetilde{\mathbb{E}}_{\Theta(x)}^t \subset \mathbb{E}_{\mathrm{id}_n}^t$ of Variant 3.10 and x classifies the inclusion $[x] \hookrightarrow B_\Theta$. In particular, this yields a reduction $\psi_x^{\mathrm{red}}\colon \mathbb{E}_{\Theta(x)}^\otimes \to \mathrm{End}_x^{\mathrm{red}} \mathbb{E}_\Theta^\otimes$. ◁

Proposition 3.29. *Given a map $\Theta\colon B_\Theta \to B_G O(n)$ and an orbit $[x] \subset B_\Theta$, we have an equivalence*

$$\psi_x^{\mathrm{red}}\colon \mathbb{E}^{\otimes}_{\Theta(x)} \xrightarrow{\sim} \mathrm{End}_x^{\mathrm{red}}\, \mathbb{E}^{\otimes}_{\Theta}$$

Proof. Explicitly, ψ_x^{red} is the coassembly map for the lefthand diagram

$$\begin{array}{ccccc} \mathbb{E}^{\otimes}_{\Theta(x)} & \longrightarrow & \mathbb{E}^{\otimes}_{\Theta} & \longrightarrow & \mathbb{E}^{\otimes}_{\mathrm{id}_n} \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ \mathrm{Comm}^{\otimes}_G & \longrightarrow & B^{\sqcup}_{\Theta} & \longrightarrow & B_G O(n)^{\sqcup} \end{array}$$

so by [HTT, Lem. 4.4.2.1] it suffices to prove that the outer square is cartesian, i.e. we may assume $\Theta = \mathrm{id}_n$. Now, ψ is a map of one-color G-operads so it suffices to prove that it is an equivalence on S-ary structure spaces for each $S \in \mathbb{F}_H$; this is the coassembly map for the outer square of the following diagram.

$$\begin{array}{ccc} \mathrm{Emb}^{G,\mathrm{Rect}}\left(S\cdot P^V, P^V\right) & \xrightarrow{\iota} & \mathrm{Emb}^{G}\left(S\cdot P^V, P^V\right) \\ \downarrow & & \downarrow \\ \prod_{[G/H]\subset S} \mathrm{Emb}^{H,\mathrm{Rect}}\left(S\cdot P^V, P^V\right) & \xrightarrow{\iota} & \prod_{[G/H]\subset S} \mathrm{Emb}^{G}\left(P^V, P^V\right) \\ \wr & & \| \\ * & \longrightarrow & \prod_{[G/H]\subset S} \mathrm{Emb}^{G}\left(P^V, P^V\right) \end{array}$$

Indeed, the bottom square is (homotopy)-cartesian as its vertical arrows are equivalences and the upper by Corollary 2.40, so [HTT, Lem. 4.4.2.1] implies the claim. □

Corollary 3.30. *Pullback along the maps $\psi_x\colon \mathbb{E}^{\otimes}_{\Theta(x)} \to \mathrm{Res}^G_{\mathrm{stab}(x)} \mathbb{E}^{\otimes}_{\Theta}$ yields a natural G-limit presentation*

$$\underline{\mathrm{Alg}}_{\mathbb{E}_\Theta}(\mathcal{C}) \simeq \underline{\lim}_{x\in B_\Theta} \underline{\mathrm{Alg}}_{\mathbb{E}_{\Theta(x)}}(\mathcal{C}).$$

Proof. The disintegration and assembly procedure of Proposition 1.31 provides a natural G-colimit presentation

$$\mathbb{E}^{\otimes}_{\Theta} \simeq \underline{\mathrm{colim}}_{x\in B_\Theta} \mathbb{E}^{\otimes}_{\Theta(x)};$$

since right G-adjoints turn G-colimits into G-limits, this yields a G-limit presentation of G-operads

$$\underline{\mathrm{Alg}}^{\otimes}_{\mathbb{E}_\Theta}(\mathcal{C}) \simeq \underline{\lim}_{x\in B_\Theta} \underline{\mathrm{Alg}}^{\otimes}_{\mathbb{E}_{\Theta(x)}}(\mathcal{C});$$

the result follows by noting that $U\colon \underline{\mathrm{Op}}_G \to \underline{\mathrm{Cat}}_G$ is a right G-adjoint (hence it preserves G-limits) and it takes $\underline{\mathrm{Alg}}^{\otimes}_{\mathcal{O}}(-)$ naturally to $\underline{\mathrm{Alg}}_{\mathcal{O}}(-)$. □

3.2.2. *Additivity with tangential structure.* We begin to prove additivity by constructing a map.

Proposition 3.31. *There exists a map of G-operads $\widetilde{\varphi}\colon \mathbb{E}^{\otimes}_{\mathrm{id}_k} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_{\mathrm{id}_\ell} \to \mathbb{E}^{\otimes}_{\mathrm{id}_{k+\ell}}$ such that, for all orbits*

$$(V,W)\colon [G/H]\times[G/K] \to B_G O(k)\times B_G O(k'),$$

the associated diagram commutes

$$\begin{array}{ccccccc} \mathbb{E}^{\otimes}_V \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_W & \hookrightarrow & \mathbb{E}^{\otimes}_{\mathrm{id}_k} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_{\mathrm{id}_{k'}} & \hookrightarrow & B_G O(k)^{G-\sqcup} \overset{\mathrm{BV}}{\otimes} B_G O(k')^{G-\sqcup} & \simeq & (B_G O(k)\times B_G O(k'))^{G-\sqcup} \\ \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \widetilde{\varphi}} & & \downarrow & & \downarrow{\scriptstyle \oplus^{\sqcup}} \\ \mathbb{E}^{\otimes}_{V\oplus W} & \hookrightarrow & \mathbb{E}^{\otimes}_{\mathrm{id}_{k+k'}} & \hookrightarrow & B_G O(k+k')^{G-\sqcup} & = & B_G O(k+k')^{G-\sqcup} \end{array}$$

where the middle horizontal arrows classify the equivalences of Proposition 3.26, the top right equivalence is Corollary 1.27, and the left horizontal arrows are induced by $\psi_{(-)}$.

Proof. Since all elements of the bottom row are G-operads, it suffices to construct a commutative diagram of topolgoical G-preoperads rewrite heavily

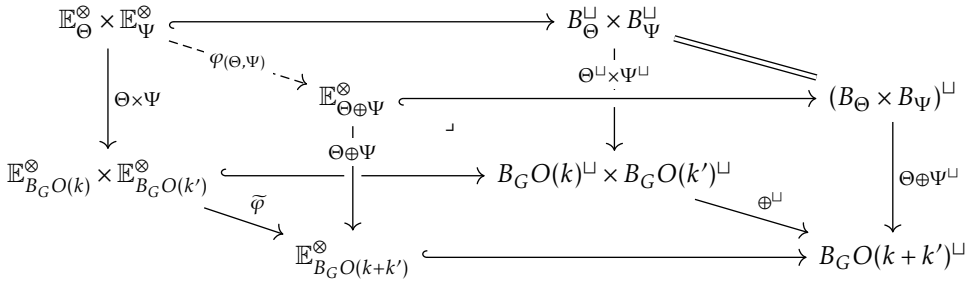
$$\begin{array}{ccccc}
\mathbb{E}^{\otimes}_{\Theta(x)} \times \mathbb{E}^{\otimes}_{\Psi(y)} & \longrightarrow & \mathbb{E}^{\otimes}_{B_G O(k)} \times \mathbb{E}^{\otimes}_{B_G O(k')} & \longrightarrow & \left(B_G \mathrm{Emb}^{\mathrm{Aff}}_{k}\right)^{G-\sqcup} \times \left(B_G \mathrm{Emb}^{\mathrm{Aff}}_{\ell}\right)^{G-\sqcup} \\
\downarrow{\scriptstyle \varphi} & & \vdots{\scriptstyle \widetilde{\varphi}} & & \downarrow \\
\mathbb{E}^{\otimes}_{\Theta(x)\oplus\Psi(y)} & \longrightarrow & \mathbb{E}^{\otimes}_{B_G O(k+k')} & \longrightarrow & \left(B_G \mathrm{Emb}^{\mathrm{Aff}}_{k+\ell}\right)^{G-\sqcup}
\end{array}$$

Now, note that the bottom left horizontal arrow has action maps classified by the subspace inclusions

$$\mathrm{Emb}^{H}\left(\coprod_{U}^{S} R_U, R\right) \hookrightarrow \prod_{U}^{S} \mathrm{Emb}^{H}(R_U, R)$$

and is essentially injective; moreover, the top arrow is an external product of two such arrows. In particular, we can replace the desired right square (up to equivalence) with one whose horizontal arrows are point-set monic in the sense of Proposition 1.42 and verify the conditions there to consturct $\widetilde{\varphi}$ such that the right squhare commutes. Since the right horizontal arrows are U-equivalences by definition, condition (a) is immediate. Moreover, condition (b) follows by unwinding definitions, as does commutativity of the outer diagram;commutativity of the left diagram then follows by the fact that point-set monics are monomorphisms in $\mathrm{Cat}_{\mathrm{Top}}$. □

Construction 3.32. Given $\Theta\colon B_\Theta \to B_G O(k)$ and $\Psi\colon B_\Psi \to B_G O(\ell)$ a pair of G-tangential structures, there is a *Dunn map* of G-preoperads so that the following diagram commutes.



In fact, the bottom and right face form a G-functor

$$\begin{aligned}
\underline{\mathcal{S}}_{G,/B_G O(k)} \times \underline{\mathcal{S}}_{G,/B_G O(k')} \longrightarrow & \underline{\mathrm{Fun}}_G\left(\mathrm{Infl}^G_e \Delta^1 \times \Delta^1, \underline{\mathrm{PreOp}}_G\right) \times_{\underline{\mathrm{Fun}}\left(\mathrm{Infl}^G_e \Delta^1, \underline{\mathrm{PreOp}}_G\right)} \underline{\mathrm{Fun}}_G\left(\mathrm{Infl}^G_e \Delta^1 \times \Delta^1, \underline{\mathrm{PreOp}}_G\right) \\
\simeq & \underline{\mathrm{Fun}}_G\left(\mathrm{Infl}^G_e \Delta^1 \times \Delta^1 \sqcup_{\Delta^1} \Delta^1 \times \Delta^1, \underline{\mathrm{PreOp}}_G\right)
\end{aligned}$$

and the formation of pullbacks of the bottom and top cospan extends this to a G-functor

$$\underline{\mathcal{S}}_{G,/B_G \mathrm{Top}(k)} \times \underline{\mathcal{S}}_{G,/B_G \mathrm{Top}(k')} \longrightarrow \underline{\mathrm{Fun}}_G\left(\mathrm{Infl}^G_e \left(\Delta^1\right)^3, \underline{\mathrm{PreOp}}_G\right);$$

evaluation at the dashed arrow yields a commutative diagram of G-functors

$$\begin{array}{ccccc}
\underline{\mathcal{S}}_{G,/B_G \mathrm{Top}(k)\times B_G O(k')} & \xleftarrow{\times} & \underline{\mathcal{S}}_{G,/B_G O(k)} \times \underline{\mathcal{S}}_{G,/B_G O(k')} & \xrightarrow{(\mathbb{E}_{(-)},\mathbb{E}_{(-)})} & \underline{\mathrm{PreOp}}_G \times \underline{\mathrm{PreOp}}_G \\
\downarrow{\scriptstyle \mathbb{E}_{(-)}} & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \times} \\
\underline{\mathrm{PreOp}}_G & \xleftarrow[\mathrm{ev}_1]{} & \underline{\mathrm{Fun}}_G\left(\mathrm{Infl}^G_e \Delta^1, \underline{\mathrm{PreOp}}_G\right) & \xrightarrow[\mathrm{ev}_0]{} & \underline{\mathrm{PreOp}}_G
\end{array}$$

In other words, we've constructed a natural transformation of G-functors $\varphi\colon \mathbb{E}^{\otimes}_{\Theta} \times \mathbb{E}^{\otimes}_{\Psi} \to \mathbb{E}^{\otimes}_{\Theta\oplus\Psi}$. ◁

Now, our precise form of Corollary B shows that this is a natural equivalence.

Theorem 3.33. *The natural transformation $\varphi\colon \mathbb{E}^{\otimes}_{\Theta} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_{\Psi} \to \mathbb{E}^{\otimes}_{\Theta\oplus\Psi}$ is an equivalence.*

Proof. It follows by two-out-of-three that φ always induces an equivalence $U\mathbb{E}_X \times U\mathbb{E}_Y \to U\mathbb{E}_{X\times Y}$. Moreover, for all pairs of orbits $(x,y) \subset X \times Y$, Propositions 3.29 and 3.31 and Construction 3.32 constructs an equivalence of arrows

$$\begin{array}{ccc} \mathbb{E}^{\otimes}_{T_x} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_{T_y} & \xrightarrow{\quad\varphi\quad} & \mathbb{E}^{\otimes}_{T_x\oplus T_y} \\ \wr\downarrow & & \wr\downarrow \\ \mathrm{End}^{\mathrm{red}}_x(\mathbb{E}^{\otimes}_X) \overset{\mathrm{BV}}{\otimes} \mathrm{End}^{\mathrm{red}}_y(\mathbb{E}^{\otimes}_Y) & \xrightarrow{\varphi} & \mathrm{End}^{\mathrm{red}}_{(x,y)}(\mathbb{E}^{\otimes}_{X\times Y}) \end{array}$$

hence φ induces an equivalence of reduced endomorphism H-operads over pairs of orbtis, and the result follows by Corollary 1.32. □

3.2.3. *The relationship with manifold topology.* The following proposition follows by unwinding definitions.

Proposition 3.34. *The topological full G-suboperad $\widetilde{\mathbb{E}}^t_{\mathrm{id}_n} \subset \mathbf{Mfld}^{n,\sqcup}_G$ spanned by disks is equivalent to $\mathbb{E}^t_{\mathrm{id}_n}$.*

Now, define the G-symmetric monoidal full subcategory

$$\underline{\mathrm{Disk}}^{\Theta,\sqcup}_G := \{\Theta\text{-framed multidisks}\} \subset \underline{\mathrm{Mfld}}^{\Theta,\sqcup}_G;$$

this participates in a diagram of G-operads

$$\begin{array}{ccccc} \mathbb{E}^{\otimes}_{\Theta} & \longrightarrow & \underline{\mathrm{Disk}}^{\Theta,\sqcup}_G & \hookrightarrow & \underline{\mathrm{Mfld}}^{\Theta,\sqcup}_G \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{E}^{\otimes}_{\mathrm{id}_n} & \hookrightarrow & \underline{\mathrm{Disk}}^{n,\sqcup}_G & \hookrightarrow & \underline{\mathrm{Mfld}}^{n,\sqcup}_G \end{array}$$

under the equivalence between $X^{\sqcup}$ and the full G-suboperad of $\underline{\mathcal{S}}^{\sqcup}_{G,/X}$ spanned by arrows with contractible domain; that is, this observation and Proposition 3.34 witness $\mathbb{E}^{\otimes}_{\Theta}$ as the full G-suboperad of $\underline{\mathrm{Mfld}}^{\Theta,\sqcup}_G$ spanned by Θ-framed disks, which factors through $\underline{\mathrm{Disk}}^{\Theta,\sqcup}_G$. In fact, we find the following.

Corollary 3.35. *The fully faithful inclusion $\mathbb{E}^{\otimes}_{\Theta} \hookrightarrow \underline{\mathrm{Disk}}^{\Theta,\sqcup}_G$ classifies an equivalence $\mathrm{Env}_G\mathbb{E}_{\Theta} \xrightarrow{\sim} \underline{\mathrm{Disk}}^{\Theta,\sqcup}_G$. In particular, Theorem 3.33 yields an equivalence*

$$\underline{\mathrm{Disk}}^{\Theta,\sqcup}_G \square \underline{\mathrm{Disk}}^{\Psi,\sqcup}_G \simeq \underline{\mathrm{Disk}}^{\Theta\oplus\Psi,\sqcup}_G.$$

Proof. Under Proposition 1.33, we're tasked with constructing an equifibration $\underline{\mathrm{Disk}}^{\Theta,\sqcup}_G \to \underline{\mathbb{F}}^{\sqcup}_G$ whose fiber over $*_H$ is the H-equivariant disks; in fact, the underlying homotopy type of an element of $\underline{\mathrm{Disk}}^{\Theta,\sqcup}_G$ is an element of $\underline{\mathbb{F}}^{\sqcup}_G$, so we acquire an essentially unique factorization

$$\begin{array}{ccc} \underline{\mathrm{Disk}}^{\Theta,\sqcup}_G & \longrightarrow & \underline{\mathcal{S}}^{\sqcup}_{G,/B_\Theta} \\ {\scriptstyle \underline{\pi}_0}\downarrow & & \downarrow \\ \underline{\mathbb{F}}^{\sqcup}_G & \hookrightarrow & \underline{\mathcal{S}}^{\sqcup}_G \end{array}$$

We claim that $\underline{\pi}_0$ is an equifibration, i.e. the following squares are cartesian

$$\begin{array}{ccc} \mathrm{Disk}^{\Theta}_K & \longrightarrow & \mathbb{F}_K \\ {\scriptstyle H\times_K(-)}\downarrow & & {\scriptstyle \mathrm{Ind}^H_K}\downarrow \\ \mathrm{Disk}^{\Theta}_H & \longrightarrow & \mathbb{F}_H \end{array} \qquad \begin{array}{ccc} \left(\mathrm{Disk}^{\Theta}_H\right)^2 & \longrightarrow & \mathbb{F}^2_H \\ \sqcup\downarrow & & \sqcup\downarrow \\ \mathrm{Disk}^{\Theta}_H & \longrightarrow & \mathbb{F}_H \end{array}$$

Let $F\left(\mathrm{Disk}^{\Theta}_H\right)^2 \to \underline{\mathbb{F}}^2_H \times_{\mathbb{F}_H} \mathrm{Disk}^{\Theta}_H$ be the coassembly functor, which takes

$$(D,D') \mapsto \left(\left(\underline{\pi}_0 D, \underline{\pi}_0 D'\right), D \sqcup D'\right).$$

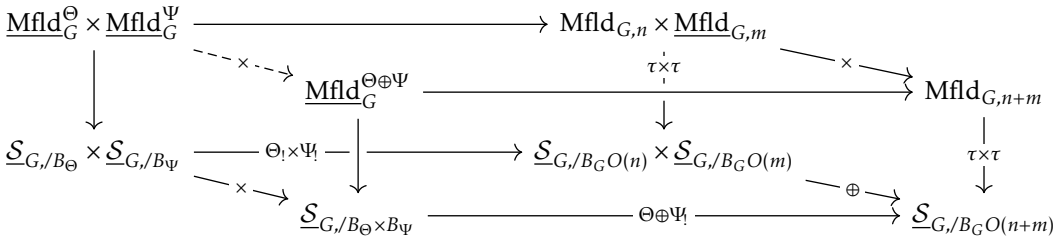
We claim that F is fully faithful and essentially surjective. Essential surjectivity is easy; simply take $D \to S \sqcup S'$ to the pair $(D \times_{S\sqcup S'} S, D' \times_{S\sqcup S'} S')$. Moreover, fully faithfulness follows quickly from Observation 2.9 and commutativity of the defining pullback diagram of Eq. (10) with products.

The other claim is similar; this time, we have functor $\mathrm{Disk}_H^\Theta \to \mathrm{Disk}_K^\Theta \times_{\mathbb{F}_K} \mathbb{F}_H$, noting that every such multidisk has a canonical map $D \to \underline{\pi}_0 D = \mathrm{Ind}_K^H S$, which corresponds canonically with a K-equivariant multidisk under Observation 2.9. □

It is worthwhile to try to understand this construction a bit more directly; we'll do so via *products*.

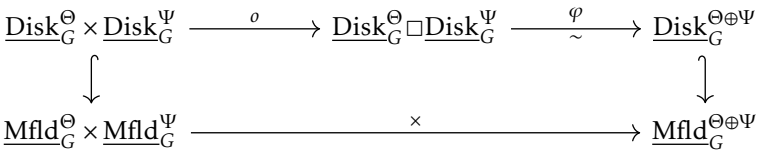
Construction 3.36. The product construction yields a map of coefficient systems of topological categories $\times\colon \mathbf{Mfld}_G^2 \to \mathbf{Mfld}_G$; we refer to the associated map of G-∞-categories as $\times\colon \underline{\mathrm{Mfld}}_G^2 \to \underline{\mathrm{Mfld}}_G$.

Note that this restricts to a G-functor $\times\colon \underline{\mathrm{Mfld}}_G^n \times \underline{\mathrm{Mfld}}_G^m \longrightarrow \underline{\mathrm{Mfld}}_G^{n+m}$. Given G-tangential structures $\Theta\colon B_\Theta \to B_G O(n)$ and $\Psi\colon B_\Psi \to B_G O(m)$, we define the *structured* product operation as the dashed arrow

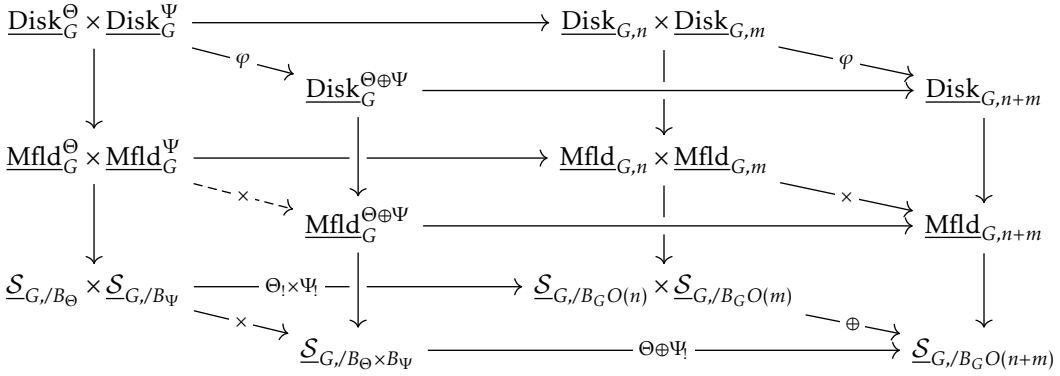


Now, using unit $\varnothing \in \underline{\mathrm{Mfld}}_G^{*_G \simeq B_G O(0)}$, this endows $\underline{\mathrm{Mfld}}^{\mathrm{all}} \coloneqq \coprod_\Theta \underline{\mathrm{Mfld}}_G^\Theta$ with an $\mathbb{A}_2$-monoidal G-category structure. ◁

Proposition 3.37. *The following diagram of G-categories commutes:*



Proof. It suffices to show that the solid subdiagram of the following commutes and apply the universal property for the lower front pullback square.



Now, the outer cubical diagram, the back faces, and the bottom righthand face commute. We're left with the upper righthand face; that is, we are reduced to the case that Θ and Ψ are identity.

In this case, Proposition 3.31 constructs φ as the nerve of an explicit topological G-preoperad map

$$\mathbb{E}^t_{B_G O(n)} \wr \mathbb{E}^t_{B_G O(m)} \to \mathbb{E}^t_{B_G O(n+m)}.$$

Using Proposition 1.61 we're left with making sure that this acts on object and morphisms by taking products; indeed, this follows by unwinding definitions. □

3.3. **Additivity for embedding calculus right-modules.** We now take on Corollary C.

Definition 3.38. Given $\mathcal{O}^\otimes$ a G-operad, the *G-∞-category of right $\mathcal{O}$-modules* is the presheaf G-∞-category

$$\underline{\mathrm{RMod}}_{\mathcal{O}} \coloneqq \underline{\mathrm{Fun}}_G\left(\mathrm{Env}_G(\mathcal{O}), \underline{\mathcal{S}}_G\right).$$ ◁

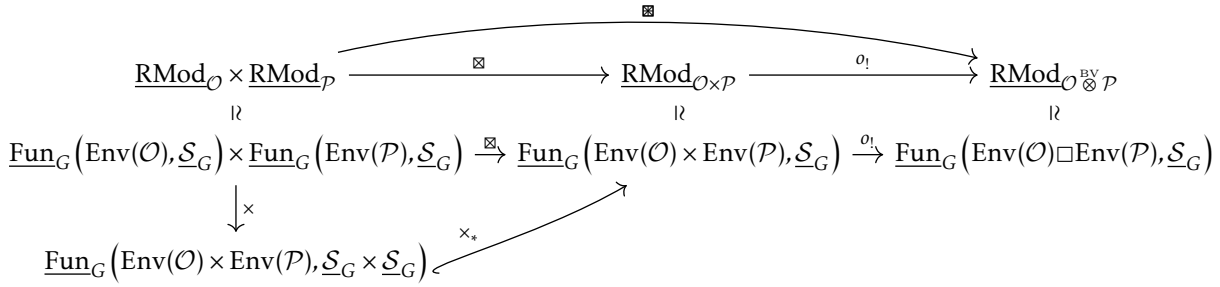
An essential example of this is the candidate *input* to Θ-framed embedding calculus.

Construction 3.39. Under the equivalence $\underline{\mathrm{Disk}}_G^{\Theta,\sqcup} \simeq \mathrm{Env}_G \mathbb{E}_\Theta^\otimes$ of Corollary 3.35, the restricted Yoneda lemma yields a G-functor

$$E^\Theta_{(-)} \colon \underline{\mathrm{Mfld}}_G^\Theta \hookrightarrow \underline{\mathrm{Fun}}_G\left(\underline{\mathrm{Mfld}}_G^\Theta, \underline{\mathcal{S}}_G\right) \xrightarrow{\iota^*} \underline{\mathrm{Fun}}_G\left(\underline{\mathrm{Disk}}_G^\Theta, \underline{\mathcal{S}}_G\right) \simeq \underline{\mathrm{RMod}}_{\mathcal{O}}. \qquad \triangleleft$$

To find an additivity phenomenon, we define the following.

Construction 3.40. Given G-operads $\mathcal{O}^\otimes$ and $\mathcal{P}^\otimes$ together with a G-symmetric monoidal category $\mathcal{C}$, we define the *external Boardman-Vogt tensor product of $\mathcal{O}$- and $\mathcal{P}$-modules* as the composite operation



This is analogous to Dwyer-Hess-Knudsen's *derived lifted Boardman-Vogt tensor product* [DHK18, Def. 3.10]. $\triangleleft$

Theorem 3.41. *There exists a natural equivalence* $E^\Theta_M \boxtimes E^\Psi_M \xrightarrow{\sim} \varphi^* E^{\Theta\oplus\Psi}_{M\times N}$.

We will prove this by hypercodescent, for which we need distributivity.

Lemma 3.42. $\boxtimes$ *preserves G-colimits separately in each variable.*

Proof. $\times \colon \underline{\mathcal{S}}_G \times \underline{\mathcal{S}}_G \to \underline{\mathcal{S}}_G$, does by [NS22, Prop. 3.25] which implies the same for $\boxtimes$ by [Sha23, Prop. 9.17]. To conclude, it suffices to note that $o_!$ preserves G-colimits since it is a G-left adjoint [Sha23, Thm. 10.3]. $\square$

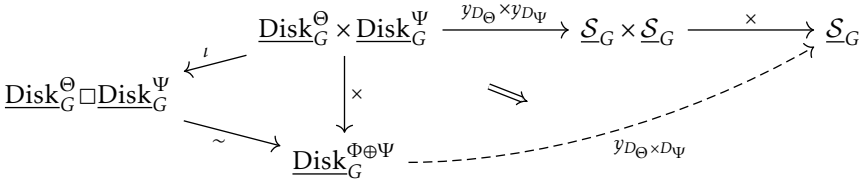
Now, we once again relitigate hypercodescent, this time for $E^\Theta_{(-)}$.

Proposition 3.43 (Filtered Weiss G-hypercodescent of $E^\Theta_{(-)}$)**.** *Given a Θ-framed G-manifold M and a complete Weiss G-cover $\mathcal{U} \subset \underline{\mathrm{Open}}(M)$ such that $\mathcal{U}^H$ is filtered for all $H \subset G$, the assembly map yields an equivalence*

$$\mathrm{colim}_{U\in\mathcal{U}} E^\Theta_U \xrightarrow{\sim} E^\Theta_M.$$

Proof. Recall that G-colimits in functor G-categories are computed pointwise; we proved that the pointwise assembly maps are equivalences in Corollary 2.41 $\square$

Proof of Theorem 3.41. We first prove this in the case that M, N are multi-disks, i.e. we require that the following is a G-left Kan extension diagram



Applying [Sha23, Prop. 10.10] we may check fiberwise that it is a left Kan extension diagram, in which case this follows from $\mathbb{A}_2$-monoidality of the Yoneda embedding (adapt [HA] or apply [Hin23, Prop. 8.4.s]).

To conclude, we bootstrap from the disk case using the Weiss G-hypercodescent and distributivity of Lemma 3.42 and Proposition 3.43 along the disjoint-union closed (hence filtered) complete G-Weiss covers

$\underline{\mathrm{Disk}}^{\Theta}(M) \to M$ and $\underline{\mathrm{Disk}}^{\Theta}(M) \times \underline{\mathrm{Disk}}^{\Psi}(N) \to M \times N$ of Example 2.42:

$$\begin{aligned} E^{\Theta}_M \boxtimes E^{\Psi}_N &\simeq \operatorname*{colim}_{(D_M, D_N) \in \underline{\mathrm{Disk}}^{\Theta}(M) \times \underline{\mathrm{Disk}}^{\Psi}(N)} E^{\Theta}_{D_M} \boxtimes E^{\Psi}_{D_N} \\ &\simeq \operatorname*{colim}_{(D_M, D_N) \in \underline{\mathrm{Disk}}^{\Theta}(M) \times \underline{\mathrm{Disk}}^{\Psi}(N)} E^{\Theta}_{D_M} \boxtimes E^{\Psi}_{D_N} \\ &\simeq \operatorname*{colim}_{(D_M, D_N) \in \underline{\mathrm{Disk}}^{\Theta}(M) \times \underline{\mathrm{Disk}}^{\Psi}(N)} E^{\Theta \oplus \Psi}_{D_M \times D_N} \\ &\simeq E^{\Theta \oplus \Psi}_{M \times N}. \end{aligned}$$

□

3.4. **Additivity with corners.**

3.4.1. *Rectilinear Swiss-cheese embeddings.* With only minor variations, we now generalize the above work to higher-order equivariant swiss cheese operads. First, colors.

Definition 3.44. Given d a natural number, we write $\underline{\mathscr{P}}(d)$ for the constant G-poset with values the power set $\mathscr{P}([d])$ on a set of d-elements. We refer to an element of $\underline{\mathscr{P}}(d)$ as a *depth* $\leq d$ *face datum*, of $\underline{\Sigma}_{\underline{\mathscr{P}}(d)}$ as a *depth* $\leq d$ *face multidatum*, a $\underline{\mathscr{P}}(d)$-profile as a *depth* $\leq d$ *face profile*, and a $\underline{\mathscr{P}}(d)$-multiprofile as a *depth* $\leq d$ *face multi-profile*; when d is clear from context, we will omit the phrase *depth* $\leq d$. ◁

These are face data for neat (multi-)embeddings between the following G-manifolds with corners.

Definition 3.45. Fix $d \in \mathbb{N}$ and $C \subset [d]$. The V*-thickened* C*-orthant polydisk* is the G-submanifold with corners

$$I_{V,D,d} \coloneqq D^V \times [0,1)^C \times (-1,1)^{[d]-C} \subset P^{V \oplus \mathbb{R}^{\oplus D}}.$$

More generally, given a depth $\leq d$ face multidatum $\mathbf{C}$, we define the V*-thickened* $\mathbf{C}$*-orthant multi-polydisk*

$$I_{V,\mathbf{C},d} \simeq \coprod_{H/K \subset \pi\mathbf{C}} \mathrm{Ind}^H_K I_{\mathrm{Res}^G_K V, C_{H/K}, d}. \qquad \triangleleft$$

Note that face mutliprofiles consist of face multidata $(\mathbf{C};\mathbf{D})$ together with a map $f : \pi\mathbf{C} \to \pi\mathbf{D}$ such that, for all $[G/H] \in \pi\mathbf{C}$, we have an inclusion $C_{[G/H]} \subset D_{f[G/H]}$. We visualize $\mathscr{P}([d])$ as the face data of the basic manifold with corners $[0,1)^d$ (or more generally $I_{V,[d],[d]}$) and the inclusions $C_{[G/H]} \subset D_{f([G/H])}$ as indicating that the latter is *deeper* than the former in the face structure.

As in the proof of Theorem A, it will be convenient to begin with an affine model, which we choose as a convenient subspace of rectilinear embedding spaces.

Definition 3.46. Given $(\mathbf{C};\mathbf{D})$ a depth $\leq d$ face multi-profile lying over f, we define the topologial subspace of $(\mathbf{C};\mathbf{D})$-swiss cheese multi-embeddings

$$\mathrm{Emb}^{H,\mathrm{SC},V,d}_f(\mathbf{C};\mathbf{D}) \subset \mathrm{Emb}^{H,\mathrm{Rect}}_{\pi f}\left(\pi\mathbf{C} \cdot P^{V \oplus \mathbb{R}^{\oplus d}}, \pi\mathbf{D} \cdot P^{V \oplus \mathbb{R}^{\oplus d}}\right)$$

as those which restrict componentwise to (neat) orthant polydisk embeddings

$$I_{V,C_{[H/K]},d} \longrightarrow I_{V,C_{[H/K]},d} \subset I_{V,D_{f[H/K]},d} \subset P^{V \oplus \mathbb{R}^{\oplus d}}.$$

In the case that $\mathbf{C}'$ is a single color, f is determined completely by $\mathbf{C}$, so we omit it. ◁

Remark 3.47. The condition of Definition 3.46 can be interpreted concretely as follows:

- The embedding preserves sends points of the boundary $\partial I_{V,C_{[H/K]},d}$ to axis hyperplanes corresponding with their face data, and
- the interior of $I_{V,C_{[H/K]},d}$ is sent to the *positive* orthant (i.e. $2^{-|C|}$th space) cut out by the hyperplanes bounding $I_{V,C_{[H/K]},d}$.

In particular, in the case that $|D| = 1$, disks without boundary land in the open positive half space and disks with boundary send bounding hyperplane to axis hyperplane and interior to open positive half space. This is illustrated in Section 3.4.1. ◁

These can be deformed into standard form wherein only components with the same face data interact:

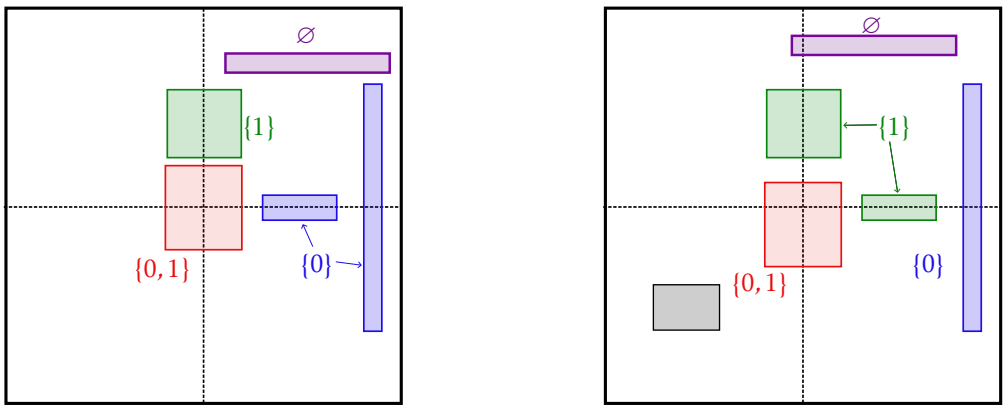


Figure 2. The figure on the left is a depth ≤ 2 0-swiss cheese embedding and the figure on the right exhibits failure of several conditions: the $\varnothing$-marked square is not swiss-cheese embedded as it has empty face data but intersects an axis-hyperplane, the lower $\{1\}$-marked square is not swiss-cheese embedded as its center does not lie on the 1-hyperplane and it intersects the 0-hyperplane, the $\{0,1\}$-marked square is not swiss-cheese embedded as its center does not lie on the 1-hyperplane and does not lie in the nonnegative orthant, and the unmarked gray square is not swiss cheese embedded regardless of its face data, as its center does not lie in the nonnegative orhant.

Definition 3.48. We say that a configuration $c\colon S \hookrightarrow P^V$ is a *standardized C-orthant configuration* if it factors through the standard embedding

$$s_{V,C,C',D}\colon D^V \times \{0\} \times (0,1)^{D-C} \simeq D^V \times \{0\} \times \left(\frac{3}{4},1\right)^{D-C} \hookrightarrow V \times (-1,1)^C \times (0,1)^{D-C}.$$

Note that c is standardized for at most one C; if this holds, we simply say c is *standardized*.

We say that a swiss cheese embedding

$$\iota\colon P^{V\oplus\mathbb{R}^{\oplus D}} \longrightarrow \cdot P^{V\oplus\mathbb{R}^{\oplus D}}$$

is *center-standardized* if its center configuration is standardized, *separated* if it moreover factors through the standard embedding

$$s_{V,C,C',D}\colon D^V \times (-1,1)^C \times (0,1)^{D-C'} \simeq D^V \times \left(-\frac{1}{2},\frac{1}{2}\right)^C \times \left(\frac{1}{2},1\right)^{D-C} \hookrightarrow V \times (-1,1)^{C'} \times (0,1)^{D-C'},$$

and *standardized* if its projected embedding

$$(-1,1)^{[d]-C} \to (-1,1)^{[d]-C}$$

is equidiameter of diameter $\min_{i\in[d]-C} \frac{\mathrm{t}_{V_i}(\iota)}{4}$.

More generally, we say that a swiss cheese multi-embedding $I_{V,\mathbf{C},D} \hookrightarrow I_{V,\mathbf{C}',D}$ is center-separated (resp. separated, standardized) if its restriction to each path orbit is center-separated (separated, standardized). ◁

We set the temporary notation

$$\mathrm{Emb}^{H,\mathrm{SC},\mathrm{Std},V,d}(\mathbf{C};D) \subset \mathrm{Emb}^{H,\mathrm{SC},\mathrm{Sep},V,d}(\mathbf{C};D) \subset \mathrm{Emb}^{H,\mathrm{SC},\mathrm{CSep},V,d}(\mathbf{C};D) \subset \mathrm{Emb}^{H,\mathrm{SC},V,d}(\mathbf{C};D) \tag{11}$$

for the subspaces of standardized embeddings, separated embeddings, and center-separated embeddings.

Proposition 3.49. *Each of the inclusions in Eq. (11) admits a deformation retract.*

Proof. To deform onto center-separated embeddings requires some care: we would like to *linearly translate* all embeddings until they have separated centers, but this will cause intersecting images. Instead, given this center trajectory, let $w\colon \mathrm{Emb}^{H,\mathrm{SC},V,d}(\mathbf{C};D) \to (0,1]$ be defined as the minimum number such that this trajectory applied to $\iota \circ e_{0,w(\iota)}$ preserves embeddings; then, $\iota \circ e_{0,tw(\iota)+(1-t)}$ provides a deforamation retract onto the subspace of embeddings compatible with this trajectory, and then the trajectory onto separated embeddings.

Now, to deform from separated centers from separated embeddings is easy; we simply dilate diameters linearly, stopping the dilation on each component when the component is separated.

Standardization is similar; this time, one notes that a separated swiss cheese embedding is cocentric with shared non-corner diameters with precisely one standardized embedding, and linear interpolation to this provides the desired deformation. □

Note that projection and rescaling yields a map

$$p\colon \mathrm{Emb}^{H,\mathrm{SC},V,d}\left(\coprod_{C\subset D} S_C\cdot C, D\right)\longrightarrow \prod_{C\subset D}\mathrm{Emb}^{H,\mathrm{Rect}}\left(S_C\cdot P^{V\times\mathbb{R}^{\oplus[d]-C}}, P^{V\oplus\mathbb{R}^{\oplus[d]-C}}\right) \tag{12}$$

Now, this admits an inverse map gotten by "filling in" the remaining diameters for each embedding with $\min_{i\leq[d]-C}\frac{t_{V_i}(\iota)}{4}$. That is, we have the following.

Proposition 3.50. *The map p of Eq. (12) is a homeomorphism.*

Corollary 3.51. *Composing centers with the standardization retract furnishes a weak equivalence*

$$\mathfrak{c}\colon \mathrm{Emb}^{H,\mathrm{SC},V,d}\left(\coprod_{C\subset D} S_C\cdot C, D\right)\xrightarrow{\sim}\prod_{C\subset D}\mathrm{Conf}^H_{S_C}\left(V\oplus\mathbb{R}^{[d]-C}\right).$$

Proof. In sight of Propositions 2.26, 3.49 and 3.50, we may simply apply two-out-of-three to the sequence

$$\prod_{C\subset D}\mathrm{Emb}^{H,\mathrm{Rect}}\left(S_C\cdot P^{V\times\mathbb{R}^{\oplus[d]-C}}, P^{V\oplus\mathbb{R}^{\oplus[d]-C}}\right)\hookrightarrow \mathrm{Emb}^{H,\mathrm{SC},V,d}\left(\coprod_{C\subset D} S_C\cdot C, D\right)\xrightarrow{\mathfrak{c}}\prod_{C\subset D}\mathrm{Conf}^H_{S_C}\left(V\oplus\mathbb{R}^{[d]-C}\right).$$ □

Example 3.52. Given a one-depth face profile $(S\cdot C; D)$, an orthant configuration $c\colon S\hookrightarrow V\times\mathbb{R}^{[d]-C}$, and a number $(0,1)$, we define the *equidiameter swiss cheese map* to have $\mathfrak{t}_{V_i}=\varepsilon=\mathfrak{t}_j$ for all i and all $j\in[d]-C$, and $\mathfrak{t}_j=\frac{\varepsilon}{4}$ for $j\in C$. Taking products, this outlines a map

$$\begin{aligned} e\colon \mathbb{R}_{>0}\times\prod_{C\subset D}\mathrm{Conf}^H_{S_C}\left(V\oplus\mathbb{R}^{[d]-C}\right)&\xrightarrow{\Delta}\prod_{C\subset D}\mathbb{R}_{>0}\times\mathrm{Conf}^H_{S_C}\left(V\oplus\mathbb{R}^{[d]-C}\right)\\ &\xrightarrow{e}\prod_{C\subset D}C^\infty\left(S_C\cdot P^V, P^V\right)\\ &\simeq C^\infty\left(\coprod_{C\subset D}S_C\cdot P^V, P^V\right)\end{aligned}$$

In the case that $e_{c,\varepsilon}$ is a standardized swiss cheese embedding, we refer to this as an *equidiameter swiss cheese embedding*. Note that, for a fixed orthant configuration $\left(c_C=\left(x_1^C,\dots,x_{n_C}^C\right)\right)_{C\subset D}$, $e_{c,\varepsilon}$ is an equidiameter swiss cheese embedding if and only if $\varepsilon\leq\varepsilon^{\mathrm{SC},\max}(c_C)$, where

$$\varepsilon^{\mathrm{SC},\max}((c_C))=\min_{C\subset D}\varepsilon^{\max}(c_C)$$ ◁

Construction 3.53. Given $V_\bullet$ a polyrepresentation and $d\in\mathbb{N}$, we define the topological $\pi_0\underline{\mathscr{P}}(d)$-symmetric sequence

$$\mathbb{E}^t_{V_\bullet,d}(\mathbf{C};D):=\mathrm{Emb}^{H,\mathrm{SC},V,d}(\mathbf{C};D).$$

By restriction of Observations 2.8 and 2.9 and Lemma 2.10, the restriction of γ endows this with the structure of a $\pi_0\underline{\mathscr{P}}(d)$-colored genuine topological G-operad, which we give the same name. ◁

We will prove that this doesn't depend on the decomposition in Proposition 3.55. First, we consider prefactorization algebras.

3.4.2. *Swiss cheese prefactorization algebras.* We refer to the images of $(\mathbf{C};\mathbf{D})$-indexed standardized swiss cheese embeddings as

$$\underline{\mathrm{SCDisk}}_{\mathbf{C}}(V,d)_{\mathbf{D}} \subset \underline{\mathrm{RPDisk}}\left(V\oplus\mathbb{R}^d\right);/$$

taking a union over $(f,\mathbf{C})$, we simply write $\underline{\mathrm{SCDisk}}(V,d)_{\mathbf{D}}$. Spread this notation everywhere!

Definition 3.54. A (V,d) *swiss cheese G-precover* is a choice of G-subposets $\mathcal{U}(D)\subset \underline{\mathrm{SCDisk}}^{V,d}(D)$ for all $D\subset[d]$. $\{\mathcal{U}(D)\}$ is called a *complete* (V,d) *swiss cheese G-Weiss cover* if moreover, for all $U\in\mathcal{U}_{\mathbf{C}}(D)$ and subordinate stadardized orthant configurations

$$\iota\colon S\hookrightarrow I_{V,\mathbf{C},d}\subset P^{V\oplus\mathbb{R}^{\oplus d}}\simeq U,$$

there exists some $U'\in\mathcal{U}(D)$ admitting a factorization

$$S\hookrightarrow U'\hookrightarrow U.$$ ◁

$\{\mathcal{U}(D)\}$ is further called *equidiameterizable* if, for all choices $\{S_C\}$, there exists a one-parameter family of maps

$$\varepsilon_s\colon(0,1)\to\mathrm{Map}\left(\prod_{C\subset D}\mathrm{Conf}^H_{S_C}\left(V\times\mathbb{R}^{[d]-C}\right),\mathbb{R}_{>0}\right)$$

such that $\lim_{s\to0}\varepsilon_s=0$ and, for all $\mathcal{U}(D)$-admissible ι, all $s\in(0,1)$, and all $t\in[0,1]$, the linear interpolation

$$h_{s,t}\iota(v):=t\cdot e_{\mathfrak{c}(\iota),\varepsilon_s(\mathfrak{c}(\iota))}(v)+(1-t)\cdot\iota(v)$$

is a $\mathcal{U}(D)$-admissible embedding We refer to such ε as *equidiameter data.*

Proposition 3.55. *If $\mathcal{U}$ is an equidiameterizable (V,d) swiss cheese G-cover, then taking centers yields a commutative diagram of weak equivalences*

$$\begin{array}{ccccc}
B\mathcal{U}^H_{\coprod_{C\subset D}S_C\cdot C;D} & \xrightarrow{\alpha_{\mathcal{U}}} & \mathrm{Emb}_f^{H,\mathrm{SC},V,d,\mathcal{U}}\left(\coprod_{C\subset D}S_C\cdot C;D\right) & \xrightarrow{\mathfrak{c}} & \prod_{C\subset D}\mathrm{Conf}^H_{S_C}(V\times\mathbb{R}^{[D]-C}) \\
\downarrow & & \wr\downarrow & & \| \\
B\underline{\mathrm{SCDisk}}_{\coprod_{C\subset D}S_C\cdot C}(V,d)^H_D & \xrightarrow{\alpha} & \mathrm{Emb}_f^{H,\mathrm{SC},V,d}\left(\coprod_{C\subset D}S_C\cdot C;D\right) & \xrightarrow{\mathfrak{c}} & \prod_{C\subset D}\mathrm{Conf}^H_{S_C}(V\times\mathbb{R}^{[D]-C})
\end{array}$$

Proof. This is analogous to Proposition 2.28. First note that $\underline{\mathrm{SCDisk}}(V,d)$ has equidiameter datum

$$\varepsilon_s((c_C))=s\cdot\varepsilon^{\mathrm{SC},\max}((c_C))$$

so applying two-out-of-three, it suffices to prove the top row consists of weak equivalences.

Exactly as before, the composite

$$\prod_{C\subset D}\mathrm{Conf}^H_{S_C}(V\times\mathbb{R}^{[D]-C})\xrightarrow{(\mathrm{id},\varepsilon)}\prod_{C\subset D}\mathrm{Conf}^H_{S_C}(V\times\mathbb{R}^{[D]-C})\times\mathbb{R}_{>0}\xrightarrow{e}C^\infty\left(\coprod_C\pi S_C\cdot P^V,P^V\right)$$

restricts to a section of $\mathfrak{c}$, and h_t yields a deformation retract onto its image, so $\mathfrak{c}$ is an equivalence.

Again by two-out-of-three, we're left with proving that $\mathfrak{c}\circ\alpha_{\mathcal{U}}$ is an equivalence; this is a staightforward application of Lemma 2.18. □

As before, we acquire invariance under change of decomposition.

Proposition 3.56. *If $V_\bullet$ and $V'_\bullet$ are two polyrepresentations with the same direct sum, then there exists a zigzag of equivalences*

$$\mathbb{E}^t_{V,d}\simeq\mathbb{E}^t_{V'_\bullet,d}.$$

Proof. This is analogous to Corollary 3.5. This time, we let $\mathcal{U}\subset\underline{\mathrm{SCDisk}}(V)$ consist of equidiameter (V,d) swiss cheese embeddings whose associated equidiameter (V',d)-swiss cheese map is an equidiameter swiss cheese embedding, and we define $\mathcal{U}'\subset\underline{\mathrm{SCDisk}}(V')$ symmetrically.

The $\mathcal{U}$-admissible embeddings are closed under composition, and in particular, defines a pair of wide topological G-suboperad inclusions

$$\mathbb{E}^t_{V,d,\mathcal{U}}\hookrightarrow\mathbb{E}^t_{V,d}\qquad\text{and}\qquad\mathbb{E}^t_{V',d,\mathcal{U}'}\hookrightarrow\mathbb{E}^t_{V',d}$$

which are weak equivalences by Proposition 3.55 (using the tautological equidiameter data). Moreover, there exists an obvious equivalence $\mathbb{E}^t_{V,d,\mathcal{U}} \simeq \mathbb{E}^t_{V',d,\mathcal{U}'}$. □

Example 3.57. Given $C \subset C'$ an inclusion of face data, the *maximal swiss cheese embedding from C to C'* is the rescaling embedding

$$\begin{aligned}\mathrm{Res}^G_H P^V \times [0,1)^C \times (-1,1)^{C'-C} \times (-1,1)^{[d]-C'} &\simeq \mathrm{Res}^G_H P^V \times [0,1)^C \times (0,1)^{C'-C} \times (-1,1)^{[d]-C'} \\ &\hookrightarrow \mathrm{Res}^G_H P^V \times [0,1)^{C'} \times (-1,1)^{[d]-C'}.\end{aligned}$$

This yields a (V,d) swiss cheese G-precover $\underline{\mathscr{P}}(d) \subset \underline{\mathrm{SCDisk}}(V,d)$. ◁

We will need to be precise about the claim that this these are maximal. To do so, we define the G-poset

$$\underline{\mathrm{SCDisk}}_*(V,d) \coloneqq \bigcup_{C \subset D} \underline{\mathrm{SCDisk}}_C(V,d)_D.$$

The inclusion of maximal embeddings factors through a poset map $i \colon \underline{\mathscr{P}}(d) \longrightarrow \underline{\mathrm{SCDisk}}_*(V,d)$. The following lemma is obvious, but crucial.

Lemma 3.58. *i is right adjoint to the map $\alpha \colon \underline{\mathrm{SCDisk}}_*(V,d) \to \underline{\mathscr{P}}(d)$ classifying face data.*

Construction 3.59. Given a (V,d) swiss cheese G-cover $\mathcal{U}$, we define the G-subpreoperad $\mathbb{P}^{\otimes}_{\mathcal{U}} \subset \mathcal{U}^{\otimes}$ to have colors $\mathcal{U}_*$ and $f \colon \mathbf{C} \to \mathbf{D}$ multi-operations given by

$$\mathrm{Mul}^f_{\mathbb{P}_{\mathcal{U}}}(U';U) \simeq \begin{cases} * & U' \in \iota^* \mathcal{U}_{\mathbf{C}}; \\ \varnothing & \text{otherwise.} \end{cases}$$

That is, there is an $i \colon U' \to U$-operation if and only if, under the diffeomorphism $\iota \colon U \hookrightarrow I_{V,\mathbf{D},d}$, i corresponds with a $\mathcal{U}$-admissible multi-embedding. We define $\mathbb{P}^{\otimes}_{V,d} \coloneqq \mathbb{P}^{\otimes}_{\underline{\mathrm{SCDisk}}(V,d)}$.

Now, similarly to Variant 3.10, we let $\mathcal{C}^t \subset \mathbf{Mfld}^{\sqcup}_G$ be the topological full G-suboperad spanned by the image of $\underline{\mathrm{SCDisk}}_*(V,d)$. We define the wide topological G-suboperad $\widetilde{\mathbb{E}}^t_{V,d} \subset \mathcal{C}^t$ by operations

$$\mathrm{Emb}^{H,\mathrm{SC},V,D}(U';U) \subset \mathrm{Emb}^H(U',U) \simeq \mathrm{Emb}^H\left(S \cdot P^V, P^V\right)$$

as those embeddings which become swiss cheese embeddings under the standard pre-postcomposition homeomorphism. As before, we acquire a cospan

$$\mathbb{P}^t_{V,d} \xrightarrow{\alpha} \widetilde{\mathbb{E}}^t_{V,d} \xleftarrow{\phi} \mathbb{E}^t_{V,d}.$$ ◁.

Proposition 3.60. *The map $\phi \colon \mathbb{E}^t_{V,d} \to \widetilde{\mathbb{E}}^t_{V,d}$ is an equivalence.*

Proof. By the same proof as Proposition 3.11, we find that $U\phi$ is an equivalence and ϕ induces equivalences on structure spaces. □

Taking nerves, this yields a composite functor $\alpha_{\mathcal{U}} \colon \mathbb{P}^{\otimes}_{\mathcal{U}} \to \mathbb{P}^{\otimes}_{V,d} \to \mathbb{E}^{\otimes}_{V,d}$.

Proposition 3.61. *If $\mathcal{U}$ is an equidiameterizable and contains $\underline{\mathscr{P}}(d)$, then $\alpha_{\mathcal{U}}$ is a weak approximation.*

Proof. As in the proof of Proposition 3.21, we find that Proposition 3.55 implies Condition (WA-a) under Proposition 1.56 for the $\mathcal{T}$-category $\mathcal{C} = \underline{\mathscr{P}}(d)$. Moreover, Condition (WA-b) follows by restricting Lemma 3.58. □

3.4.3. *Decomposable swiss cheese embeddings.* We define the collection of $\mathbf{C}$-*octant decomposable swiss cheese embeddings* as

$$\underline{\mathrm{SCDisk}}^{V|W,d_V|d_W}_{\mathbf{C}}(D) \coloneqq \underline{\mathrm{SCDisk}}^{V,d_V}(\mathbf{C}) \,|\, \underline{\mathrm{SCDisk}}^{W,d_W}_{\mathbf{C}}(D);$$

we define the G-precovers of *decomposable swiss cheese embeddings* as

$$\underline{\mathrm{SCDisk}}^{V|W,d_V|d_W}(D) \coloneqq \bigcup_{\mathbf{C} \to D} \underline{\mathrm{SCDisk}}^{V|W,d_V|d_W}_{\mathbf{C}}(D).$$

The following lemma will occasionally be useful.

Lemma 3.62. *Swiss cheese embeddings are decomposable if and only if their underlying little disk embeddings are decomposable, i.e. we have an equality of G-precovers*

$$\underline{\mathrm{SCDisk}}_D(V \mid W, d_V \mid d_W) = \underline{\mathrm{SCDisk}}_D(V \oplus W, d_V + d_W) \cap \underline{\mathrm{RPDisk}}\left(V \oplus \mathbb{R}^{\oplus d_V} \mid W \oplus \mathbb{R}^{\oplus d_W}\right)$$

Proof. It's clear from construction that the left hand side is contained in the righthand side, so we're tasked with showing that a decomposable little disk embedding $\iota\colon S \cdot P^{V\oplus W\oplus\mathbb{R}^{\oplus d_V+d_W}} \to P^{V\oplus W\oplus\mathbb{R}^{\oplus d_V+d_W}}$ which is standardized siwss cheese has decomposition which is standardized swiss cheese in both the V and W direction; but this follows straightforwardly by uwinding the definitions. □

Proposition 3.63. $\left\{\underline{\mathrm{SCDisk}}^{V|W,d_V|d_W}(D)\right\}$ *is equidiameterizable; hence the map* $\mathbb{P}^{\otimes}_{V|W,d_V|d_W} \to \mathbb{E}^{\otimes}_{V\oplus W,d_V+d_W}$ *is a weak approximation.*

Proof. This time, we claim that the function

$$\varepsilon_s\left(\left(c_{C_V\sqcup C_W}\right)\right) \coloneqq s \cdot \min_{C\subset D} \varepsilon^{V\oplus\mathbb{R}^{[d_V]-C_V}|W\oplus\mathbb{R}^{[d_W]-C_W}}\left(c_{C_V\sqcup C_W}\right)$$

is an equidiameter datum. Indeed, this follows by Lemmas 2.29 and 3.62 and Proposition 2.21. □

3.4.4. *The additivity theorem with corners.* We begin with the relevant maps.

Construction 3.64. Under Proposition 1.64, to construct a map $\varphi\colon \mathbb{E}^t_{V,d_V} \wr \mathbb{E}^t_{W,d_W} \longrightarrow \mathbb{E}^t_{V\oplus W,d_V+d_W}$ lying over the equivalence $\underline{\mathscr{P}}(d_V)\times\underline{\mathscr{P}}(d_W) \simeq \underline{\mathscr{P}}(d_V+d_W)$, we are tasked with constructing maps

$$\varphi\colon \mathrm{Emb}^{H,\mathrm{SC},V,d_V}_g(\mathbf{C};\mathbf{D}) \times \mathrm{Emb}^{H,\mathrm{SC},W,d_W}_f(\mathbf{A};\mathbf{B}) \longrightarrow \mathrm{Emb}^{H,\mathrm{SC},V\oplus W,d_V+d_W}_{g\circ f}(\varphi(\mathbf{C},\mathbf{A});\varphi(\mathbf{D},\mathbf{B}));$$

subject to naturality and restriction-stability conditions; the external product map restricts to a map with this signature and satisfying these conditions. Moreover, as before, this recipe extends to a similar construction non $\widetilde{\mathbb{E}}_{V,d_V}$. ◁

Theorem 3.65. *The induced map* $\varphi_{\mathbb{E}}\colon \mathbb{E}^{\otimes}_{V,d_W} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_{W,d_W} \longrightarrow \mathbb{E}^{\otimes}_{V\oplus W,d_V+d_W}$ *is an equivalence.*

As in the proof of Theorem A, we prove this by first verifying additivity for prefactorization algebras.

Lemma 3.66. *There exists an equivalence* $\varphi_{\mathbb{P}}$ *making the following diagram commute*

$$\begin{array}{ccc} \mathbb{P}^t_{V,d_V} \wr \mathbb{P}^t_{W,d_W} & \xrightarrow{\varphi_{\mathbb{P}}} & \mathbb{P}^t_{V|W,d_V|d_W} \\ \downarrow & & \downarrow \\ \widetilde{\mathbb{E}}^t_{V,d_V} \wr \widetilde{\mathbb{E}}^t_{W,d_W} & \longrightarrow & \widetilde{\mathbb{E}}^t_{V\oplus W,d_V+d_W} \end{array}$$

Proof. This is analogous to Lemma 3.23; once more, we use the restriction of the map $\varphi_{\underline{\mathrm{RPDisk}}}$ of Proposition 2.20 to be $U\varphi_{\mathbb{P}}$, which is an equivalence by restriction of Proposition 2.20. □

Proof of Theorem 3.65. Once more, we apply the bookkeeping theorem, Theorem 1.1; Condition (BK-a) is Proposition 3.63, Condition (BK-b) is Lemma 3.66, and Condition (BK-c) follows essentially by definition. □

3.4.5. *Swiss cheese as unital left modules.* We now incorporate LMod into this picture.

Construction 3.67. Let LMod^t be the topological operad called **LM** in [HA, Def. 4.2.1.1]. Note that this admits a natural topological suboperad inclusion

$$\mathrm{LMod}^t \hookrightarrow \mathbb{E}^t_{0,1},$$

under we acquire a restricted topological Dunn map

$$\psi^t\colon \mathrm{LMod}^t \wr \mathbb{E}^t_0 \hookrightarrow \mathbb{E}^t_{0,1} \wr \mathbb{E}^t_0 \xrightarrow{\sim} \mathbb{E}^t_{0,1}.$$

In particular, this presents a map

$$\psi\colon \mathrm{LMod}^{\otimes} \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_0 \longrightarrow \mathbb{E}_{0,1}.$$ ◁

Proposition 3.68. ψ *is an equivalence; in particular, taking iterated inverses yields an equivalence*

$$\mathbb{E}^{\otimes}_{V,d} \simeq \mathbb{E}^{\otimes}_V \overset{\mathrm{BV}}{\otimes} \mathbb{E}^{\otimes}_{0,d} \simeq \mathbb{E}^{\otimes}_V \overset{\mathrm{BV}}{\otimes} \mathrm{LMod}^{\otimes d}.$$

Proof. Let $\mathrm{LModDisk}_{[1]} \subset \mathrm{SCDisk}(0,1)_{[1]}$ be the precover consisting of swiss cheese embeddings $[0,1) \sqcup \ell \cdot (-1,1) \hookrightarrow [0,1)$. Writing $\mathrm{LModDisk}_{\varnothing} = \mathrm{SCDisk}(0,1)_{\varnothing}$, we acquire a $(0,1)$ swiss cheese precover LModDisk, and we write $\mathbb{P}^{\otimes}_{\mathrm{LMod}} \coloneqq \mathbb{P}^{\otimes}_{\mathrm{LModDisk}}$.

Note that the composite map $\mathbb{P}^{t}_{\mathrm{LMod}} \to \mathbb{P}^{t}_{0,1} \to \mathbb{E}^{t}_{0,1}$ factors through a map $\alpha_{\mathrm{LMod}} \colon \mathbb{P}_{t}^{\mathrm{LMod}} \to \mathrm{LMod}^{t}$, yielding a commutative diagram of G-preoperad maps

$$\begin{array}{ccc} \mathbb{P}^{\otimes}_{\mathrm{LMod}} \wr \mathbb{P}^{\otimes}_{0} & \xrightarrow{\psi_{\mathbb{P}}} & \mathbb{P}^{\otimes}_{0|0,0|1} \\ \downarrow & & \downarrow \\ \mathrm{LMod}^{\otimes} \wr \mathbb{E}^{\otimes}_{0} & \longrightarrow & \mathbb{E}^{\otimes}_{0,1} \end{array}$$

Now, once again using Theorem 1.1, and Proposition 3.63 it suffices to prove that $\psi_{\mathbb{P}}$ is an L_{Op_G}-equivalence and α_{LMod} is a weak approximation, after which the result follows by Theorem 1.1.

To prove that α_{LMod} is a weak approximation, note that this follows by the same the proof as Proposition 3.61; the same likewise follows for $\psi_{\mathbb{P}}$ being an equivalence using the evident restriction of Proposition 2.20. □

3.4.6. *Framed manifolds with boundary.* For the rest of this section, we fix $G = e$ the trivial group.

Construction 3.69. Let $\mathrm{Mfld}_{d,\partial,\sqcup}$ be the symmetric monoidal topological category of smooth d-manifolds with boundary and neat embeddings (as in [KKM25, Rmk. 3.4]), and let

$$B^{\partial}O(d) \subset \mathrm{Disk}^{d,\partial,\sqcup} \subset \mathrm{Mfld}^{d,\partial,\sqcup}$$

be the full subcategory (resp. symmetric monoidal subcategory) spanned by $\mathbb{R}^n$ and $\mathbb{R}^n \times [0,\infty)$. Let

$$\tau \colon \mathrm{Mfld}^{d,\partial,\sqcup} \to \mathrm{Fun}\left(B^{\partial}O(d)^{\mathrm{op}}, \mathcal{S}\right)^{\sqcup}$$

be the (symmetric monoidal) restricted Yoneda functor. Given a right fibration $\Theta^{\partial} \colon B^{\partial}_{\Theta} \to B^{\partial}O(d)$, we define the symmetric monoidal ∞-category of Θ^{∂}-*framed manifolds* as the pullback

$$\begin{array}{ccc} \mathrm{Mfld}^{\Theta^{\partial},\sqcup} & \longrightarrow & \mathrm{Mfld}^{d,\partial,\sqcup} \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Fun}\left(B^{\partial,\mathrm{op}}_{\Theta}, \mathcal{S}\right) & \longrightarrow & \mathrm{Fun}\left(B^{\partial}O(d)^{\mathrm{op}}, \mathcal{S}\right) \end{array}$$

In the case of the trivial right fibration $\Delta^1 \to B^{\partial}O(d)$, we simply write $\mathrm{Mfld}^{d-\mathrm{fr},\partial,\sqcup}$, and we define $\mathrm{Disk}^{\Theta^{\partial},\sqcup}$ similarly. ◁

Proposition 3.70. *For $d \geq 1$, there exist full suboperad inclusions*

$$\mathbb{E}^{\otimes}_{d-1,1} \longrightarrow \mathrm{Disk}^{d-\mathrm{fr},\partial,\sqcup} \longrightarrow \mathrm{Mfld}^{d-\mathrm{fr},\partial,\sqcup}$$

with composite essential image given by $\left\{[0,1) \times \mathbb{R}^{d-1}, \mathbb{R}^{d}\right\}$ and witnessing $\mathrm{Disk}^{d-\mathrm{fr},\partial,\mathrm{fr}}$ as the symmetric monoidal envelope of $\mathbb{E}^{\otimes}_{d-1,1}$. Hence there exists an equivalence

$$\mathrm{Disk}^{d-\mathrm{fr},\partial,\sqcup} \square \mathrm{Disk}^{k-\mathrm{fr},\sqcup} \simeq \mathrm{Disk}^{(d+k)\mathrm{fr},\partial,\sqcup},$$

where $\square$ denotes the tensor product of symmetric monoidal ∞-categories.

Proof. By Proposition 1.33, it suffices to construct an *equifibration* $\pi \colon \mathrm{Disk}^{d-\mathrm{fr},\partial,\mathrm{fr}} \to \mathbb{F}^{\sqcup}$ in the sense of [BHS22] together with an equivalence between $\mathbb{E}^{\otimes}_{d,1}$ and the full suboperad spanned by multidisks mapping to $*$. Now, since $\mathrm{Disk}^{d-\mathrm{fr},\partial,\sqcup}$ is unital (i.e. has initial unit $\varnothing$), it suffices to provide a *functor* $\mathrm{Disk}d\mathrm{fr},\partial \to \mathbb{F}$ and verify that it sends tensor products to coproducts. We choose the homotopy-coherent nerve of the topological functor $\pi_0 \colon \mathbf{Disk}^{\partial,\mathrm{fr}}_{n+1} \to \mathbb{F}$, which obviously satisfies this property.

To show that this is an equifibration, we must verify that the induced square

$$\begin{array}{ccc} \mathrm{Disk}^{d-\mathrm{fr},\partial} \times \mathrm{Disk}^{d-\mathrm{fr},\partial} & \xrightarrow{\sqcup} & \mathrm{Disk}^{d-\mathrm{fr},\partial} \\ \downarrow & & \downarrow \\ \mathbb{F} \times \mathbb{F} & \xrightarrow{\sqcup} & \mathbb{F} \end{array}$$

is cartesian; this follows by an identical argument to Corollary 3.35. Moreover, the indicated full suboperad is spanned by $\left\{[0,1) \times \mathbb{R}^n, \mathbb{R}^{n+1}\right\}$, so we're left with constructing a fully faithful operad map $\mathbb{E}_{n,1}^{\otimes} \to \mathrm{Disk}_{n+1}^{\partial,\mathrm{fr}}$ with this as its essential image. By to do so, we're tasked with constructing equivalences

$$\mathrm{Emb}^{\mathrm{SC}}(I_{n,C,D}, I_{n,C',D}) \to \mathrm{Emb}^{\partial,\mathrm{fr}}(I_{n,C,D}, I_{n,C',D})$$

compatible with composition. To formulate the map, we simply choose framings given by derivatives; now, to show this is an equivalence, we observe that it extends to a diagram

$$\begin{array}{ccc} \mathrm{Emb}^{\mathrm{SC}}(k \cdot \mathbb{R}^{d+1} \sqcup \ell \cdot \mathbb{R}^d \times [0,\infty), \mathbb{R}^d \times [0,\infty)) & \longrightarrow & \mathrm{Emb}^{\partial,\mathrm{fr}}(k \cdot \mathbb{R}^{d+1} \sqcup \ell \cdot \mathbb{R}^d \times [0,\infty), \mathbb{R}^d \times [0,\infty)) \\ \uparrow & & \uparrow \\ \mathrm{Emb}^{\mathrm{Rect}}(k \cdot \mathbb{R}^{d+1}, \mathbb{R}^{d+1}) \times \mathrm{Emb}^{\mathrm{Rect}}(\ell \cdot \mathbb{R}^d, \mathbb{R}^d) & \longrightarrow & \mathrm{Emb}^{\mathrm{fr}}(k \cdot \mathbb{R}^{d+1}, \mathbb{R}^{d+1}) \times \mathrm{Emb}^{\mathrm{fr}}(\ell \cdot \mathbb{R}^d, \mathbb{R}^d) \end{array}$$

whose vertical arrows admit deformation retracts, by Proposition 3.49 and its evident framed analog; the result then follows from Corollary 3.35 and two out of three. □

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